

# SEED STENCILS FOR DIFFUSION THROUGH MATERIAL EDGES TOO STEEP FOR THE GRID

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ABSTRACT. A diffusion coefficient can vary smoothly on the scale of the grid, jump across an interface, or change smoothly over a width  $\delta$  smaller than the node spacing  $h$ . Through such a sub-grid edge, standard high-order stencils that sample the coefficient at the nodes are first order until the grid resolves the edge, with a knee at  $h \approx \delta$ , and the interface-aware stencils built for a jump, applied at the edge's centre, sit on an error floor of order  $\delta$  and leave it once the grid resolves the edge. We keep the nodes and replace the polynomial basis of the stencils that see the edge by *seeds*. The seeds are the profiles of solutions polynomial in time, continued through the edge by ordinary differential equations along the normal or along the edge's own coordinates; in one dimension they are the classical formal powers of the Sturm–Liouville operator, and as  $\delta \rightarrow 0$  they reduce to the interface-aware stencils of a jump. On scattered nodes the resulting RBF-FD stencils are fourth order at every  $\delta$  tested, elliptic and parabolic, with no switch between regimes. We compare them with the coefficient treatments of diffusion practice. The comparison is against coefficient sampling on the same nodes and against our implementation of the cell harmonic and arithmetic means, a widened edge and, in one dimension, the conservative scheme with exact face conductances, on one contrast and one test problem per dimension. Through an edge the grid does not resolve none of them is better than second order; the conservative scheme is exact at the one-dimensional equilibrium and, on the sinusoidal medium at  $\delta = 0$ , ahead of the seeds below 801 nodes, where the seeds' line is pre-asymptotic; on the scattered nodes none improves on sampling by more than a factor of 1.75 in the RMS norm, and the seeds lie 2.3 to 4.7 orders below the best of them over 1250 to 40,000 nodes.

## 1. INTRODUCTION

Heat conduction through a composite material is governed by

$$(1) \quad u_t = \nabla \cdot (\alpha \nabla u) \equiv Lu$$

and, at equilibrium, by  $Lu = 0$ , with a conductivity  $\alpha > 0$  that changes sharply between materials. At the two ends of the range of media the numerical picture is settled. Where  $\alpha$  varies smoothly on the scale of the grid, a high-order stencil samples it at the nodes and keeps its order. Where  $\alpha$

jumps across an interface, the temperature and the flux  $\alpha \partial_n u$  are continuous while the gradient is not, a standard stencil across the interface is first order, and the remedies rebuild the stencils or the elements that the interface cuts so that they satisfy the interface conditions (Section 1.1). One of them, the interface-aware RBF-FD stencils of [Mar16, MF17b] that this paper extends, keeps fourth order on scattered nodes, elliptic and parabolic, with a few rows of nodes placed across each interface. Between the two ends lies a third case: an edge that is smooth, so that no interface condition applies, but changes over a width  $\delta$  smaller than the node spacing  $h$ . Every material edge is smooth on some scale, a model that resolves an edge on a fine grid presents it as sub-grid on a coarser one, and any regularisation of a jump produces one.

A standard stencil through a sub-grid edge samples  $\alpha$  on its two sides, sees a jump, and treats the coefficient as smooth between its nodes, which is the first-order failure of the jump case. It recovers once the grid resolves the edge. In the one-dimensional ramp problem of Section 4.2, the error of the standard fourth-order operator through a 9 : 1 edge of width  $\delta = 0.01$  or 0.0025 falls by 100 to 200 times between  $h = 2\delta$  and  $h = \delta/2$ , where a first-order line falls by four. On scattered nodes in two dimensions (Section 6.2) its elliptic error stays at 0.86 to 0.94 of its error at the jump while  $h \gtrsim 6\delta$ , falls to 0.081 to 0.084 of it at  $h \approx \delta$  for the three widths that reach it, and converges at the fifth order of the bulk stencils below. We call the turn at  $h \approx \delta$  the knee.

The interface-aware stencils do not reach this regime either. Applied with the centre of the edge as the interface and the two sides' coefficients as if  $\delta$  were 0, they converge to the jump problem's solution, which is not the smooth problem's: the resistance  $\int dx/\alpha$  of a tanh edge between conductivities  $a$  and  $b$  falls short of the jump's by  $c\delta$ ,  $c = (a - b) \ln(a/b)/(2ab)$ , so their error sits on a floor of order  $\delta$  while the grid does not resolve the edge. Once it does, the rebuilt rows keep enforcing a kink that the solution lacks, and their error grows: through a 9 : 1 edge with  $\delta = 0.01$  it reaches 0.11 at 1600 nodes in the one-dimensional equilibrium of Section 4.3. On scattered nodes they beat the standard stencils only while  $h \gtrsim 3\delta$ , and by at most a factor of 6. Using them means knowing  $\delta$  and switching them off.

This paper keeps everything about the scheme except the basis of the stencils that see the edge. In place of the monomials a stencil is made exact on *seeds*, the  $t = 0$  profiles of solutions of (1) that are polynomial in time. With a constant  $\alpha$  these profiles are the polynomials, the initial values of the heat polynomials [RW59]. In one dimension, with  $L = \partial_x \alpha \partial_x$  and anchored at a stencil's evaluation point  $x_e$ , they form the chain

$$(2) \quad \phi_0 = 1, \quad \alpha \phi_1' = \alpha(x_e), \quad L\phi_k = k(k-1) \alpha(x_e) \phi_{k-2} \quad (k \geq 2),$$

with  $\phi_k(x_e) = 0$  for  $k \geq 1$  and  $\phi_k'(x_e) = 0$  for  $k \geq 2$ , integrated through the edge as a first-order system of ordinary differential equations that never differentiates  $\alpha$  (Section 3.1). The seed  $\phi_1$  has a constant flux, and  $\phi_2$  is

the profile of a solution whose rate of change  $u_t$  is constant; in general the even seeds carry  $\partial_t^k u = \text{const}$  and the odd ones  $\alpha \partial_x \partial_t^k u = \text{const}$ , which are the interface conditions of a jump. Read across a jump, the chain is the translated polynomial basis of the interface-aware stencils (Section 3.2), so one construction covers the jump, the sub-grid edge and the resolved edge. In one dimension it is classical (Section 1.1). It is also the construction of our companion paper for the wave equation [Mar26], specialised to a single field: the two interface conditions of diffusion, on  $u$  and on its flux, are both conditions on  $u$ , so one chain serves every stencil.

In two dimensions the seeds replace the polynomial augmentation of the interface-aware RBF-FD stencils, and the Gaussian part and the saddle-point solve stay as they were. Where  $\alpha$  varies along the normal only, as across a straight edge, seeds polynomial in the tangential coordinate turn the chain into a triangular system of ordinary differential equations in the normal coordinate, one short march per stencil (Section 5.1). Where the edge is curved or  $\alpha$  varies along it, a profile frozen along the normal is inconsistent at  $O(1)$  through an unresolved edge, and the chain is marched in the curve's own coordinates instead (Section 5.3). A resistive layer thinner than the stencil, the insulating ring of [MF17b], needs its exact width, its own series and seeds of one degree higher that carry the flux (Section 5.4). An imperfect contact costs strong-form schemes a derivative in the jump conditions, as Lombard and Piroux observed for elastic waves [LP03, LP06]; the flux seeds add the functions of one degree higher that carry the flux, and nothing else.

We measure the seeds in one and two dimensions, on elliptic and parabolic problems, against three baselines: the standard stencils with  $\alpha$  sampled at the nodes; the interface-aware stencils of the jump applied at the edge's centre; and the coefficient treatments of diffusion practice, which change the coefficient and keep the scheme: harmonic and arithmetic means of  $\alpha$  over a cell (over a disc on scattered nodes), the edge widened to the grid, and, in one dimension, the conservative scheme with exact face conductances. On constant pieces the one-dimensional equilibrium cannot rank these methods, since its solution has a constant flux and lies in the span of the first two seeds, on which the seeds and the exact face conductances are both exact (Section 3.5); the one-dimensional ranking is the parabolic problem's. In two dimensions the tangential variation of the solution takes it out of that span, and the equilibrium is a real test.

### Contributions.

- (1) The seeds for the diffusion operator on the unchanged scattered nodes of the interface-aware RBF-FD stencils, which they reduce to as  $\delta \rightarrow 0$ : the one-dimensional chain (Section 3), its two-dimensional form for a straight edge in the normal coordinate, for a curved or tangentially varying edge in the curve's own coordinates, and for a resistive layer thinner than the stencil (Section 5). To our knowledge

no earlier scheme builds the basis of stencils on an unchanged scattered node set through a smooth material edge thinner than the node spacing by marching ordinary differential equations; its ingredients (the exact face conductance, the formal powers, bases from local solutions of the equation) are each classical, and in one dimension so is the construction.

- (2) In one dimension (Section 4): the knee of the standard operator, a drop of 100 to 200 times between  $h = 2\delta$  and  $h = \delta/2$  at a 9 : 1 edge and of 770 to 980 times at the 10 : 1 edges of the sinusoidal medium; the jump construction on its floor, the distance of order  $\delta$  between the jump's solution and the smooth one, while  $h \gtrsim 2\delta$  on the two-constant medium, and growing once the grid resolves the edge; and the seeds, fourth order at every  $\delta$  on one line: on the ramp problem of the two-constant medium they lie within 0.1 to 0.3%, 1 to 1.5% and 3 to 5% of the jump's line at  $\delta = 0.0025$ , 0.01 and 0.04.
- (3) On scattered nodes through a flat band (Section 6): the seeds are fourth order at every  $\delta$  tested, elliptic and parabolic, with elliptic fits of 4.23 to 4.48 over 1250 to 40,000 nodes; the elliptic errors at the five widths, the jump included, differ by factors of 1.2 to 2.4 at each node count, and at 40,000 nodes and  $\delta = 0.0025$  the seeds' error is  $7.99 \times 10^{-9}$  against the standard operator's  $8.36 \times 10^{-5}$ . That is no deterioration as  $\delta \rightarrow 0$  on these node sets, not a proven  $\delta$ -uniform constant. The rule is to seed every row whose stencil sees an edge within  $20\delta$ , with no threshold in  $\delta$ . The seeded rows' least diagonal dominance never falls below the jump-aware rows', the seeded matrices' condition estimates and GMRES counts stay bounded as  $\delta \rightarrow 0$ , and their spectra have no eigenvalue in the right half-plane.
- (4) Through the curved band of case 2 the tangential chain is fourth order, within 1.4 to 3.9 times of the flat seeds' RMS error from 5000 nodes at every  $\delta$  (Section 6.5). On the insulating ring thinned at a fixed resistance from  $s = 10^3$  to  $10^{11}$  the seeds are exact on the matched radial profile to rounding ( $1.3 \times 10^{-15}$  to  $2.1 \times 10^{-15}$ ) and converge at fourth order (fits 4.43 to 4.71) at every  $s$ ; through a smooth ring whose width and edges are both below  $h$  they converge at fourth order on the nodes away from it (Section 6.6).
- (5) The comparison with the coefficient treatments (Sections 4.5 and 6.7). The comparison is against coefficient sampling on the same nodes and against our implementation of the cell harmonic and arithmetic means, a widened edge and, in one dimension, the conservative scheme with exact face conductances, on one contrast and one test problem per dimension. Through an edge the grid does not resolve none of them is better than second order; the conservative scheme is exact at the one-dimensional equilibrium and, on the sinusoidal medium at  $\delta = 0$ , ahead of the seeds below

801 nodes, where the seeds' line is pre-asymptotic; on the scattered nodes none improves on sampling by more than a factor of 1.75 in the RMS norm, and the seeds lie 2.3 to 4.7 orders below the best of them over 1250 to 40,000 nodes.

- (6) An open implementation in Python, written with the AI tools disclosed at the end of the paper, with a test for every numerical routine, in which every figure and table of this paper is drawn by a committed script from committed results files (<https://github.com/bradleypmartin/rbf-elliptic-parabolic-interfaces-2026>).

**1.1. Relation to prior work.** In one dimension the construction is classical in every part, and we claim nothing for it. The first seed, the constant-flux function  $\alpha_e \int d\xi/\alpha$ , is over one cell the exact face conductance of Tikhonov and Samarskii's homogeneous schemes [TS62, Sam01] and, for a piecewise-constant coefficient, the harmonic-mean interface conductivity of finite-volume heat conduction [Pat80]; the conservative scheme built on it, exact at the source-free equilibrium, is second order [PXL26]. The whole chain is the representation of Sturm–Liouville solutions by recursive integrals, as power series in the spectral parameter [Kra08, KP10], taken from the anchor; for a constant coefficient its time-polynomial solutions are the heat polynomials [RW59, Wid75]. Three-point schemes of any order for  $(ku)'$  with a piecewise-smooth  $k$ , whose coefficients come from Cauchy problems integrated across each cell, are the exact and truncated difference schemes [SM90, GHMK11]; three-point schemes of any order exist across a jump too [Gar89], one of them built by carrying Taylor data across the step [Suj10]. In two dimensions, local solutions of the equation that satisfy the interface conditions are the basis of the Trefftz difference schemes of FLAME, on irregular stencils too [Tsu06, Tsu10], and of the tailored finite point methods [Hua09, HZ12]; Galerkin methods build their bases from local solves [BCO94, HW97, CGH10, OZ07, MP14] and reach high order through coefficients the mesh does not resolve [Mai21, HP23], and the quasi-Trefftz spaces, one of them for the heat equation, build polynomial local solutions by Taylor expansion [IGMPS25, GPS26], which cannot reach through an edge thinner than the stencil [Mar26]. For a jump, the interface-aware RBF-FD stencils this work generalises [Mar16, MF17b] stand beside the immersed interface method and unfitted finite elements [LL94, LI06, HH02, BCH<sup>+</sup>15], now of fourth and sixth order [FZ20, FHM24], all of which take the jump as given. A resistive layer thinner than the grid is an imperfect interface in the limit [SP74, PP13, Has01], which Section 5.4 takes up. The standing alternative changes the coefficient rather than the stencil: harmonic or arithmetic means over a cell [Pat80, EIL01, GA92], averaging for layering finer than the grid [MDH<sup>+</sup>99, DDH03], or a regularised coefficient [TE03, TE06]. Those methods take the cell or the width from the grid; here the width is a property of the medium, below the node spacing, and the aim is the scheme's full order through it. We found no prior instance of stencils

on an unchanged scattered node set whose basis is continued through a sub-grid smooth edge by ordinary differential equations, nor of such stencils reducing to a jump construction as  $\delta \rightarrow 0$ ; the search is logged in the repository’s `LITERATURE.md`. We ran the cell means, a widened edge and, in one dimension, the conservative scheme with exact face conductances, as our implementation of each on one problem per dimension: through an edge the grid does not resolve none is better than second order, and on the scattered nodes none improves on sampling the coefficient by more than a factor of 1.75 in the root-mean-square error. We claim nothing beyond that measurement.

**Outline.** Section 2 fixes the problems, the discretisations and the interface-aware stencils of a jump that the seeds extend. Section 3 constructs the seeds in one dimension and proves what can be proved about them. Section 4 measures them against the standard operator, the jump construction and the coefficient treatments. Section 5 builds the seeds on scattered nodes for straight, curved and doubly sub-grid features. Section 6 measures them there, elliptic and parabolic, with the matrices they make. Section 7 records the limitations and what we would do next.

## 2. SETTING

*Notation.* The symbols the sections share, fixed here:

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|                                 |                                                                                                                    |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------|
| $n, h$                          | node count and spacing: $h = 2/(n - 1)$ in one dimension and the row spacing $1/\text{round}(0.95\sqrt{n})$ in two |
| $\alpha; \alpha^-, \alpha^+; L$ | conductivity; the pieces on the two sides of an edge; $L = \nabla \cdot (\alpha \nabla \cdot)$                     |
| $\delta; \sigma$                | width of a tanh edge, $\delta = 0$ the jump; the blend weight $\sigma(z) = \frac{1}{2}(1 + \tanh z)$               |
| $s$                             | the thinness of the ring of [MF17b], $1/s$ its width                                                               |
| $x_e; (x', y')$                 | a stencil’s evaluation point (anchor); its local frame, $y'$ normal to the edge                                    |
| $\phi_k$                        | the one-dimensional seeds (2)                                                                                      |
| $D_x, D_y; A$                   | first-derivative matrices; $\text{diag } \alpha$ at the nodes                                                      |
| naive, direct                   | $D_x A D_x (+D_y A D_y)$ ; $\alpha \nabla^2 u + \nabla \alpha \cdot \nabla u$ on one stencil                       |
| jump-aware                      | the interface-aware stencils of Section 2.4                                                                        |
| the $\delta = 0$ construction   | the same, applied at a smooth edge’s centre                                                                        |
| seeds                           | the operator whose rows that see an edge are rebuilt on seeds                                                      |

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**2.1. The problem and its interface conditions.** On a domain  $\Omega$  in one or two dimensions we solve (1) with Dirichlet data on  $\partial\Omega$ , and its equilibrium  $Lu = 0$ , the elliptic problem. The conductivity  $\alpha > 0$  is one smooth function on each side of every interface  $\Gamma$ . Across a jump the temperature and the heat flux are continuous at all times, hence so are all their time derivatives,

and  $\partial_t^k u = L^k u$  turns these into conditions on spatial derivatives,

$$(3) \quad [L^k u] = 0, \quad [\alpha \partial_n L^k u] = 0 \quad \text{on } \Gamma, \quad k = 0, 1, 2, \dots,$$

where  $[\cdot]$  is the difference of the two sides' limits,  $n$  the normal to  $\Gamma$ , and each side's  $L$  is taken with that side's  $\alpha$  [Mar16, MF17b].

*A smooth edge.* With an edge width  $\delta > 0$  the jump at an interface is replaced by a tanh transition between the two pieces,

$$(4) \quad \alpha = (1 - \sigma(d/\delta)) \alpha^- + \sigma(d/\delta) \alpha^+, \quad \sigma(z) = \frac{1}{2}(1 + \tanh z),$$

where  $d$  is the signed distance to the interface,  $x - x_c$  in one dimension and the distance along the normal through the foot point on the curve in two, positive on the side of  $\alpha^+$ . The pieces keep their own variation, and  $\delta = 0$  is the jump. A band between two interfaces is blended twice, from the outside piece into the inside one across the lower interface and back across the upper, so that a band wider than its edges has two independent transitions; Section 5.4 defines the smooth form of a layer thinner than its edges. Beyond  $19\delta$  from its centre an edge is its pieces to within two units in the last place.

*Three regimes.* For  $\delta \gg h$  a stencil sees a smooth coefficient. For  $\delta = 0$  it sees a jump, and (3) applies. For  $0 < \delta \lesssim h$  it sees what looks to it like a jump between two of its nodes, but no interface condition applies, since  $\alpha$  is smooth, and the smooth problem and the jump problem have different solutions. They differ at first order in  $\delta$ : in one dimension the resistance  $\int dx/\alpha$  of a tanh edge between constant pieces  $a$  and  $b$  falls short of the jump's by

$$(5) \quad c\delta, \quad c = \frac{(a-b)\ln(a/b)}{2ab},$$

which is  $8.79\delta$  for the 9 : 1 edge of Section 2.2 and is positive for either ordering of  $a$  and  $b$ .

*Norms and orders.* In one dimension an error is the relative  $\ell^2$  error at the nodes,  $\|e\|_2/\|u\|_2$ , and a rate is  $\log_2$  of the ratio of consecutive errors as the node count doubles. In two dimensions an error is the root-mean-square (RMS) error over all nodes, the Dirichlet nodes included, and an order is the least-squares slope of  $\log e$  against  $\log h$  over the node counts named. A maximum-norm error is named as such wherever one is quoted, and every caption says which norm it shows.

**2.2. One dimension.** The one-dimensional problem is (1) with  $L = \partial_x \alpha \partial_x$  on  $[-1, 1]$  with Dirichlet ends. A grid has  $n$  equispaced nodes with both ends on nodes, so  $h = 2/(n-1)$ . Two media are used, each at  $\delta = 0, 0.04, 0.01$  and  $0.0025$ :

- the *two-constant medium*,  $\alpha = 1/9$  for  $x < 0$  and  $1$  for  $x > 0$ , the single interface of the time-dependent test in the code behind [Mar16], which the dissertation does not report; the edge lies midway between two nodes when  $n$  is even;

- the *sinusoidal medium* of [Mar16, eq. (75)],  $\alpha = 0.1 + 0.4 \sin 2\pi x$  on the layer  $[0, \frac{1}{2}]$  and 1 elsewhere, whose two edges both have the contrast 10 : 1, since the sinusoid vanishes at both; both edges fall on nodes when  $n - 1$  is a multiple of 4.

Each medium carries two problems: the equilibrium with  $u(-1) = 1$  and  $u(1) = 0$ , and the *ramp problem*, which starts from  $u = 0$ , holds  $u(1, t) = 0$ , raises  $u(-1, t)$  from 0 to 1 over  $0 \leq t \leq 1$  by the  $C^\infty$  step  $e^{-1/t}/(e^{-1/t} + e^{-1/(1-t)})$ , and is read at  $t = 2$ . (The original code ramps with a logistic function, whose value and derivatives at  $t = 0$  are small but not zero; the mismatch with the zero start at the corner would cap a fourth-order study.) Section 4.1 gives the reference solutions.

The *naive* operator is  $D_x A D_x$  [Mar16, eq. (76)], with  $D_x$  the fourth-order first-derivative matrix of Fornberg’s weights [For88], one-sided at the ends, and  $A = \text{diag } \alpha(x_j)$ , which gives nine-point rows in the interior. It is the form in which a standard stencil sees a jump at all: a direct discretisation of  $\alpha u_{xx} + \alpha' u_x$  with the coefficients sampled at the nodes is blind to the jumps, and its error on the sinusoidal medium stays at 0.22 as the grid is refined. Time stepping is the fourth-order backward differentiation formula (BD4) with  $\Delta t = h$ , one sparse LU factorisation per operator and the Dirichlet values imposed at every step; its three starting values come from classical Runge–Kutta sub-steps at half the stability limit.

**2.3. Two dimensions.** The two-dimensional problems are the three test cases of [MF17b], (1) on the unit strip  $0 \leq y \leq 1$ , periodic in  $x$  with period 1, with Dirichlet data on  $y = 0$  and  $y = 1$ . The conductivity is one smooth piece in a band between two interfaces and another outside it.

- *Case 1*, a flat band  $0.6 \leq y \leq 0.8$  with  $\alpha = 0.2$  in it and 1 outside. With  $u = \sin 2\pi x$  on  $y = 1$  and  $u = 0$  on  $y = 0$  the equilibrium is  $\sin 2\pi x v(y)$  with  $v$  piecewise exponential, and the parabolic problem is marched to  $t = 0.1$  against the exact solution  $e^t \sin 2\pi x w(y)$  of the same form [MF17b].
- *Case 2*, both interfaces bent to  $y = 0.6 + 0.02 \sin 2\pi x$  and  $y = 0.8 + 0.02 \sin 2\pi x$ ,<sup>1</sup> with  $\alpha = 0.2 + 0.1 \sin 2\pi x \sin 2\pi y$  in the band and the boundary data of case 1.
- *Case 3*, a ring  $0.35 - 1/s \leq r \leq 0.35$  about  $(\frac{1}{2}, \frac{1}{2})$  with  $\alpha = 1/(1.5s) + \sin 2\pi x \sin 2\pi y/(3s)$  in it and 1 outside, around a cooling circle  $r = 0.05$  whose inside is removed, with  $u = \sin 6\pi x$  on  $y = 0$  and  $y = 1$  and  $u = 0$  on the circle. At  $s = 10^3$  this is the insulating ring of [MF17b], 0.001 wide at a contrast of about 1500 : 1; the extremizing sweep of that paper runs  $s$  up to  $10^{11}$ , thinning the ring at a fixed resistance, its width over its conductivity, of  $1.5/(1 + \frac{1}{2} \sin 2\pi x \sin 2\pi y)$ .

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<sup>1</sup>The published texts do not state the curves; these are the ones of the original code.

With  $\delta > 0$  each interface of cases 1 and 2 becomes an edge (4) in the signed normal distance to its curve; the ring's smooth form is Section 5.4's. On case 1 the medium is then the one-dimensional one in  $y$ , and the reference solutions stay separable (Section 6.1).

*Node sets.* A set of  $n$  nodes is built as in [MF17b]. Rows hold  $m = \text{round}(0.95\sqrt{n})$  nodes per unit length, so their spacing is  $h = 1/m$ . Six fixed rows straddle each interface, three on each side at normal distances  $(\frac{1}{2} + j\frac{\sqrt{3}}{2})h$ ,  $j = 0, 1, 2$ , the middle row on each side staggered by half a spacing along the curve, so that the rows cross the interface in a hexagonal pattern; a fixed row lies on each Dirichlet curve; and the remaining nodes are placed by 100 steps of a simulated repulsion from a random start, kept clear of the straddling rows. The ring is thinner than  $h$  at every node count used, so one set of six rows straddles its midline, and both of its interfaces pass between the innermost pair. Unless a spread over node sets is quoted, each node count has one node set, from a fixed random seed.

*RBF-FD.* RBF-FD [FF15a, FF15b] builds each row of a differentiation matrix from a stencil. For a stencil of nodes  $x_1, \dots, x_{N_s}$  about an evaluation node  $x_e$ , the weights  $w$  of a linear operator  $\mathcal{L}$  solve

$$(6) \quad \begin{bmatrix} \Phi & P \\ P^\top & 0 \end{bmatrix} \begin{bmatrix} w \\ w_* \end{bmatrix} = \begin{bmatrix} \mathcal{L} \varphi(\|x - x_i\|)|_{x_e} \\ \mathcal{L} p_j(x)|_{x_e} \end{bmatrix},$$

with  $\Phi_{il} = \varphi(\|x_i - x_l\|)$  for the Gaussian  $\varphi(r) = e^{-(\varepsilon r)^2}$ ,  $P_{ij} = p_j(x_i)$  over the bivariate monomials  $p_j$  up to a degree  $p$  in coordinates scaled by the stencil's radius, and  $w_*$  discarded. The constraint rows make the weights exact on the polynomials, which take over from the Gaussians under refinement [FFBB16, BFFB17]; the shape parameter is  $\varepsilon = 0.4/d$ , with  $d$  the distance from  $x_e$  to its nearest neighbour. Stencils have 42 nodes and degree 5 in the bulk, and 30 nodes and degree 4 within  $3/\sqrt{n}$  of a Dirichlet curve and wherever a 42-node stencil would reach across an interface. Of the latter, the interface-aware construction rebuilds the rows whose 30 nodes cross an interface (Section 2.4); the seeds rebuild the rows whose stencils see an edge (Section 5.2).

*Operators and solvers.* The naive operator is  $D_x A D_x + D_y A D_y$ , the two-dimensional form of [Mar16, eq. (76)], with every node's rows of  $D_x$  and  $D_y$  built on its own stencil, so that it reaches the neighbours of the neighbours. The direct operator applies  $\alpha \nabla^2 u + \nabla \alpha \cdot \nabla u$  at each node through one stencil; it is blind to a jump and serves as the control where the grid resolves the edge. Dirichlet rows are the identity. The elliptic systems are solved directly with SuperLU; the iterative study applies GMRES and BiCGSTAB to the interior system, with and without an incomplete LU factorisation and the diagonal-dominance preconditioner of [Mar16, App. B]. The parabolic problem of case 1 is marched by BD4 with  $\Delta t = h$  from the exact solution's history, 3 to 19 steps to  $t = 0.1$ .

**2.4. Interface-aware stencils for a jump.** The seeds extend the interface-aware stencils of [Mar16, MF17b], whose wave-equation form is in [MFSC15, MF17a]. We summarise them in the form the later sections use; the derivations are in those references.

*One dimension.* Expand  $u$  on each side of an interface  $x_c$  in a polynomial of degree four about  $x_c$ , with the Taylor coefficients of that side's  $\alpha$  entering through multiplication matrices on the coefficient vectors, and impose (3) on  $L^k u$  for  $k = 0, 1, 2$  and on the flux for  $k = 0, 1$ . These five conditions give a square *continuity matrix* per side,  $C^- a^- = C^+ a^+$  on the coefficient vectors  $a^\pm$ . Keeping the monomials on the evaluation node's side and carrying them across with  $(C^+)^{-1} C^-$  gives the *translated basis*: piecewise polynomials, one per monomial, that satisfy the five conditions. The weights of a five-node stencil that straddles  $x_c$  reproduce  $L$  at the evaluation node on this basis, a  $5 \times 5$  solve per stencil, and a stencil that straddles both edges of a thin layer translates twice.

*Two dimensions.* A stencil whose 30 nodes lie in more than one region is given a local frame at the point of each interface nearest its centre,  $x'$  along the tangent and  $y'$  along the normal, in units of the stencil's radius, and the interface is expanded there as  $y' = f(x')$ . The continuity conditions equate, along the interface, the first  $5 - 2k$  coefficients in  $x'$  of  $L^k u$  and the first  $4 - 2k$  of the normal flux  $\alpha(\partial_{y'} - f' \partial_{x'}) L^k u$ , fifteen conditions for degree four, one per bivariate monomial; the curvature enters through  $f$ , in the curve along which the coefficients are read and in the normal derivative. The translated basis replaces the monomials  $p_j$  in (6). The Gaussians are *warped*: in the frame of the evaluation node's side, the normal coordinate of every node is replaced by a piecewise-linear one whose slope changes by the factor  $\alpha^-/\alpha^+$  across each interface, so that a Gaussian's value and its  $\alpha \partial_n$  are both continuous there and the RBF part of the stencil satisfies the flux condition to first order.

This construction, with warped Gaussians, is what we call *jump-aware*. Applied with the centre of a smooth edge as the interface and the pieces' Taylor data as if  $\delta$  were 0, it is the  $\delta = 0$  *construction* of Sections 4.3 and 6.2. It is also the  $\delta \rightarrow 0$  limit of the seeds, which reproduce it at  $\delta = 0$  (Sections 3.2 and 5.2).

**2.5. Scope of the implementation.** The experiments run on a reimplementation of the 2016–2017 methods in Python (NumPy and SciPy, with a test for every numerical routine), written for this study. Every published figure of the one- and two-dimensional heat problems of [Mar16, MF17b] but the cornered interface's has a counterpart in it, and every published convergence line is reproduced at its published rate or better. Where the numbers part company the repository's port notes say so. Two differences bear on this paper: the iterative solvers do not break down on the ring here (BiCGSTAB converges at every node count), and the extremizing sweep does not lose convergence at  $s = 10^{11}$  (every  $s$  is fourth order down to its

reference's own error). The 2016 two-dimensional code is not preserved, so neither difference could be traced to it.

The implementation leaves out the cornered interface of [Mar16, Sec. 5.4.5], which is approximated there by circular segments at second order. Its two-dimensional media are a band between two curves, periodic in  $x$ , so no edge meets the domain's boundary. The bulk stencils are degree five and the stencils that see an edge degree four, seeded or interface-aware. Each case has one contrast: 9 : 1 and 10 : 1 in one dimension, 5 : 1 on cases 1 and 2, and on the ring about 1500 : 1 at  $s = 10^3$ , growing with  $s$  at a fixed resistance. Every smooth edge is a tanh. Section 7 returns to these restrictions.

### 3. SEEDS FOR THE DIFFUSION OPERATOR IN ONE DIMENSION

Fix a stencil of the grid of Section 2.2: five nodes  $x_1 < \dots < x_5$  and an evaluation point  $x_e$ , one of them. The standard weights are exact on the monomials  $(x - x_e)^k$ ,  $k \leq 4$ , and the interface-aware weights of Section 2.4 on the translated basis. This section builds the basis for a smooth edge, shows that the translated basis is its limit as  $\delta \rightarrow 0$ , proves that every stencil solve is nonsingular, and says what the truncation error is. Each ingredient is classical in one dimension (Section 1.1), and the construction is that of our companion paper for the wave equation [Mar26], specialised to the diffusion operator.

**3.1. Seeds as the profiles of solutions polynomial in time.** Look for solutions of  $u_t = Lu$ ,  $L = \partial_x \alpha \partial_x$ , that are polynomial in time,  $u = \sum_{j=0}^J t^j g_j(x)$ . Matching powers of  $t$  gives  $(j+1)g_{j+1} = Lg_j$  and  $Lg_J = 0$ , so the profile at  $t = 0$  satisfies  $L^{J+1}g_0 = 0$ . The heat equation spends one power of  $L$  on each power of  $t$  where the wave equation spends two, but the profiles fill the same nested kernels of  $L$ . With a constant  $\alpha$  they are the polynomials: the time-polynomial solutions of  $u_t = \alpha u_{xx}$  are the heat polynomials

$$v_k(x, t) = \sum_{0 \leq j \leq k/2} \frac{k!}{j!(k-2j)!} x^{k-2j} (\alpha t)^j$$

[RW59, Wid75], whose profiles at  $t = 0$  are the monomials. The seeds are the same profiles for a variable  $\alpha$ , anchored at the evaluation point.

**Definition 3.1** (Seeds). Let  $\alpha > 0$  and  $\alpha_e = \alpha(x_e)$ . The *seeds* at  $x_e$  are the functions  $\phi_0, \phi_1, \dots$  of the chain (2):  $\phi_0 = 1$ ,  $\alpha \phi'_1 = \alpha_e$  and  $L\phi_k = k(k-1)\alpha_e \phi_{k-2}$  for  $k \geq 2$ , with  $\phi_k(x_e) = 0$  for  $k \geq 1$  and  $\phi'_k(x_e) = 0$  for  $k \geq 2$ .

**Lemma 3.2.** *For every  $k$  the function*

$$(7) \quad u_k(x, t) = \sum_{0 \leq j \leq k/2} \frac{k!}{j!(k-2j)!} (\alpha_e t)^j \phi_{k-2j}(x)$$

solves  $u_t = Lu$ , and  $u_k(\cdot, 0) = \phi_k$ . With a constant  $\alpha$ ,  $\phi_k = (x - x_e)^k$  and  $u_k(x, t) = v_k(x - x_e, t)$ .

*Proof.* Write  $c_j = k!/(j!(k-2j)!)$ . By the chain,  $Lu_k = \sum_j c_j (k-2j)(k-2j-1) \alpha_e^{j+1} t^j \phi_{k-2j-2}$  and  $\partial_t u_k = \sum_j (j+1) c_{j+1} \alpha_e^{j+1} t^j \phi_{k-2j-2}$ , and the two agree because  $(j+1) c_{j+1} = c_j (k-2j)(k-2j-1)$ . With  $\alpha$  constant,  $L(x - x_e)^k = \alpha k(k-1)(x - x_e)^{k-2}$  and the anchor conditions fix the two constants of integration, so  $\phi_k = (x - x_e)^k$  by induction; with  $\alpha_e = \alpha$ , (7) is then  $v_k(x - x_e, t)$ .  $\square$

Read one at a time, the seeds say what each monomial becomes through the edge. The seed  $\phi_1$  solves  $L\phi_1 = 0$ : it is the equilibrium of constant flux,  $\phi_1(x) = \alpha_e \int_{x_e}^x d\xi/\alpha(\xi)$ , and what stays smooth through the edge is its flux  $\alpha\phi'_1$ , not its slope. The seed  $\phi_2$  is the profile of  $u_2 = \phi_2 + 2\alpha_e t$ , whose rate of change is constant,  $u_t = 2\alpha_e$ ; the constant matters, since  $u_t = 0$  would only return  $\ker L$ . In general, by (7),  $\partial_t^m u_{2m} = (2m)! \alpha_e^m$  is constant and  $\partial_t^m u_{2m+1} = (2m+1)! \alpha_e^m \phi_1$  has a constant flux: the even seeds carry  $\partial_t^m u = \text{const}$  and the odd ones  $\alpha \partial_x \partial_t^m u = \text{const}$ . Taken in the order  $\phi_0, \dots, \phi_4$  these are the five conditions of (3) from which the translated basis of a jump is built, on  $u$ , its flux,  $u_t = Lu$ , the flux of  $u_t$ , and  $u_{tt} = L^2 u$ . In the wave equation the same alternation passes between two fields; in diffusion both kinds of condition bear on  $u$ , so one chain serves every stencil.

In one dimension the seeds are a classical object. Integrating the chain twice from the anchor gives  $\phi_k(x) = k(k-1) \alpha_e \int_{x_e}^x \alpha(s)^{-1} \int_{x_e}^s \phi_{k-2}(r) dr ds$ , and unrolling the recursion,

$$(8) \quad \phi_k = k! \alpha_e^{\lceil k/2 \rceil} I_k, \quad I_k(x) = \int_{x_e}^x \frac{ds_1}{\alpha(s_1)} \int_{x_e}^{s_1} ds_2 \int_{x_e}^{s_2} \frac{ds_3}{\alpha(s_3)} \cdots,$$

where  $I_0 = 1$  and  $I_k$  has  $k$  nested integrals whose weights alternate between  $1/\alpha$  and 1, starting from  $1/\alpha$  outermost. These are the recursive integrals,  $X^{(k)}$  for odd  $k$  and  $\tilde{X}^{(k)}$  for even  $k$ , by which Kravchenko and Porter write the solutions of  $(pu')' + qu = \lambda ru$  as power series in the spectral parameter [Kra08, KP10], here with  $q = 0$ ,  $r = 1$ ,  $p = \alpha$ , the particular solution  $u_0 = 1$  and the integrals taken from the anchor. The seeds are thus the formal powers of the Sturm–Liouville operator  $L$  anchored at the evaluation point, and the coefficients of those series are the time-polynomial profiles, with  $\lambda$  in the role of  $\partial_t$ . The repository’s tests check the identity against the ODE march below to a relative  $10^{-9}$  through a tanh edge.

The weights  $w_i$  of a stencil approximate  $Lu$  at  $x_e$  from the nodal values by reproducing  $L$  on the first five seeds,

$$(9) \quad \sum_{i=1}^5 w_i \phi_k(x_i) = (L\phi_k)(x_e) = \begin{cases} 2\alpha_e, & k = 2, \\ 0, & k = 0, 1, 3, 4, \end{cases}$$

since  $L\phi_k = k(k-1) \alpha_e \phi_{k-2}$  vanishes at  $x_e$  for every  $k$  but 2. These are the conditions the interface-aware stencils impose on the translated basis, and

with a constant  $\alpha$  they return  $\alpha$  times the standard second-derivative weights [For88], which the implementation reproduces to rounding. Stencils exact on a local space other than the polynomials go back to fitted operators and to HODIE [AS55, Il'69, LR80, RST08]. Three-point schemes whose coefficients come from Cauchy problems integrated across a piecewise-smooth coefficient are the exact and truncated schemes [SM90, GHMK11], and the seed stencils are a five-point variant of them.

*First-order form.* The chain is integrated as a first-order system in each seed and its flux  $\psi_k = \alpha \phi'_k$ ,

$$(10) \quad \phi'_k = \psi_k / \alpha, \quad \psi'_k = k(k-1) \alpha_e \phi_{k-2},$$

with  $\phi_0(x_e) = 1$ ,  $\psi_1(x_e) = \alpha_e$  and every other initial value zero, so that  $\alpha$  is never differentiated. The ten states march together from  $x_e$  outward to each side of the stencil. Across a jump  $\phi$  and  $\psi$  are continuous, which is  $L$  read in the weak sense: the march stops at the jump and reads  $\alpha$  on each side from the piece that owns it, so a jump is marched like any other width, and a stencil that straddles both edges of a thin layer marches through both.

*Stencil coordinate.* In the coordinate  $\xi = (x - x_e)/h_s$ , with  $h_s = \max_i |x_i - x_e|$ , the functions  $\phi_k(x)/h_s^k$  satisfy the same chain with  $\alpha(x_e + h_s \xi)$  in place of  $\alpha$ . They are close to  $\xi^k$ , and the matrix  $A_{ki} = \phi_k(x_i)/h_s^k$  of (9) is a perturbed Vandermonde matrix; the weights are  $w = \tilde{w}/h_s^2$ , with  $A\tilde{w}$  equal to the right-hand side of (9). For the centred stencil,  $\xi_i = 0, \pm\frac{1}{2}, \pm 1$ , the condition number of  $A$  is 23.5 with a constant  $\alpha$ . On the stencil of Figure 1, half a cell from a 9 : 1 edge, it is 90 at  $\delta = 0$  and 138 at  $\delta = h/2$ , and 24 again once  $\delta \geq 10h$ , where the seeds are close to the monomials.

*The march and the seeded rows.* The march is an explicit adaptive Runge–Kutta method of order eight (SciPy’s DOP853) at relative tolerance  $10^{-13}$  and absolute tolerance  $10^{-15}$ , with segment boundaries at each edge centre and at  $10\delta$  on either side of it, so that the step never has to find the edge inside a long step over flat material. A row is seeded when the span of its stencil meets  $(x_c - 20\delta, x_c + 20\delta)$  for an edge centre  $x_c$ , beyond which the edge is its pieces to the last bit; every other row is the standard one. At  $\delta = 0$  the seeded rows are exactly those that straddle the jump; at  $\delta = 0.0025$  and  $0.04$  on 100 nodes they number 8 and 84; once  $20\delta$  covers the domain, every row is seeded. No threshold in  $\delta$  switches the seeds on or off: a resolved edge costs them nothing (Section 4.4), and at  $\delta = 0$  they are the jump’s weights (Section 3.2). Rows whose nodes sit at the same offsets from  $x_e$  march as one system, so an operator costs some sixteen calls of the integrator whatever the number of seeded rows: 0.11 s for the 1284 seeded rows of  $\delta = 0.04$  on 1600 nodes. The march has one floor of its own. With  $L_h$  the discrete operator, the seeded rows’ residual on the exact equilibrium,  $h^2 \max |L_h u|$ , is at most  $2.5 \times 10^{-16}$  at 200 nodes while the march crosses the edge in a few steps ( $\delta \geq h/10$ ), and  $2.4 \times 10^{-13}$  and  $5.8 \times 10^{-13}$  at  $\delta = h/100$  and  $h/1000$ , where it takes many. The equilibrium solution shows it once

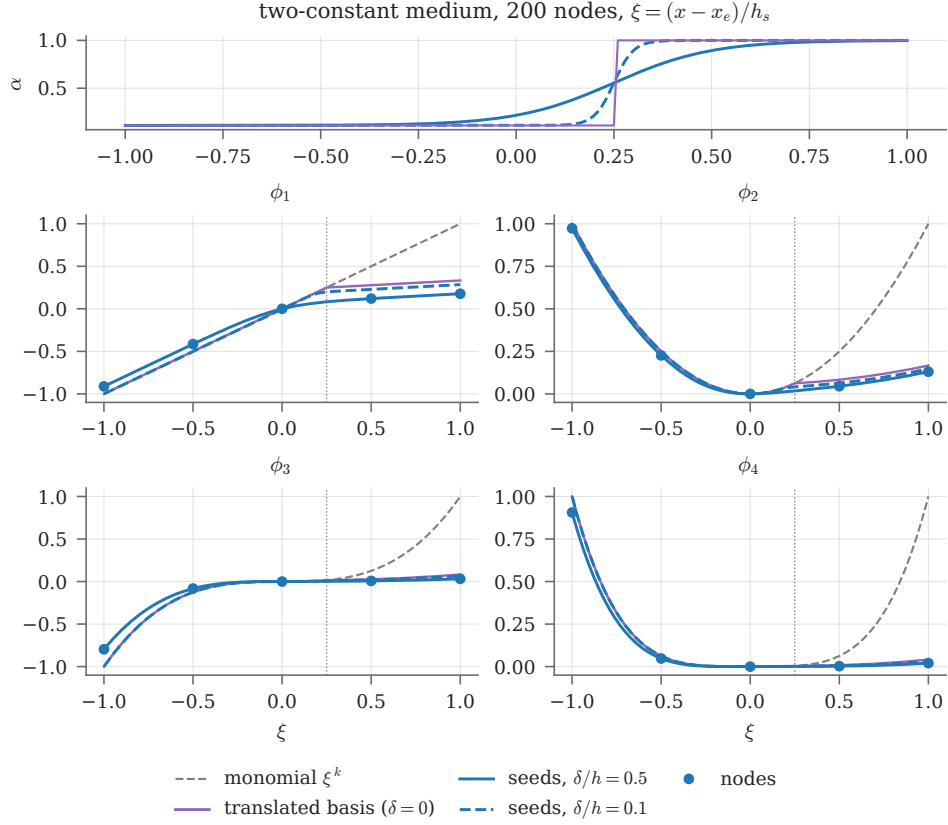


FIGURE 1. The seeds  $\phi_1, \dots, \phi_4$  of one five-point stencil of the two-constant medium at 200 nodes, whose evaluation point  $x_e$  lies half a cell to the left of the 9 : 1 edge (dotted), in the stencil coordinate  $\xi = (x - x_e)/h_s$  with  $h_s = 2h$ ;  $\alpha$  across the stencil on top. Grey, dashed: the monomials  $\xi^k$ . Purple: the translated basis of the jump ( $\delta = 0$ ). Blue: the seeds at  $\delta = h/2$  (solid, with their values at the nodes) and  $\delta = h/10$  (dashed). Each seed is multiplied by  $(\alpha_0/\alpha_e)^{\lceil k/2 \rceil}$ , with  $\alpha_0$  the jump's value at  $x_e$  and  $\alpha_e$  the edge's, so that the lines compare across  $\delta$ ; the factor is the normalisation of (2), which the weights never see. On the node's side of the jump its seeds are the monomials, and across it the slope of  $\phi_1$  falls ninefold; the seeds of a smooth edge bend through it instead of kinking, and approach the translated basis at first order in  $\delta/h$ .

$\delta \lesssim h/40$ , at  $10^{-11}$  to  $4 \times 10^{-11}$ . A collocation integration of the same chain would remove the floor; we have not built one.

**3.2. Nested kernels and the jump as the limit.** Since  $L$  sends each seed two steps down the chain,  $\text{span}\{\phi_0, \dots, \phi_k\}$  is the  $(k+1)$ -th space of

$$(11) \quad \{1\} \subset \ker L \subset \ker \partial_x L \subset \ker L^2 \subset \ker \partial_x L^2 \subset \dots,$$

where  $\ker \partial_x L^m$  holds the  $g$  with  $L^m g$  constant. The spaces depend on neither the anchor nor the normalisation, so the weights do not either; only the basis does.

For a jump, read  $L$  in the weak sense,  $\phi$  and  $\alpha\phi'$  continuous across it. Then  $\ker L$  is spanned by 1 and the piecewise-linear function whose slope changes by the factor  $\alpha^-/\alpha^+$  across the jump, the translated  $x$  of Section 2.4. On constant pieces  $L\phi_2 = 2\alpha_e$  makes  $\phi_2'' = 2\alpha_e/\alpha^\pm$  on the two sides, with  $\phi_2$  and its flux continuous, which is the continuity of  $u_t$  and of its flux; and so on up the chain, each condition  $\partial_t^m u = \text{const}$  or  $\alpha \partial_x \partial_t^m u = \text{const}$  being one of (3). For constant pieces the chain across a jump is therefore the translated basis, the anchor conditions picking one basis of its span, and the interface-aware stencils are the seed stencils at  $\delta = 0$ : one construction with two implementations, algebra for a jump and an ODE march for a smooth edge. On the stencil of Figure 1 the march reproduces the translated basis to  $3 \times 10^{-15}$  function by function, and its weights to  $7 \times 10^{-16}$ . The same device, Taylor data carried across a step to make the weights, appears in photonics [Suj10], and three-point schemes of any prescribed order across a jump in the coefficient are Gartland's [Gar89].

As  $\delta \rightarrow 0$  the seed weights approach the jump's at first order in  $\delta/h$ . On the stencil of Figure 1,  $\max_i |w_i - w_i^{\text{jump}}| / \max_i |w_i^{\text{jump}}|$  is 84, 46, 11, 1.1 and 0.11 percent at  $\delta/h = 1, \frac{1}{2}, 0.1, 0.01$  and 0.001, and the seeds approach the translated basis at the same rate. A stencil that straddles both edges of a layer two cells thick,  $\alpha = 1 | 0.1 | 1$ , marches through both: at  $\delta = 0$  its weights are those of the jump construction translated twice, to  $7.6 \times 10^{-15}$ , and at  $\delta/h = 0.1, 0.01$  and 0.001 they differ from them by 22, 1.9 and 0.18 percent.

When the pieces vary the limit is not exact. The interface-aware construction expands each side's  $\alpha$  to degree four about the interface and truncates the operator there (Section 2.4), while the seed  $\phi_2$  solves  $(\alpha\phi_2')' = 2\alpha_e$  exactly,  $\phi_2 = (x - x_e)^2 - \frac{2}{3} (\alpha'(x_e)/\alpha_e) (x - x_e)^3 + \dots$ , so that within a stencil the two five-dimensional spaces differ at relative order  $h\alpha'/\alpha$ . On the sinusoidal medium, whose layer has  $\alpha'/\alpha = 25$  at its edge, the seed and jump weights of the row whose evaluation point is that edge differ at  $h = 0.01$  by 1.25 times the largest weight at  $\delta = 0$  and by 1.26 to 2.24 times for  $\delta/h$  from 0.001 to 1, and the difference falls with  $h$ . Neither space is wrong, each being exact on its own five functions; on this medium the seed operator's error is the smaller, by 6 to 44 times at equilibrium and 6 to 38 times on the ramp problem at  $\delta = 0$  over 101 to 1601 nodes. There the seeds are measured against the reference solution, never against the jump construction.

**3.3. Well-posedness.** The stencil solve is nonsingular for every edge width, the jump included, and for every set of distinct nodes, not only the ones we used.

**Proposition 3.3.** *Let  $\alpha$  be positive and bounded away from zero on an interval  $I$  containing  $x_e$ , and continuous on  $I$  except at finitely many jumps, across which  $L$  is read in the weak sense. Then for every  $k$  the seeds  $\phi_0, \dots, \phi_k$  form a complete Chebyshev system on  $I$ : interpolation in their span at any  $k + 1$  distinct points of  $I$  is uniquely solvable. Where  $\alpha$  is smooth, as through a tanh edge between smooth pieces, they form an extended complete Chebyshev system. In particular (9) has a unique solution for every  $\delta \geq 0$  and every five distinct nodes in  $I$ .*

*Proof.* By (8) the seeds are nonzero multiples of  $I_0, \dots, I_k$ , nested integrals whose weights  $v_1 = 1/\alpha$ ,  $v_2 = 1$ ,  $v_3 = 1/\alpha, \dots$  (outermost first) are positive. Such a family is the canonical basis of the kernel of the operator in Pólya form  $\partial_x \frac{1}{v_k} \partial_x \dots \frac{1}{v_2} \partial_x \frac{1}{v_1} \partial_x$ , which is  $\partial_x L^m$  for  $k = 2m$  and  $L^{m+1}$  for  $k = 2m + 1$ , and whose kernel is the  $(k + 1)$ -th space of (11). Suppose  $g = \sum_{j \leq k} c_j I_j$  vanishes at  $k + 1$  distinct points of  $I$ . Then  $g' = v_1 g_1$ , where  $g_1 = \sum_{1 \leq j \leq k} c_j I'_{j-1}$  is a combination of the  $k$  nested integrals  $I'_0, \dots, I'_{k-1}$  with the weights  $v_2, v_3, \dots$ , and  $g_1$  is continuous. Between two consecutive zeros  $a < b$  of  $g$ ,  $\int_a^b v_1 g_1 = 0$  with  $v_1 > 0$ , so  $g_1$  vanishes in  $(a, b)$ : it has  $k$  distinct zeros. By induction on  $k$ ,  $g_1 \equiv 0$ , so  $c_1 = \dots = c_k = 0$ , and then  $g = c_0$  vanishes, so  $c_0 = 0$ . Only the positivity and integrability of the weights entered, so the argument holds across finitely many jumps. Where  $\alpha$  is smooth the operator is disconjugate on  $I$  and the family is an extended complete Chebyshev system [P622, KS66, Cop71].  $\square$

When  $x_e$  lies on a jump, as it does for the rows whose evaluation node falls on an edge of the sinusoidal medium,  $\alpha_e$  is the value of the piece that owns  $x_e$ ;  $\phi_1$  then has slope 1 on that side and  $\alpha_e/\alpha$  on the other, and nothing else changes, since  $\alpha_e$  enters the seeds only as a factor. The argument is the one the companion paper makes for the two fields of the wave equation [Mar26], shortened by (8).

**3.4. Truncation error.** Interpolation in an extended complete Chebyshev system has a Newton form whose remainder carries the operator that annihilates the space where the polynomial remainder carries the  $(k + 1)$ -th derivative [Mü73]. For the five seeds that operator is  $M = \partial_x L^2$ , and the local truncation error of a seed stencil on a smooth function  $g$  is

$$(12) \quad \tau(g) = \sum_{i=1}^5 w_i g(x_i) - (Lg)(x_e) = O(h^3 M g),$$

with a constant that the remainder expresses through  $1/\alpha$  inside the stencil. That is third order, not fourth: a second derivative on five points is generically third order, and the centred standard stencil gains its fourth

only because its  $h^3$  term vanishes by symmetry, which an edge breaks. The interface-aware rows of a jump show the same. On the function of degree five in their translated basis, which lies outside the space they reproduce, the  $h^3$  term is 0.32, 0.084 and 172 (in the stencil coordinate) for a 9 : 1 jump half a cell and one and a half cells from the evaluation node and at the node itself, and zero to rounding for equal pieces.

The global order is still four. A local error of order  $h^3$  on a fixed number of rows costs one order less in the solution than the same error on every row, since the inverse of the discrete operator has entries of order  $h$ , so the solution error is  $O(h^4)$ . This is what the interface-aware stencils reach at a jump (rates 3.85 to 4.19, elliptic and parabolic, in our port of [Mar16]), and it is the phenomenon Beale and Layton prove for interface corrections, where an  $O(h)$  truncation error at the interface leaves the solution second order [BL06]. The bound (12) has two readings, and Section 4 is built on them.

*Equilibrium.* If  $Lu = 0$  then  $Mu = 0$ , and more than that:  $u$  has a constant flux, so on every seeded stencil it lies in  $\text{span}\{\phi_0, \phi_1\}$ , and (9) reproduces  $Lu$  there exactly, at any  $\delta$  and any  $h$  (Section 3.5).

*Time dependence.* If  $u_t = Lu$  then  $L^2u = u_{tt}$  and  $Mu = \partial_x u_{tt}$ , the  $x$ -derivative of another solution of the heat equation. Since  $u_{tt}$  has a continuous flux,  $\partial_x u_{tt}$  is bounded independently of  $\delta$  and changes across the edge by the ratio of the pieces, whatever the width. The standard stencil's error is driven by  $u^{(5)}$  instead, which grows like  $\delta^{-4}$  inside the edge, and the rows of  $D_x A D_x$  that see an unresolved edge miss a fixed fraction of the jump in slope, an error of order  $1/h$  in  $Lu$ . The bound does not make the constant in (12) independent of  $\delta/h$ , since the remainder involves  $1/\alpha$  across the stencil. Measured on a separable solution  $e^{ct}v(x)$  of the two-constant medium from 100 to 400 nodes, the seeded rows' residual is third order (rates 3.06 to 3.78), its constants at  $\delta = 0.01$  and 0.0025 within a factor of two of each other, and the standard rows' residual is fourth order (3.94 to 4.01).

Diffusion smooths, so for  $t > 0$  every  $\partial_t^k u$  is itself a solution and the second reading holds for the whole chain. And since  $\partial_x(\alpha u_x) = u_t$ , the flux changes across an edge of width  $\delta$  only by  $\int u_t dx = O(\delta)$ : in diffusion a sub-grid edge is close to a jump at all times. That is why the  $\delta = 0$  construction is a serious baseline here (Section 4.3), and why the seeds' margin over it is set by  $\delta$ .

**3.5. The elliptic problem in one dimension.** The one-dimensional equilibrium has a constant flux  $\alpha u' = B$ , so  $u = u(-1) + B \int_{-1}^x d\xi/\alpha$ , and two constructions are exact on it at every  $h$  and every  $\delta$  when the pieces are constant.

The first is the seed operator. Its seeded rows are exact by the first reading of Section 3.4, and its standard rows see a linear function on a constant piece. At  $\delta = 0$  it is the interface-aware operator of the jump.

The second is the conservative three-point scheme with the face conductances

$$(13) \quad a_{i+1/2} = h / \int_{x_i}^{x_{i+1}} \frac{d\xi}{\alpha(\xi)},$$

whose discrete flux  $a_{i+1/2}(u_{i+1} - u_i)/h$  equals  $B$  at the exact nodal values whatever  $\alpha$  is, so that the exact solution satisfies every discrete equation. It is the homogeneous scheme of Tikhonov and Samarskii [TS62, Sam01], whose face coefficient is the conductance of the constant-flux seed over one cell. For a piecewise-constant  $\alpha$  it is the harmonic-mean interface conductivity of finite-volume heat conduction, which Patankar derives from the same steady one-dimensional flux [Pat80, §4.2-3] and which is the exact interblock transmissivity of groundwater models for steady one-dimensional flow through blocks of uniform transmissivity [GA92]. Pan, Xu and Li call the scheme the harmonic average method and prove it second order [PXL26]. The exactness belongs to the conservative form: the same harmonic mean sampled at the nodes and used as  $A$  in  $D_x A D_x$ , the like-for-like changed medium of Section 4.5, leaves a residual above 0.5 on the exact equilibrium of the 9 : 1 jump at 100 nodes, whether the mean is taken over one cell or two.

So on constant pieces the one-dimensional equilibrium cannot rank the seeds against the conservative scheme. It is the sanity check and the limit case, and the one-dimensional ranking is the parabolic problem's, where the conservative scheme is second order and the seeds are fourth (Section 4.5). Both exactness results are classical, and so is the conservative scheme's second order; we observe them and claim none of them. Where the pieces vary, as in the sinusoidal medium, the equilibrium does rank the two, for a reason that has nothing to do with the edge: the conservative scheme stays exact, while the seed operator's standard rows inside the layer keep the error of a smooth coefficient, and it is exact only where the seeded rows cover the layer (Section 4.4).

#### 4. ONE-DIMENSIONAL RESULTS

This section measures the naive operator, the  $\delta = 0$  construction and the seeds, and then the coefficient treatments, on the two media of Section 2.2 at  $\delta = 0, 0.04, 0.01$  and  $0.0025$ . Node counts double from 50 to 1600 on the two-constant medium and from 101 to 1601 on the sinusoidal one, whose edges then fall on nodes;  $h = \delta$  falls at  $n = 51, 201$  and  $801$  for the three widths. Errors are relative  $\ell^2$  errors at the nodes and rates are per doubling of  $n$  (Section 2.1). The ramp problem is marched by BD4 with  $\Delta t = h$ , so its errors are those of the fully discrete scheme; Section 4.4 separates the time integrator's share.

**4.1. Media and references.** With  $\delta > 0$  each medium blends its pieces across each edge by (4), the two edges of the sinusoidal medium in turn, so

that the pieces' weights remain a partition of unity; at  $\delta = 0$  the blend is the jump to the last bit. Both references cut  $[-1, 1]$  into elements on which  $\alpha$  is smooth up to their ends: the pieces for a jump, and for an edge also the points  $x_c \pm m\delta$ ,  $m = 1, 3, 9, 27$ , so that outside the core  $|x - x_c| \leq \delta$  each element spans at most a factor of three in distance from the centre and is resolved by a polynomial of modest degree.

The equilibrium reference is  $u = u(-1) + B \int_{-1}^x d\xi/\alpha$ , with  $B$  fixed by  $u(1)$  and the integral taken by Gauss quadrature on the elements. Through a tanh edge between constants  $a$  and  $b$  the integrand has the antiderivative

$$\delta \left[ \frac{z}{a} + \frac{a-b}{2ab} \ln(a + b e^{2z}) \right], \quad z = \frac{x - x_c}{\delta},$$

which the quadrature reproduces to  $10^{-14}$ ; subtracting the jump's  $z/a$  and  $z/b$  on the two sides leaves the deficit (5) as  $z \rightarrow \pm\infty$ . The parabolic reference is Chebyshev collocation on the same elements, 20 points on each and elements wider than 0.1 split, with  $u$  and  $\alpha u_x$  matched at every element end, integrated to  $t = 2$  by SciPy's implicit Radau method at relative and absolute tolerances  $10^{-9}$  and  $10^{-11}$ ; the absolute tolerance sits above the round-off of the collocation system on elements of width of order  $\delta$ , which a tighter tolerance would make the step control chase. Against a second resolution, 24 points and elements to 0.05, it agrees to  $5 \times 10^{-12}$  on the two-constant medium and  $6 \times 10^{-11}$  on the sinusoidal one, at worst over the four widths. That is round-off, not truncation, and the finest errors of the seeds sit on it (Section 4.4).

Against each  $\delta > 0$  line the figures also draw the *floor*  $\|u_0 - u_\delta\|_2/\|u_\delta\|_2$  at the nodes, the distance between the jump problem's solution  $u_0$  and the smooth one  $u_\delta$ , from the two references. It is what a method that converges to the jump's solution reaches (Section 4.3).

**4.2. The knee.** Figure 2 gives the three operators at every width, and Table 1 the naive operator's and the construction's errors on the ramp problem. At  $\delta = 0$  the naive operator is first order, with rates of 0.98 to 1.02 on the two-constant medium and 1.00 to 1.01 on the sinusoidal one from 201 nodes. While  $h \geq 4\delta$  its lines at  $\delta > 0$  stay with the jump's, at 0.43 to 1.05 times its error and at rates of 0.9 to 1.6 that depend on where the nodes happen to sample the edge: its rows that see the edge treat it as a jump between two of their nodes, the first-order failure of Section 3.4.

The knee is at  $h \approx \delta$ . Between  $h = 2\delta$  and  $h = \delta/2$  the ramp problem's error falls by 117 and 199 times on the two-constant medium at  $\delta = 0.01$  and 0.0025, and by 771 and 977 times on the sinusoidal medium, where a first-order line falls by four. On the two-constant medium, whose edge lies midway between two nodes, most of the fall comes in the one doubling from  $h = \delta$  to  $\delta/2$ , at rates of 5.9 to 6.8; on the sinusoidal medium, whose edges lie on nodes, it comes a doubling earlier, at 5.2 to 5.4. From  $h = \delta/2$  on, the nodes resolve the tanh and the lines are fourth order, at rates of 3.95

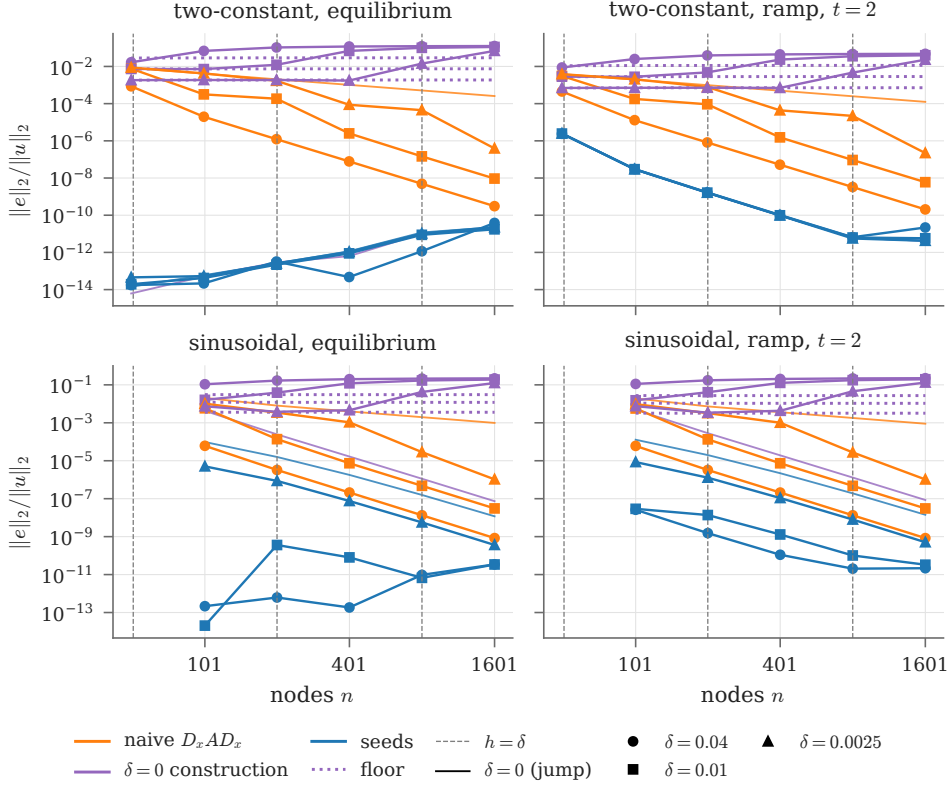


FIGURE 2. The naive operator  $D_x A D_x$  (orange), the  $\delta = 0$  construction (purple) and the seeds (blue) against the node count  $n$  on the two-constant medium (top) and the sinusoidal medium (bottom), at equilibrium (left) and on the ramp problem at  $t = 2$  (right); errors are  $\|e\|_2/\|u\|_2$  at the nodes. Markers give  $\delta$ , and lines without markers are the jump,  $\delta = 0$ . Dotted: the floor  $\|u_0 - u_\delta\|_2/\|u_\delta\|_2$  of each  $\delta > 0$ . Dashed verticals:  $h = \delta$ . At equilibrium on the two-constant medium the seeds, and at  $\delta = 0$  the construction, are exact, and their lines are the direct solve's rounding.

to 4.23 up to the largest count. The equilibrium has the same knee at the same counts: the knee belongs to the operator, not to the problem.

**4.3. The jump construction on a smooth edge.** The  $\delta = 0$  construction converges to the jump problem's solution, so while it converges the floor bounds its error from below. It has two regimes (Figure 2, purple, and Table 1).

*Unresolved.* While the grid does not resolve the edge, the construction sits on the floor. At  $\delta = 0.0025$  on the two-constant medium its equilibrium error at 50, 100 and 200 nodes is the floor to three digits, and wherever  $h \geq 2\delta$

TABLE 1. The naive operator and the  $\delta = 0$  construction on the ramp problem at  $t = 2$ :  $\|e\|_2/\|u\|_2$  at the nodes, against the Chebyshev reference at each  $\delta$ . *floor*: the two references' difference  $\|u_0 - u_\delta\|_2/\|u_\delta\|_2$  at the largest count. Last line: the constant  $c$  of the resistance deficit (5), measured from the two quadratures at every  $\delta$  and in closed form.

| $n$                                                                                  | naive $D_x A D_x$      |                       |                       | $\delta = 0$ construction |                       |                       |
|--------------------------------------------------------------------------------------|------------------------|-----------------------|-----------------------|---------------------------|-----------------------|-----------------------|
|                                                                                      | $\delta = 0.04$        | $\delta = 0.01$       | $\delta = 0.0025$     | $\delta = 0.04$           | $\delta = 0.01$       | $\delta = 0.0025$     |
| <i>two-constant medium 1/9   1, ramp problem at <math>t = 2</math></i>               |                        |                       |                       |                           |                       |                       |
| 50                                                                                   | $4.50 \times 10^{-4}$  | $3.09 \times 10^{-3}$ | $3.85 \times 10^{-3}$ | $8.80 \times 10^{-3}$     | $2.82 \times 10^{-3}$ | $6.97 \times 10^{-4}$ |
| 100                                                                                  | $1.29 \times 10^{-5}$  | $1.79 \times 10^{-4}$ | $2.04 \times 10^{-3}$ | $2.51 \times 10^{-2}$     | $2.78 \times 10^{-3}$ | $7.12 \times 10^{-4}$ |
| 200                                                                                  | $8.19 \times 10^{-7}$  | $9.25 \times 10^{-5}$ | $7.90 \times 10^{-4}$ | $3.91 \times 10^{-2}$     | $4.79 \times 10^{-3}$ | $7.16 \times 10^{-4}$ |
| 400                                                                                  | $5.22 \times 10^{-8}$  | $1.54 \times 10^{-6}$ | $4.36 \times 10^{-5}$ | $4.45 \times 10^{-2}$     | $2.34 \times 10^{-2}$ | $7.01 \times 10^{-4}$ |
| 800                                                                                  | $3.28 \times 10^{-9}$  | $9.32 \times 10^{-8}$ | $2.19 \times 10^{-5}$ | $4.67 \times 10^{-2}$     | $3.54 \times 10^{-2}$ | $4.65 \times 10^{-3}$ |
| 1600                                                                                 | $2.05 \times 10^{-10}$ | $5.96 \times 10^{-9}$ | $2.20 \times 10^{-7}$ | $4.77 \times 10^{-2}$     | $3.99 \times 10^{-2}$ | $2.30 \times 10^{-2}$ |
| floor                                                                                |                        |                       |                       | $1.14 \times 10^{-2}$     | $2.87 \times 10^{-3}$ | $7.18 \times 10^{-4}$ |
| <i>sinusoidal medium, ramp problem at <math>t = 2</math></i>                         |                        |                       |                       |                           |                       |                       |
| 101                                                                                  | $6.06 \times 10^{-5}$  | $5.67 \times 10^{-3}$ | $9.61 \times 10^{-3}$ | $1.12 \times 10^{-1}$     | $1.53 \times 10^{-2}$ | $7.51 \times 10^{-3}$ |
| 201                                                                                  | $3.25 \times 10^{-6}$  | $1.34 \times 10^{-4}$ | $3.23 \times 10^{-3}$ | $1.73 \times 10^{-1}$     | $4.09 \times 10^{-2}$ | $3.41 \times 10^{-3}$ |
| 401                                                                                  | $2.11 \times 10^{-7}$  | $7.36 \times 10^{-6}$ | $1.01 \times 10^{-3}$ | $2.04 \times 10^{-1}$     | $1.27 \times 10^{-1}$ | $4.29 \times 10^{-3}$ |
| 801                                                                                  | $1.33 \times 10^{-8}$  | $4.76 \times 10^{-7}$ | $2.74 \times 10^{-5}$ | $2.18 \times 10^{-1}$     | $1.77 \times 10^{-1}$ | $4.60 \times 10^{-2}$ |
| 1601                                                                                 | $8.36 \times 10^{-10}$ | $3.06 \times 10^{-8}$ | $1.03 \times 10^{-6}$ | $2.24 \times 10^{-1}$     | $1.97 \times 10^{-1}$ | $1.32 \times 10^{-1}$ |
| floor                                                                                |                        |                       |                       | $2.71 \times 10^{-2}$     | $1.06 \times 10^{-2}$ | $3.20 \times 10^{-3}$ |
| two-constant medium: deficit $c\delta$ , $c = 8.78890$ measured, 8.78890 closed form |                        |                       |                       |                           |                       |                       |

on that medium it lies at 0.93 to 1.00 times the floor, at equilibrium and on the ramp problem. The floor is first order in  $\delta$ ,  $0.73$  to  $0.75\delta$  at equilibrium and  $0.29\delta$  on the ramp problem at  $t = 2$ , since the edge's resistance falls short of the jump's by  $c\delta$  with  $c = 8.79$  (5), which the two quadratures reproduce to six digits at every width; the flux through the medium changes by  $c\delta/F_0 = 0.88\delta$  relative to the jump's,  $F_0 = 10$  being the jump's total resistance. On the sinusoidal medium the floor is not yet proportional to  $\delta$  at these widths: it is  $0.76$ ,  $1.20$  and  $1.44\delta$  at equilibrium for  $\delta = 0.04$ ,  $0.01$  and  $0.0025$ , and the measured deficit is  $0.44$ ,  $0.74$  and  $0.91$  of the limit  $20.7\delta$  of its two  $10 : 1$  edges. The sinusoid has  $\alpha'/\alpha = 25$  where it meets the edges, so that at  $\delta = 0.01$  it changes by 75% within  $3\delta$  of the edge, and the pieces the edge blends are not the constants that (5) assumes. There the coarse rows also carry the sinusoid's own error, and the construction lies at  $1.05$  to  $2.3$  times the floor where  $h \geq 2\delta$ .

*Resolved.* Once the grid resolves the edge the construction leaves the floor and its error grows. Its plain rows are then fourth-order rows on a smooth coefficient, but its three or four rebuilt rows still impose the translated basis, a kink of slope ratio  $\alpha^+/\alpha^-$  at the edge's centre, on a solution whose slope hardly changes across the centre cell, and the discrete solution takes the

TABLE 2. The seeds on the ramp problem at  $t = 2$ :  $\|e\|_2/\|u\|_2$  at the nodes, with the rate per doubling of  $n$  in parentheses. Errors of a few times  $10^{-12}$  on the two-constant medium and a few times  $10^{-11}$  on the sinusoidal one are the references' own floor (Section 4.1), where the rates mean nothing.

| $n$                                                                         | $\delta = 0$                  | $\delta = 0.04$                | $\delta = 0.01$               | $\delta = 0.0025$             |
|-----------------------------------------------------------------------------|-------------------------------|--------------------------------|-------------------------------|-------------------------------|
| <i>two-constant medium <math>1/9   1</math>, ramp problem, error (rate)</i> |                               |                                |                               |                               |
| 50                                                                          | $2.49 \times 10^{-6}$         | $2.39 \times 10^{-6}$          | $2.46 \times 10^{-6}$         | $2.48 \times 10^{-6}$         |
| 100                                                                         | $2.95 \times 10^{-8}$ (6.40)  | $3.05 \times 10^{-8}$ (6.29)   | $2.99 \times 10^{-8}$ (6.36)  | $2.95 \times 10^{-8}$ (6.39)  |
| 200                                                                         | $1.61 \times 10^{-9}$ (4.19)  | $1.69 \times 10^{-9}$ (4.17)   | $1.64 \times 10^{-9}$ (4.19)  | $1.62 \times 10^{-9}$ (4.19)  |
| 400                                                                         | $9.76 \times 10^{-11}$ (4.05) | $1.03 \times 10^{-10}$ (4.04)  | $9.89 \times 10^{-11}$ (4.05) | $9.74 \times 10^{-11}$ (4.05) |
| 800                                                                         | $6.11 \times 10^{-12}$ (4.00) | $6.54 \times 10^{-12}$ (3.97)  | $6.32 \times 10^{-12}$ (3.97) | $5.61 \times 10^{-12}$ (4.12) |
| 1600                                                                        | $4.18 \times 10^{-12}$ (0.55) | $2.18 \times 10^{-11}$ (-1.74) | $5.76 \times 10^{-12}$ (0.13) | $4.22 \times 10^{-12}$ (0.41) |
| <i>sinusoidal medium, ramp problem, error (rate)</i>                        |                               |                                |                               |                               |
| 101                                                                         | $1.28 \times 10^{-4}$         | $2.61 \times 10^{-8}$          | $2.92 \times 10^{-8}$         | $8.50 \times 10^{-6}$         |
| 201                                                                         | $1.99 \times 10^{-5}$ (2.68)  | $1.53 \times 10^{-9}$ (4.09)   | $1.37 \times 10^{-8}$ (1.09)  | $1.28 \times 10^{-6}$ (2.73)  |
| 401                                                                         | $2.20 \times 10^{-6}$ (3.18)  | $1.11 \times 10^{-10}$ (3.79)  | $1.30 \times 10^{-9}$ (3.40)  | $1.08 \times 10^{-7}$ (3.57)  |
| 801                                                                         | $1.91 \times 10^{-7}$ (3.53)  | $2.05 \times 10^{-11}$ (2.43)  | $1.02 \times 10^{-10}$ (3.68) | $7.95 \times 10^{-9}$ (3.77)  |
| 1601                                                                        | $1.42 \times 10^{-8}$ (3.75)  | $2.16 \times 10^{-11}$ (-0.08) | $3.31 \times 10^{-11}$ (1.62) | $5.06 \times 10^{-10}$ (3.97) |

kink. On the two-constant medium the error then follows  $h/\delta$ , not  $\delta$ : at  $h \approx \delta/2$  it is 0.0685 to 0.0687 for all three widths, which is 2.4, 9.2 and 37 times the floor at  $\delta = 0.04$ , 0.01 and 0.0025. At 1600 nodes it reaches 0.11 at equilibrium for  $\delta = 0.01$ , and for  $\delta = 0.04$  it is 0.125 at equilibrium and 0.048 on the ramp problem (0.22 for both on the sinusoidal medium). Between the two regimes, at  $h \approx 2\delta$ , the partly resolved plain rows pull the solution toward the smooth one, and the error dips just below the floor, to 0.93 of it at  $\delta = 0.0025$ .

So treating the edge as a jump is the right baseline exactly where the grid cannot see the edge, and the worst of the three operators where it can. Using it means knowing  $\delta$ , switching it off near  $h = 2\delta$ , and accepting a floor that falls only as  $\delta$ . At  $\delta = 0$  it is the interface-aware operator of [Mar16], exact at equilibrium on the two-constant medium and fourth order on the sinusoidal one (rates 3.88 to 4.12).

**4.4. The seeds.** *The two-constant medium.* Table 2 and the blue lines of Figure 2 give the seeds. On the two-constant medium their error is one line for every  $\delta$ . At  $\delta = 0$  they are the interface-aware operator (Section 3.2). On the ramp problem their errors at  $\delta = 0.0025$  lie within 0.1 to 0.3% of the  $\delta = 0$  line at every count to 400, at  $\delta = 0.01$  within 1 to 1.5%, and at  $\delta = 0.04$ , where  $\delta \approx h$  at 50 nodes and the smooth solution is a different function, within 3 to 5%. The rates at  $\delta = 0$  are 6.4, 4.2 and 4.05, then 4.0 to 4.1 at every width, from  $2.5 \times 10^{-6}$  at 50 nodes to  $6 \times 10^{-12}$  at 800, where

every width reaches the references' floor. At equilibrium the seeds are exact at every width (Remark 4.1).

*Time and space.* These are the errors of the fully discrete scheme at  $\Delta t = h$ , and part of each is the time integrator's. With  $\Delta t = h/8$  the spatial error alone at  $\delta = 0$  is  $1.47 \times 10^{-8}$ ,  $1.05 \times 10^{-9}$ ,  $7.40 \times 10^{-11}$  and  $4.5 \times 10^{-12}$  at 100 to 800 nodes, rates 3.8, 3.8 and 4.0, with the  $\delta = 0.0025$  line within 0.3% of it and the  $\delta = 0.01$  line within 1.8%. The fourth order and the one line are the space discretisation's. The time error is about half the total at 100 nodes and a quarter from 200, and it is where the first rate of 6.4 comes from.

*A resolved edge costs nothing.* Across the knee, where the naive operator changes order and the construction leaves its floor, the seeds' line does not move: it is the same fourth-order line at every  $h/\delta$  from 16 down to  $1/8$ , below which the references' floor takes over, with no threshold in  $\delta$  (Section 3.1). At 200 nodes and  $\delta = 0.0025$ , where  $h = 4\delta$ , the seeds' error on the ramp problem is  $1.6 \times 10^{-9}$ , against the naive operator's  $7.9 \times 10^{-4}$  and the construction's  $7.2 \times 10^{-4}$ . The construction's floor falls as  $\delta$ , so the seeds' margin over it is set by  $\delta$  and closes at  $\delta = 0$ , where on constant pieces the two are one operator.

*The sinusoidal medium.* At equilibrium the seeded rows are exact on the sinusoidal medium too, and the seed operator is exact only where the seeded rows cover the layer. On both problems the rows whose stencils lie farther than  $20\delta$  from both edges are standard rows on the sinusoid, and their error is the line. At  $\delta = 0.04$  the reach covers the layer, every row is seeded, and the ramp problem's error is  $2.6 \times 10^{-8}$ ,  $1.5 \times 10^{-9}$  and  $1.1 \times 10^{-10}$  at 101, 201 and 401 nodes (rates 4.1 and 3.8), and then the references' floor. At  $\delta = 0.01$  the reach covers the layer at 101 nodes only; from 201 on it leaves the rows near  $x = 0.25$  standard, which carry  $1.4 \times 10^{-8}$  at 201 nodes, and the line is fourth order past them. With every row seeded instead, the  $\delta = 0.01$  errors at 201 nodes are  $1.3 \times 10^{-13}$  at equilibrium and  $1.7 \times 10^{-9}$  on the ramp problem; the reach is the operator's one parameter, and we left it at  $20\delta$ . At  $\delta = 0.0025$  and at the jump the interior of the layer is standard at every count, and the line is the sinusoid's own error under the standard fourth-order rows, which is still pre-asymptotic, the rows beside the edges seeing  $h\alpha'/\alpha = 0.5$  at 101 nodes. At  $\delta = 0$  it falls from  $1.3 \times 10^{-4}$  to  $1.4 \times 10^{-8}$  over 101 to 1601 nodes at rates rising from 2.7 to 3.75, and at  $\delta = 0.0025$  its rates rise from 2.7 to 4.0. At  $\delta = 0$  that line is still 6 to 38 times below the construction's (Section 3.2).

*Stability.* The seed operator's interior spectrum, with the Dirichlet rows removed as BD4 steps it, is real and negative on both media at  $\delta = 0$  and 0.0025, and its extreme eigenvalue is the standard stencil's  $-16/(3h^2)$  in the material with  $\alpha = 1$ , to 0.6% at 50 nodes. On the sinusoidal medium at 49 and 53 nodes, where  $h\alpha'/\alpha \approx 1$  at the layer's edges, the seed operator is stable, with largest eigenvalue  $-2.05$ , while the  $\delta = 0$  construction has

eigenvalues at +248 and +8.8; the sweep starts at 101 nodes so that all three operators share its grids.

**4.5. Coefficient treatments.** The standing alternative to a new stencil changes the coefficient and keeps the scheme (Section 1.1). We ran four such treatments on the grids and problems above, against the same references:

- T1:** the harmonic mean  $(b - a) / \int_a^b d\xi / \alpha$  of  $\alpha$  over a window  $[a, b]$  of one or two cells centred on each node, clipped at the ends of the domain and sampled into  $A$  in  $D_x A D_x$ ;
- T2:** the arithmetic mean over one cell, used the same way;
- T0:** the edge widened to  $\max(\delta, mh)$ ,  $m = 1$  or  $2$ , a coefficient regularised on the scale of the grid [TE03];
- T1-FV:** the conservative three-point scheme with the face conductances (13).

T1-FV is Pan, Xu and Li’s harmonic average method [PXL26], exact at the one-dimensional equilibrium (Section 3.5). T1 is the like-for-like changed medium: a harmonic cell mean, which as a face conductance makes the conservative scheme exact at equilibrium, sampled at the nodes of the fourth-order operator, where it does not; we found no source that uses it there. The harmonic mean is the exact conductance between blocks of uniform conductivity, and may be inferior to other means where the conductivity varies smoothly between nodes [GA92], as it does here once  $\delta > 0$ . A fifth candidate, the band-limited coefficient of the seismic literature [Mit17, KWR22], was not found in use for diffusion and is left out. The comparison is against coefficient sampling on the same nodes and against our implementation of the cell harmonic and arithmetic means, a widened edge and, in one dimension, the conservative scheme with exact face conductances, on one contrast and one test problem per dimension.

*The ranking.* Figure 3 gives every treatment on the ramp problem, and Table 3 the errors at  $\delta = 0.0025$  in the order of the ranking. On the ramp problem of the two-constant medium, at every width while  $h \geq 4\delta$ , the jump included, the order in  $\|e\|_2 / \|u\|_2$  is

$$\begin{aligned} \text{seeds} < \text{T1-FV} < \text{T1 (two cells)} < \text{naive} \gtrsim \text{T1 (one cell)} \\ &\approx \text{T2} < \text{T0} (m = 1) < \text{T0} (m = 2), \end{aligned}$$

with the middle three at most 2.02 times apart; at 200 nodes and  $\delta = 0.0025$  the errors are  $1.6 \times 10^{-9}$ ,  $8.0 \times 10^{-6}$ ,  $8.9 \times 10^{-5}$ ,  $7.9 \times 10^{-4}$ ,  $4.0 \times 10^{-4}$ ,  $4.6 \times 10^{-4}$ ,  $2.2 \times 10^{-3}$  and  $5.0 \times 10^{-3}$ . At the jump the seeds lead T1-FV by 53 times at 50 nodes and by  $2.1 \times 10^4$  at 400. On the sinusoidal medium, whose edges fall on nodes, a one-cell window straddles each edge, and at the jump the one-cell harmonic mean is 9 to 35 times below the naive operator. There T1-FV is ahead of the seeds at  $\delta = 0$  below 801 nodes and at  $\delta = 0.0025$  at 101, where the seeds’ line is the sinusoid’s pre-asymptotic

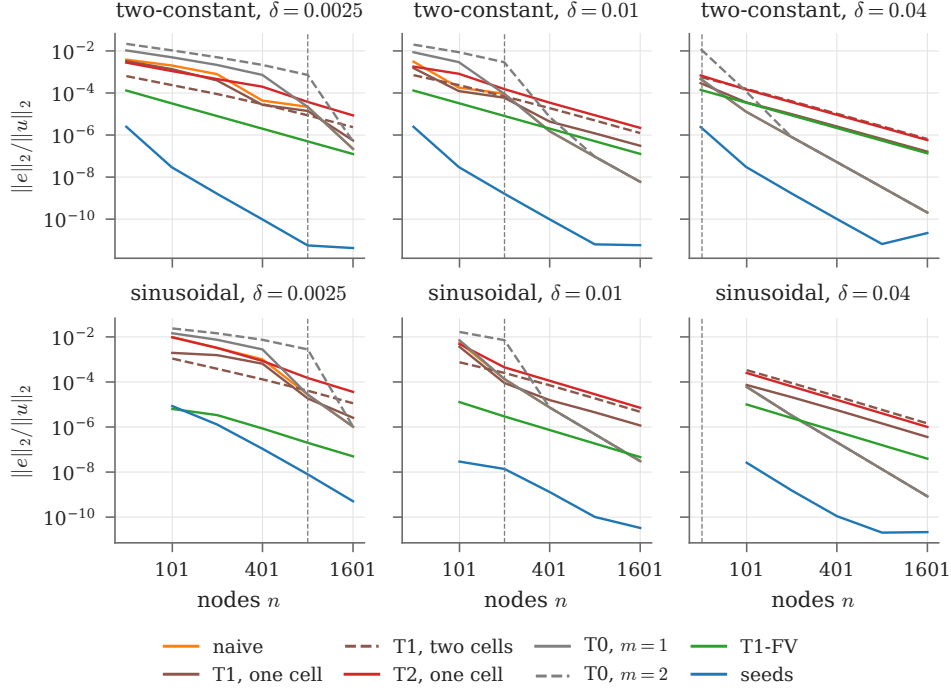


FIGURE 3. The coefficient treatments on the ramp problem at  $t = 2$ :  $\|e\|_2/\|u\|_2$  at the nodes against  $n$  on the two-constant medium (top) and the sinusoidal medium (bottom) at  $\delta = 0.0025, 0.01$  and  $0.04$ . The naive operator and the seeds are those of Figure 2, and the treatments those of Table 3. Dashed verticals:  $h = \delta$ . The seeds' flat ends are the references' floor.

one (Section 4.4) rather than the edge's; where the seeded rows cover the layer the seeds lead, by 440 times at 101 nodes and  $\delta = 0.01$ .

*Second order.* T1-FV is second order at every width, as Pan, Xu and Li prove, and on the two-constant medium with one constant: at  $\delta = 0$  its error falls from  $1.31 \times 10^{-4}$  at 50 nodes to  $1.26 \times 10^{-7}$  at 1600, rate 2.00; at  $\delta = 0.01$  and  $0.0025$  its errors lie within 1.2% of these at every count, and at  $\delta = 0.04$  they lie 7% above. It is the strongest of the treatments. Every nodal treatment caps the fourth-order operator at second order once the grid resolves the edge: for  $h \leq \delta/4$  on the two-constant medium the naive operator is fourth order and the treated ones second order (rates 1.95 to 2.00). At  $\delta = 0.04$  and 1600 nodes the naive error is  $2.05 \times 10^{-10}$ , and the treated errors are  $1.62 \times 10^{-7}$  (T1, one cell),  $5.76 \times 10^{-7}$  (T2) and  $6.50 \times 10^{-7}$  (T1, two cells), 1.2 to 4.8 times T1-FV's  $1.35 \times 10^{-7}$ . A cell mean perturbs a smooth  $\alpha$  at second order in the width of its window, and the fourth-order operator solves the perturbed problem faithfully. For the wave

TABLE 3. The treatments on the ramp problem at  $t = 2$  and  $\delta = 0.0025$ :  $\|e\|_2/\|u\|_2$  at the nodes, to two figures, in the order of the ranking of Section 4.5. T1: the harmonic mean over one or two cells; T2: the arithmetic mean over one cell; T0: the edge widened to  $\max(\delta, mh)$ ; T1-FV: the conservative scheme with the face conductances (13).

| $\ e\ _2/\ u\ _2$                                                              |                      |                      |                      |                       |                       |                       |
|--------------------------------------------------------------------------------|----------------------|----------------------|----------------------|-----------------------|-----------------------|-----------------------|
| <i>two-constant medium 1/9   1, ramp problem, <math>\delta = 0.0025</math></i> |                      |                      |                      |                       |                       |                       |
| $n$                                                                            | 50                   | 100                  | 200                  | 400                   | 800                   | 1600                  |
| seeds                                                                          | $2.5 \times 10^{-6}$ | $3.0 \times 10^{-8}$ | $1.6 \times 10^{-9}$ | $9.7 \times 10^{-11}$ | $5.6 \times 10^{-12}$ | $4.2 \times 10^{-12}$ |
| T1-FV                                                                          | $1.3 \times 10^{-4}$ | $3.2 \times 10^{-5}$ | $8.0 \times 10^{-6}$ | $2.0 \times 10^{-6}$  | $5.0 \times 10^{-7}$  | $1.3 \times 10^{-7}$  |
| T1, two cells                                                                  | $6.4 \times 10^{-4}$ | $2.4 \times 10^{-4}$ | $8.9 \times 10^{-5}$ | $3.0 \times 10^{-5}$  | $8.7 \times 10^{-6}$  | $2.4 \times 10^{-6}$  |
| naive                                                                          | $3.8 \times 10^{-3}$ | $2.0 \times 10^{-3}$ | $7.9 \times 10^{-4}$ | $4.4 \times 10^{-5}$  | $2.2 \times 10^{-5}$  | $2.2 \times 10^{-7}$  |
| T1, one cell                                                                   | $3.3 \times 10^{-3}$ | $1.4 \times 10^{-3}$ | $4.0 \times 10^{-4}$ | $2.7 \times 10^{-5}$  | $1.4 \times 10^{-5}$  | $5.5 \times 10^{-7}$  |
| T2, one cell                                                                   | $2.8 \times 10^{-3}$ | $1.1 \times 10^{-3}$ | $4.6 \times 10^{-4}$ | $2.0 \times 10^{-4}$  | $3.8 \times 10^{-5}$  | $8.6 \times 10^{-6}$  |
| T0, $m = 1$                                                                    | $1.1 \times 10^{-2}$ | $5.0 \times 10^{-3}$ | $2.2 \times 10^{-3}$ | $7.2 \times 10^{-4}$  | $2.2 \times 10^{-5}$  | $2.2 \times 10^{-7}$  |
| T0, $m = 2$                                                                    | $2.2 \times 10^{-2}$ | $1.1 \times 10^{-2}$ | $5.0 \times 10^{-3}$ | $2.2 \times 10^{-3}$  | $7.2 \times 10^{-4}$  | $5.0 \times 10^{-7}$  |
| <i>sinusoidal medium, ramp problem, <math>\delta = 0.0025</math></i>           |                      |                      |                      |                       |                       |                       |
| $n$                                                                            | 101                  | 201                  | 401                  | 801                   | 1601                  |                       |
| seeds                                                                          | $8.5 \times 10^{-6}$ | $1.3 \times 10^{-6}$ | $1.1 \times 10^{-7}$ | $8.0 \times 10^{-9}$  | $5.1 \times 10^{-10}$ |                       |
| T1-FV                                                                          | $6.4 \times 10^{-6}$ | $3.4 \times 10^{-6}$ | $8.6 \times 10^{-7}$ | $2.0 \times 10^{-7}$  | $5.0 \times 10^{-8}$  |                       |
| T1, two cells                                                                  | $1.1 \times 10^{-3}$ | $3.8 \times 10^{-4}$ | $1.3 \times 10^{-4}$ | $4.1 \times 10^{-5}$  | $1.1 \times 10^{-5}$  |                       |
| naive                                                                          | $9.6 \times 10^{-3}$ | $3.2 \times 10^{-3}$ | $1.0 \times 10^{-3}$ | $2.7 \times 10^{-5}$  | $1.0 \times 10^{-6}$  |                       |
| T1, one cell                                                                   | $2.0 \times 10^{-3}$ | $1.6 \times 10^{-3}$ | $6.5 \times 10^{-4}$ | $1.9 \times 10^{-5}$  | $2.6 \times 10^{-6}$  |                       |
| T2, one cell                                                                   | $9.7 \times 10^{-3}$ | $3.2 \times 10^{-3}$ | $8.6 \times 10^{-4}$ | $1.5 \times 10^{-4}$  | $3.6 \times 10^{-5}$  |                       |
| T0, $m = 1$                                                                    | $1.5 \times 10^{-2}$ | $7.4 \times 10^{-3}$ | $2.8 \times 10^{-3}$ | $2.7 \times 10^{-5}$  | $1.0 \times 10^{-6}$  |                       |
| T0, $m = 2$                                                                    | $2.4 \times 10^{-2}$ | $1.4 \times 10^{-2}$ | $7.4 \times 10^{-3}$ | $2.8 \times 10^{-3}$  | $1.0 \times 10^{-6}$  |                       |

equation with regularised discontinuities, Tornberg and Engquist report the same split: a second-order error component from the discontinuities under a higher-order discretisation [TE06]. So the nodal treatments share the construction's property: they help only while  $h \gtrsim \delta$ , and using them means knowing  $\delta$  and switching them off, which the seeds need not.

*The widened edge and the local defect.* T0 is the worst. The widened medium's equilibrium differs from the true one by a resistance deficit  $c(mh - \delta)$ , and T0's equilibrium error is that difference, to 0.1% for  $m = 2$  wherever  $h \geq \delta$ ; on the two-constant medium's ramp problem at  $\delta = 0$  it is 2.9 ( $m = 1$ ) and 5.7 to 5.8 ( $m = 2$ ) times the naive operator's error at every count. Widening an edge that the grid already fails to resolve adds resistance error and leaves the knee where it was. With the edge midway between nodes, a one-cell window ends on the jump at  $\delta = 0$ , and T1 (one cell) and T2 are the naive operator there to rounding. T1 over two cells converges at order 1.5 in this norm (rates 1.44 to 1.50 at  $\delta = 0$  on the two-constant medium's ramp problem), the mark of a local defect: its error is first order on the two nodes beside the edge,  $1.18 \times 10^{-3}$  to  $7.28 \times 10^{-5}$  in the maximum norm

TABLE 4. The elliptic problem:  $\|e\|_2/\|u\|_2$  at the nodes for the naive operator and the  $\delta = 0$  construction at  $\delta = 0.01$ , the seeds at three widths, and T1-FV at  $\delta = 0$  (Remark 4.1).

| $n$                                                                              | $\delta = 0.01$       |                       | seeds                 |                       |                       | T1-FV                 |
|----------------------------------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                                                                                  | naive                 | construction          | $\delta = 0$          | $\delta = 0.01$       | $\delta = 0.0025$     | $\delta = 0$          |
| <i>two-constant medium</i> $1/9 \mid 1$ , <i>equilibrium</i> , $\ e\ _2/\ u\ _2$ |                       |                       |                       |                       |                       |                       |
| 50                                                                               | $7.64 \times 10^{-3}$ | $7.35 \times 10^{-3}$ | $1.4 \times 10^{-14}$ | $1.9 \times 10^{-14}$ | $4.5 \times 10^{-14}$ | $2.7 \times 10^{-14}$ |
| 100                                                                              | $3.15 \times 10^{-4}$ | $7.21 \times 10^{-3}$ | $5.9 \times 10^{-14}$ | $4.2 \times 10^{-14}$ | $5.3 \times 10^{-14}$ | $8.2 \times 10^{-15}$ |
| 200                                                                              | $1.84 \times 10^{-4}$ | $1.23 \times 10^{-2}$ | $2.6 \times 10^{-13}$ | $2.2 \times 10^{-13}$ | $2.6 \times 10^{-13}$ | $1.2 \times 10^{-13}$ |
| 400                                                                              | $2.51 \times 10^{-6}$ | $6.86 \times 10^{-2}$ | $1.2 \times 10^{-12}$ | $8.9 \times 10^{-13}$ | $1.1 \times 10^{-12}$ | $2.1 \times 10^{-13}$ |
| 800                                                                              | $1.46 \times 10^{-7}$ | $1.00 \times 10^{-1}$ | $1.2 \times 10^{-11}$ | $8.8 \times 10^{-12}$ | $1.0 \times 10^{-11}$ | $1.7 \times 10^{-12}$ |
| 1600                                                                             | $9.34 \times 10^{-9}$ | $1.12 \times 10^{-1}$ | $2.3 \times 10^{-11}$ | $1.7 \times 10^{-11}$ | $2.3 \times 10^{-11}$ | $6.5 \times 10^{-12}$ |
| <i>sinusoidal medium</i> , <i>equilibrium</i> , $\ e\ _2/\ u\ _2$                |                       |                       |                       |                       |                       |                       |
| 101                                                                              | $5.83 \times 10^{-3}$ | $1.62 \times 10^{-2}$ | $9.7 \times 10^{-5}$  | $2.0 \times 10^{-14}$ | $5.0 \times 10^{-6}$  | $3.0 \times 10^{-14}$ |
| 201                                                                              | $1.38 \times 10^{-4}$ | $3.89 \times 10^{-2}$ | $1.6 \times 10^{-5}$  | $3.6 \times 10^{-10}$ | $8.6 \times 10^{-7}$  | $1.3 \times 10^{-13}$ |
| 401                                                                              | $7.38 \times 10^{-6}$ | $1.21 \times 10^{-1}$ | $1.8 \times 10^{-6}$  | $8.0 \times 10^{-11}$ | $7.5 \times 10^{-8}$  | $1.6 \times 10^{-13}$ |
| 801                                                                              | $4.77 \times 10^{-7}$ | $1.70 \times 10^{-1}$ | $1.6 \times 10^{-7}$  | $6.7 \times 10^{-12}$ | $5.6 \times 10^{-9}$  | $2.2 \times 10^{-12}$ |
| 1601                                                                             | $3.07 \times 10^{-8}$ | $1.90 \times 10^{-1}$ | $1.2 \times 10^{-8}$  | $3.4 \times 10^{-11}$ | $3.6 \times 10^{-10}$ | $1.4 \times 10^{-11}$ |

from 100 to 1600 nodes, and far smaller elsewhere, so that the relative  $\ell^2$  error goes as  $h n^{-1/2}$ . In the maximum norm it is first order.

*Remark 4.1* (The elliptic problem). The left column of Figure 2 and Table 4 are the sanity check and the limit case, not a ranking (Section 3.5). On the two-constant medium the seeds are exact at every width,  $1.9 \times 10^{-14}$  at 50 nodes and  $\delta = 0.01$ , growing with  $n$  to  $1.7 \times 10^{-11}$  at 1600 through the direct solve's conditioning, as the interface-aware operator's own line at  $\delta = 0$  does; T1-FV is exact too,  $10^{-14}$  to  $6.5 \times 10^{-12}$ . The naive operator shows the knee of the ramp problem, from  $7.6 \times 10^{-3}$  at  $h = 4\delta$  to  $9.3 \times 10^{-9}$  at  $h = \delta/8$  for  $\delta = 0.01$ , and the construction the same floor. On the sinusoidal medium the equilibrium does rank the two, for a reason that has nothing to do with the edge. T1-FV stays exact,  $3 \times 10^{-14}$  to  $1.4 \times 10^{-11}$  at  $\delta = 0$ , while the seed operator is exact only where its seeded rows cover the layer, at  $\delta = 0.04$  and, at 101 nodes,  $\delta = 0.01$ ; elsewhere its error is that of the standard rows inside the layer,  $9.7 \times 10^{-5}$  to  $1.2 \times 10^{-8}$  at  $\delta = 0$  over 101 to 1601 nodes and  $5.0 \times 10^{-6}$  to  $3.6 \times 10^{-10}$  at  $\delta = 0.0025$ .

**4.6. A snapshot.** Figure 4 and Table 5 put the four operators on one grid: the ramp problem at  $t = 2$  on 100 nodes of the two-constant medium with  $\delta = 0.0025$ , so that  $h = 8.08\delta$ . It is the row of the sweep at which the four lines are ordered and apart: the naive operator at  $2.04 \times 10^{-3}$ , the construction 2.9 times below it, T1-FV 22 times below that, and the seeds 1100 times below that, at  $2.95 \times 10^{-8}$ . The naive operator's error is largest beside the edge,  $1.5h$  to its left, and carries a node-to-node oscillation, the checkerboard mode of  $D_x A D_x$  excited by the sampled edge; but 84% of  $\|e\|_2^2$  lies more than  $2h$  from the edge, on the side where  $\alpha = 1/9$  and the profile

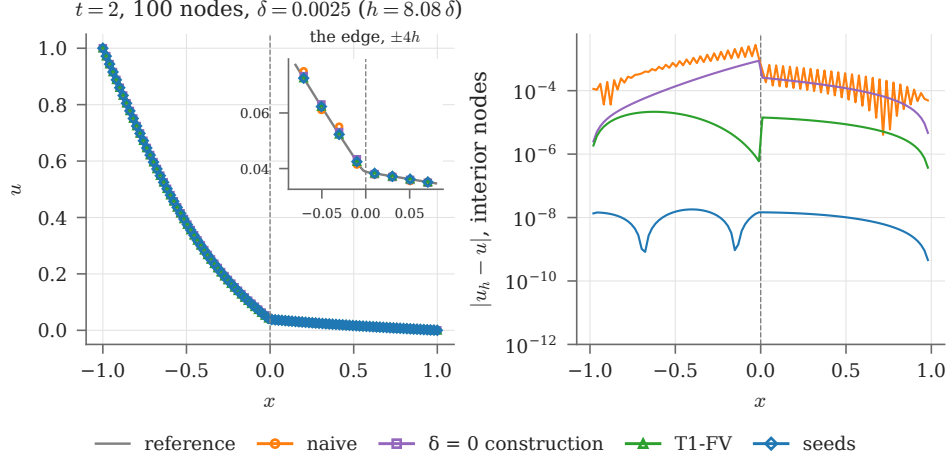


FIGURE 4. The ramp problem at  $t = 2$  on 100 nodes of the two-constant medium with  $\delta = 0.0025$  ( $h = 8.08\delta$ ). Left: the reference and the nodal solutions of the four operators, with the edge enlarged over  $\pm 4h$ . Right: the pointwise error  $|u_h - u|$  at the interior nodes. Dashed: the edge's centre.

TABLE 5. The snapshot of Figure 4:  $\|e\|_2/\|u\|_2$  at the nodes, the largest nodal error and where it lies, and the share of  $\|e\|_2^2$  from the nodes within  $2h$  of the edge.

| operator                  | $\ e\ _2/\ u\ _2$     | $\max  e $            | at $x$ | share within $2h$ |
|---------------------------|-----------------------|-----------------------|--------|-------------------|
| naive $D_x A D_x$         | $2.04 \times 10^{-3}$ | $2.78 \times 10^{-3}$ | -0.030 | 0.16              |
| $\delta = 0$ construction | $7.12 \times 10^{-4}$ | $8.79 \times 10^{-4}$ | -0.010 | 0.23              |
| T1-FV                     | $3.20 \times 10^{-5}$ | $2.16 \times 10^{-5}$ | -0.616 | 0.03              |
| seeds                     | $2.95 \times 10^{-8}$ | $1.80 \times 10^{-8}$ | -0.394 | 0.07              |

is steep. The construction's error is smooth and global, the profile shifted by the resistance deficit, and largest at the node half a cell from the edge. T1-FV's is largest far from the edge, at  $x = -0.62$ : the face conductances take the edge exactly, and what remains is second-order truncation on the curved profile. The seeds' error is the  $\delta = 0$  line's at this count, flat across the domain but for two sign changes, and about half of it is BD4's: with  $\Delta t = h/8$  it is  $1.47 \times 10^{-8}$ .

## 5. SEEDS IN TWO DIMENSIONS

A stencil of the interface group (Section 2.3) has 30 nodes and degree four, and the jump-aware construction replaces its fifteen monomials by the translated basis. Here they are replaced by seeds, the profiles at  $t = 0$  of solutions polynomial in time, anchored at the evaluation node as in Section 3 and now functions of two variables. The node sets, the stencil

sizes, the Gaussians and the saddle-point solve are kept. So is the rest of the operator: the bulk rows, the Dirichlet rows and the time stepping. The construction for a straight edge is the companion paper's for the wave equation [Mar26], specialised to one field (Sections 5.1 and 5.2); a curved or tangentially varying edge and a resistive layer thinner than the stencil each add to it (Sections 5.3 and 5.4).

**5.1. A straight feature. Frame and anchor.** A seeded stencil keeps the orientation of the jump-aware frame: at the foot point of the interface nearest the evaluation node,  $x'$  along the tangent and  $y'$  along the normal (Section 2.4). Its origin moves to the evaluation node  $x_e$ , which lies on the normal through that foot point, and lengths are measured in the stencil's radius  $h_s$ , the largest distance from  $x_e$  to a node of the stencil:

$$(14) \quad \xi = (x' - x'_e)/h_s, \quad \eta = (y' - y'_e)/h_s.$$

The material along the normal line,  $\alpha_n(\eta) = \alpha(x_e + h_s \eta n)$  with  $n$  the unit normal, is what the seeds read, and  $\alpha_e = \alpha_n(0)$ . In these coordinates  $L = h_s^{-2}(\partial_\xi \alpha \partial_\xi + \partial_\eta \alpha \partial_\eta)$ ; as in Section 3.1 we drop the factor  $h_s^{-2}$  in what follows and restore it in the weights,  $w = \tilde{w}/h_s^2$ .

*The ansatz.* Where  $\alpha$  depends on the normal coordinate alone, as it does exactly across the flat edges of case 1, the seed of the monomial  $\xi^a \eta^b$  is a polynomial in  $\xi$  whose coefficients are functions of  $\eta$ ,

$$(15) \quad \phi_{ab}(\xi, \eta) = \sum_j g_j(\eta) \xi^j, \quad j = a, a-2, \dots, a \bmod 2,$$

because  $L(g(\eta)\xi^j) = \xi^j(\alpha_n g')' + j(j-1)\alpha_n g \xi^{j-2}$ : the operator lowers the degree in  $\xi$  by two or not at all. The chain (2) read in two dimensions is

$$(16) \quad \phi_{00} = 1, \quad L\phi_{ab} = \alpha_e[a(a-1)\phi_{a-2,b} + b(b-1)\phi_{a,b-2}],$$

the constant-coefficient operator's action on the monomial with the lower monomials' seeds on the right. Collecting  $\xi^j$  and writing the flux  $\psi_j = \alpha_n g'_j$  gives a first-order pair for each seed and level,

$$(17) \quad \begin{aligned} g'_j &= \psi_j / \alpha_n, \\ \psi'_j &= \alpha_e[a(a-1)g_j^{(a-2,b)} + b(b-1)g_j^{(a,b-2)}] - (j+2)(j+1)\alpha_n g_{j+2}^{(a,b)}, \end{aligned}$$

where a superscript names the seed a level belongs to and a missing level is zero. The initial data are the monomial's at its top level,  $g_a(0) = [b=0]$  and  $\psi_a(0) = \alpha_e[b=1]$ , and zero for every other level; for  $b \geq 2$  the source generates  $\eta^b$ , as  $\phi_k(x_e) = \phi'_k(x_e) = 0$  did in one dimension. Seed  $(a, b)$  at level  $j$  needs the same level of the seeds  $(a-2, b)$  and  $(a, b-2)$  and level  $j+2$  of itself, so the system is triangular in total degree and in level, and the fifteen seeds of degree at most four are 22 levels with 44 states, marched together as one linear system that never differentiates  $\alpha$ . With a constant  $\alpha$  the seeds are the monomials, by induction on the degree as in Lemma 3.2. The proof of that lemma, applied once in each index, shows

that  $u_{ab} = \sum_{i \leq a/2, k \leq b/2} c_i^a c_k^b (\alpha_e t)^{i+k} \phi_{a-2i, b-2k}$ , with  $c_i^a = a!/(i!(a-2i)!)$ , solves the heat equation in these coordinates with  $u_{ab}(\cdot, 0) = \phi_{ab}$ : the seeds are again the profiles of solutions polynomial in time.

*Closed forms.* The linear seeds are the one-dimensional ones along the normal, stretched along the tangent:  $\phi_{10} = \xi$ ,  $\phi_{01} = \alpha_e \int_0^\eta d\eta'/\alpha_n$ , the constant-flux seed  $\phi_1$  of Section 3.1, and  $\phi_{11} = \xi\phi_{01}$ ;  $\phi_{02}$  is the seed  $\phi_2$  along the normal. The first seed that is not a one-dimensional seed times a power of  $\xi$  is

$$(18) \quad \phi_{20} = \xi^2 + g_0(\eta), \quad (\alpha_n g_0')' = 2(\alpha_e - \alpha_n), \quad g_0(0) = g_0'(0) = 0.$$

$L\xi^2 = 2\alpha_n$  is not constant across the edge, and  $g_0$  is the normal profile that restores  $L\phi_{20} = 2\alpha_e$ , the constant rate of change of a temperature quadratic along the edge. It vanishes on a constant  $\alpha$  and is of order one in stencil units across an unresolved edge (Figure 5), and no one-dimensional construction supplies it: the two-dimensional content of the seeds is this coupling of tangential monomials to normal profiles. It is also why the elliptic problem tests them in two dimensions where it could not in one (Section 3.5): case 1's equilibrium  $\sin 2\pi x v(y)$  is not in the span of the one-dimensional seeds at any resolution.

*A symmetry, and the march.* The frozen problem is invariant under shifts along  $\xi$ , and matching powers of the shift gives  $g_j^{(a,b)} = \binom{a}{j} g_0^{(a-j,b)}$ , with  $g_0^{(m,b)} = 0$  for odd  $m$ : only nine level-zero functions are independent. The 44-state march reproduces the identity to  $7 \times 10^{-16}$  at every  $\delta$  tested, from the jump to  $8h$ , and the monomials on a constant  $\alpha$  to  $1.4 \times 10^{-15}$ ; we march all 44 states because it is the coupled form that carries over to a curved or tangentially varying edge (Section 5.3). The march is that of Section 3.1: SciPy's DOP853 at the same tolerances in the stencil coordinate, from the anchor in both directions, restarted at every node's  $\eta$  so that node values are integrated and not interpolated, with segment boundaries at each edge centre and at  $10\delta$  on either side;  $\alpha$  on a segment is read from the piece that owns it, so a jump is marched like any other width, and a band thinner than the stencil is crossed in one march with both of its edges as boundaries. The true operator applied to the marched seeds by twelfth-order differences on a  $25 \times 801$  grid leaves  $L\phi_{ab}$  minus the right-hand side of (16) below  $10^{-10}$  relative, the floor of the differences.

**5.2. Stencil assembly.** *The saddle-point system.* The seeds take the monomials' place in (6):  $P_{ie} = \phi_e(\xi_i, \eta_i)$ , read off the march at the nodes, and the right-hand side is the true operator on each seed at the anchor,

$$(19) \quad (L\phi_{ab})(x_e) = 2\alpha_e [(a, b) \in \{(2, 0), (0, 2)\}] + h_s \alpha_\xi [(a, b) = (1, 0)],$$

with  $\alpha_\xi$  the tangential derivative of  $\alpha$  at the anchor. The chain gives the first term, since the right-hand side of (16) vanishes at the anchor for every seed but the two quadratics, whose lower seed is  $\phi_{00} = 1$ . The second is what the frozen normal profile leaves out:  $\partial_\xi(\alpha \partial_\xi \phi) = \alpha_\xi \phi_\xi + \alpha \phi_{\xi\xi}$ , and

at the anchor  $\phi_\xi \neq 0$  only for  $\phi_{10} = \xi$ . The normal derivative of  $\alpha$  never appears, since the seeds satisfy  $\partial_\eta \alpha_n \partial_\eta$  with the true profile. Where the jump-aware rows apply  $\alpha \nabla^2 + \nabla \alpha \cdot \nabla$  to all fifteen functions of their basis, the seeds carry  $\nabla \alpha \cdot \nabla$  through the march and need it at the anchor once, in one entry.

*The warp is the constant-flux seed.* The jump-aware stencils warp the Gaussians' normal coordinate piecewise-linearly, with its slope changed by the factor  $\alpha^-/\alpha^+$  at the interface (Section 2.4). The seeds bring the smooth version with them: the Gaussians are written in  $(\xi, \tilde{\eta})$  with

$$(20) \quad \tilde{\eta} = \phi_{01}(\eta) = \alpha_e \int_0^\eta \frac{d\eta'}{\alpha_n(\eta')},$$

read off the same march at every node. With  $\tilde{\eta}' = \alpha_e/\alpha_n$  the chain rule gives, at any point of the normal line,

$$L G(\xi, \tilde{\eta}(\eta)) = \alpha_\xi G_\xi + \alpha G_{\xi\xi} + (\alpha \tilde{\eta}')' G_{\tilde{\eta}} + \alpha \tilde{\eta}'^2 G_{\tilde{\eta}\tilde{\eta}},$$

and  $\alpha \tilde{\eta}' = \alpha_e$  is constant, the flux of  $\phi_{01}$ , so the term in  $G_{\tilde{\eta}}$ , which the jump-aware rows carry as  $\alpha_\eta G_\eta$ , cancels exactly. At the anchor

$$(21) \quad (LG)(x_e) = \alpha_e (G_{\xi\xi} + G_{\tilde{\eta}\tilde{\eta}}) + \alpha_\xi G_\xi,$$

the Gaussian's Laplacian at the warped offsets times  $\alpha_e$ , with the seeds' tangential correction. At  $\delta = 0$  on constant pieces (20) is the jump-aware warp, to  $1.8 \times 10^{-15}$  at the nodes of case 1's 576 crossing stencils at 2500 nodes. The implementation assembles the general chain-rule form and checks the cancellation along the whole marched line rather than assuming it, and takes  $\varepsilon$  from the physical offsets, as the jump-aware stencils do. Written in the frame's own  $(\xi, \eta)$  the Gaussians are plain and carry  $\alpha_\eta G_\eta$ ; Section 6.4 measures what the warp is worth.

*The jump as the limit.* At  $\delta = 0$  on constant pieces across a flat interface, the seed of  $\xi^a \eta^b$  is the monomial on the anchor's side and, on the far side, a polynomial of degree at most four, since the chain has constant coefficients and polynomial sources there. It is continuous with its flux at every  $\xi$ , and by induction down the chain so are  $L^k \phi$  and their fluxes: the fifteen conditions of the continuity matrices, all exact for constant pieces on a flat line (Section 2.4). The matrices being square and nonsingular, the far side is the translated polynomial, and the seeds and the translated basis are two bases of one space, anchored at the node and at the foot point. With the same Gaussians the seed rows are therefore the jump-aware rows, the two-dimensional form of the one construction with two implementations of Section 3.2. Over the 576 crossing stencils of case 1 at 2500 nodes the sine of the largest principal angle between the two spans is  $4.9 \times 10^{-14}$ , and the weights differ by  $1.6 \times 10^{-12}$  relative with plain Gaussians and by  $7.6 \times 10^{-12}$  with warped ones; assembled, the two operators agree to  $5.3 \times 10^{-13}$ . On a band one spacing thick, 333 crossing stencils of which 240 reach all three regions, where the jump-aware construction translates twice with a change

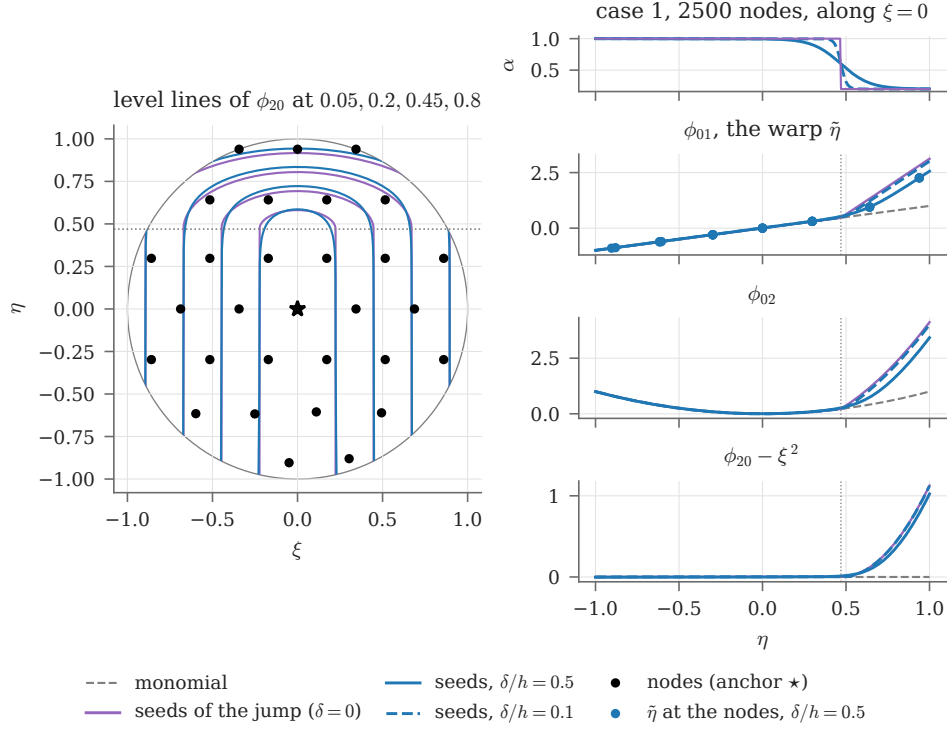


FIGURE 5. The seeds of one 30-node stencil of case 1 at 2500 nodes, anchored (star) on the second straddling row below the lower edge (dotted), in the stencil coordinates (14); the edge is at  $\eta = 0.47$  and 7 of the nodes lie across it. Left: the nodes and the level lines of  $\phi_{20}$  for the jump (purple) and for  $\delta = h/2$  (blue). Below the edge they are those of  $\xi^2$ ; above it they close, since  $g_0$  adds the normal curvature that keeps  $L\phi_{20} = 2\alpha_e$  where  $\alpha = 0.2$ . Right:  $\alpha$  and three seeds along the normal line  $\xi = 0$  against the monomials (grey, dashed): the constant-flux seed  $\phi_{01}$ , which is the Gaussians' normal coordinate  $\tilde{\eta}$  (dots: its values at the nodes for  $\delta = h/2$ ),  $\phi_{02}$ , and  $g_0 = \phi_{20} - \xi^2$ , which vanishes on a constant  $\alpha$ . Purple: the jump's seeds, which span the translated basis of the jump-aware stencil; blue:  $\delta = h/2$  (solid) and  $h/10$  (dashed). The anchor's  $\alpha_e$  is within 0.4% of the jump's at  $\delta = h/2$ , so the seeds are drawn as marched, unnormalised.

of frame between the interfaces and the march crosses both in one pass, the span distance is  $1.7 \times 10^{-14}$  and the weights differ by  $6.0 \times 10^{-13}$ . At  $\delta > 0$  the seed span leaves the jump's at first order in  $\delta/h$ , 0.46 to  $0.74\delta/h$  across four anchors, with no floor down to  $\delta = 10^{-5}h$ . On a curved interface, or

with pieces that vary along it, the two constructions differ at  $O(h)$ , and Section 6 measures each against the reference.

*Which rows are seeded.* A row is seeded when the signed distances of its 30 nodes to an edge's centre curve span a range that meets  $[-20\delta, 20\delta]$ , beyond which the edge is its pieces to the last bit (Section 2.1), or when its nodes straddle the curve; at  $\delta = 0$  the rule is the jump-aware construction's crossing test. The stencil groups follow the same rule one size up: a node whose 42-node stencil sees an edge joins the interface group and gets a 30-node stencil of degree four, so that no bulk stencil sees an unresolved edge, and of the group the members whose own 30 nodes see it are seeded. A stencil of the boundary zone that sees an edge is seeded on its own one-sided nodes, since the chain is anchored at the node and marched only where nodes are. Once  $20\delta$  covers the strip every row is seeded. No threshold in  $\delta$  switches the seeds off: a resolved edge costs them nothing but the marches (Section 6.4).

*Stencil size, conditioning and cost.* The seed stencils are the interface group's, 30 nodes and degree four. Our companion paper found degree-four seed stencils unstable across a sharp edge in the wave equation and used 19 nodes and degree three [Mar26]; the cause there was the footprint of the hyperviscosity row the wave operator needs, which has no counterpart in an implicit march of a dissipative operator, and the spectra of Section 6.3 show no eigenvalue in the right half-plane. Column-scaled, the seed block's condition number stays within 0.46 to 2.5 times the monomial block's on the same nodes at every  $\delta$  from  $10^{-5}h$  to  $8h$ ; unscaled, it carries the contrast, from 55 with the edge resolved to 212 with it unresolved on the stencil below the band. A seed row costs 1.2 to 5 times a jump-aware row, a few milliseconds.

**5.3. Curved and tangentially varying features.** *A frozen profile.* For a curved edge the natural first step, the companion paper's for the wave equation [Mar26], marches the straight-feature seeds along the true normal through the foot point and drops the curvature. The seeds' kink then lies on the tangent line instead of on the curve: a node at tangential offset  $x'$  is at normal distance  $y' - \kappa x'^2/2 + \dots$  from a curve of curvature  $\kappa$ , while its seed value assumes  $y'$ . Moving an unresolved edge by  $\Delta$  changes the values beyond it by  $\Delta$  times the jump in the slope that carries the flux, whatever  $\delta$  is: an error of  $O(\kappa h^2)$  in value and, in a row whose weights are  $O(h^{-2})$ , a truncation error of  $O(\kappa)$  that does not converge. At  $\delta = 0$  on constant pieces this frozen profile is the jump-aware construction without its curvature terms, whose flat line in [MF17b, Fig. 10] is first order. A variation of  $\alpha$  along an unresolved edge does the same another way: the ratio of fluxes the seeds impose across it changes from node to node, a kinked correction outside their span. Either term alone makes a frozen profile inconsistent at  $O(1)$  through an unresolved edge; Section 6.5 measures the two apart.

*The curve's own coordinates.* Write every point near the foot curve  $\gamma(\theta)$ , in its own parameter, as  $x = \gamma(\theta) + d n(\theta)$ , with  $n$  the unit normal and  $\kappa$  the signed curvature. These coordinates are one-to-one inside the focal distance, and the implementation refuses a stencil whose normal line reaches within a margin of it,  $\kappa_{\max}(|d_e| + 2h_s) \geq 0.8$ , with  $\kappa_{\max}$  the curve's largest curvature and  $d_e$  the anchor's distance to it. The metric is diagonal, with scale factors  $\mu = |\gamma'| (1 - \kappa d)$  along  $\theta$  and 1 along  $d$ , so that

$$(22) \quad Lu = \frac{1}{\mu} \left[ \partial_\theta \left( \frac{\alpha}{\mu} \partial_\theta u \right) + \partial_d (\alpha \mu \partial_d u) \right],$$

in divergence form with no cross term. About the anchor  $(\theta_0, d_e)$  the stencil coordinates are  $\xi = \mu_e(\theta - \theta_0)/h_s$  and  $\eta = (d - d_e)/h_s$ , with  $\mu_e = \mu(\theta_0, d_e)$ ; the operator keeps its form with  $\hat{\mu} = \mu/\mu_e$  in place of  $\mu$ , and  $\hat{\mu} = 1$  at the anchor, so that  $\xi$  is arclength along the anchor's parallel curve in stencil radii. The normal line  $\xi = 0$  is the frozen profile's, with the same  $\alpha_n$ , segment boundaries and pieces. What changes is that the foot curve is the line  $\eta = -d_e/h_s$  at every  $\xi$ , so the edge is a coordinate line and its conditions hold along one.

*Coupled levels.* Keep the ansatz (15), now with every level,  $\phi_{ab} = \sum_{j=0}^4 g_j(\eta) \xi^j$ , and expand the coefficients of (22) along the coordinate lines,  $\alpha/\hat{\mu} = \sum_k A_k(\eta) \xi^k$ ,  $\alpha\hat{\mu} = \sum_k B_k(\eta) \xi^k$  and  $\hat{\mu} = \sum_k M_k(\eta) \xi^k$ ,  $k \leq 4$ . The chain (16) is kept as it stands, multiplied through by  $\hat{\mu}$ . With  $S\phi_{ab}$  its right-hand side and  $\psi_i = \sum_k B_k g'_{i-k}$  the coefficient of  $\xi^i$  in the normal flux, collecting  $\xi^i$  gives

$$(23) \quad \begin{aligned} g'_i &= \frac{1}{B_0} \left[ \psi_i - \sum_{k \geq 1} B_k g'_{i-k} \right], \\ \psi'_i &= \alpha_e \sum_k M_k (Sg)_{i-k} - \sum_k A_k (i+2-k)(i+1) g_{i+2-k}, \end{aligned}$$

a missing level being zero. The first equation is triangular, so  $\alpha$  is still never differentiated, and  $g_i$  and  $\psi_i$  are continuous through a jump on the coordinate line, level by level, so the march reads its pieces one-sidedly as before. On a flat line with  $\alpha$  a function of  $\eta$  alone, (23) is (17) and the levels of the other parity stay zero. The initial data are those of Section 5.1; the parity trim and the shift identity do not survive.

*Every seed to level four.* The series coefficients are small:  $A_k$  and  $B_k$  are  $O(h^k)$  and  $M_k$  is  $O(h^{k+1})$  for  $k \geq 1$ . Level  $j$  of seed  $(a, b)$  is therefore  $O(h^{j-a})$  above its top and contributes  $O(h^{b+j})$  to a smooth solution, and an order count would keep it only while  $b+j \leq 4$ , 55 levels. The count hides a constant. Across a jump the far side of the seed of  $\xi^a \eta^b$  carries the flux ratio  $\alpha_e/\alpha_{\text{far}}$  to the power  $\lceil b/2 \rceil$ , and the dropped levels multiply it by the variation of  $\alpha$  along the edge. On case 2 at  $\delta = 0$  and 40,000 nodes the cut chain's error was  $3.55 \times 10^{-8}$ ; with every seed to level four it is  $7.18 \times 10^{-9}$ ,

and two more levels on every seed change nothing to two digits. The chain has 75 levels, 150 states.

*Right-hand sides, series and scope.* At the anchor only level zero of  $L\phi$  survives, and its equation is marched with every term, so the chain holds on the whole line  $\xi = 0$  and  $(L\phi_{ab})(x_e)$  is  $2\alpha_e$  on the two quadratics and zero otherwise. The entry  $h_s\alpha_\xi$  of (19) is gone: the seed of  $\xi$ , now  $\xi$  plus lower levels driven by  $A_1, A_2, \dots$ , satisfies  $L\phi_{10} = 0$ , so the march carries the tangential derivative. The warp is  $\phi_{01}$  with all its levels, and at the anchor  $(LG)(x_e) = \alpha_e(G_{\xi\xi} + G_{\eta\eta}) + A_1(0)G_\xi$ , the form of (21) with  $A_1(0) = \partial_\xi(\alpha/\hat{\mu})$  in place of  $h_s\alpha_\xi$ . The coefficients  $A_k, B_k$  and  $M_k$  come from Fornberg's weights [For88] at  $\xi = 0$  on 11 points 0.2 apart along the coordinate lines, whose positions are formed once per stencil and are exact. The foot edge's factor  $\sigma(d/\delta)$  depends on  $\eta$  alone and is applied exactly; the other edge is taken as its pieces beyond  $20\delta$  of the sampled region and otherwise read through a 17-point Chebyshev interpolant of its distance along each sample line. Between 11 points at spacing 0.2 and 15 at 0.12 the weights of 40 rows of case 2 at 1250 nodes and  $\delta = 0.0025$  move by  $3.6 \times 10^{-10}$ . At  $\delta = 0$  the other curve may meet a stencil only as a coordinate line of the foot curve's frame, a parallel line or a concentric circle; no stencil of case 2 at 1250 nodes or more crosses both of its curves. A row costs 2.5 to 3.8 times a row of the frozen profile.

**5.4. A thin resistive layer.** The ring of case 3 is  $1/s$  wide and thinner than the node spacing at every count (Section 2.3). No node lies inside it, and a stencil across its midline reaches both sides; its march crosses both circles in one pass along the radial normal, with both edges as segment boundaries. What the ring needs beyond the band is switched on only for a band whose two circles carry their gap, so that cases 1 and 2 are unchanged to the last bit.

*The width and the series.* At  $s \geq 10^{10}$  the stored radii  $0.35 - 1/s$  and  $0.35$  give the width only to  $8 \times 10^{-8}$  relative, and a march between two segment boundaries  $w/h_s$  apart, both of order one in stencil units, crosses the width only to  $\text{ulp}(1)/(w/h_s)$ , about  $10^{-7}$  at  $s = 10^{11}$ . The band therefore carries the gap  $1/s$  exactly, and the march runs in the offset from the outer circle's crossing, so that the segment it crosses is the gap to rounding. Across the ring  $1/(\alpha\hat{\mu})$  is  $1.5s$  at  $\delta = 0$ , and it multiplies the rounding of the sampled series into the small tangential levels, past the integrator's absolute tolerance; on a ring each segment's series are a piecewise Chebyshev interpolant in  $\eta$ , checked against the samples to  $10^{-13}$  of their largest coefficient, so that the integrator's stages see a smooth function.

*The smooth ring.* With  $\delta$  comparable to the width or larger, the band's composition (4) never reaches the ring's plateau and keeps only 0.41 of its contact resistance at  $\delta = w/10$  and 0.01 at  $w/4$  ( $s = 10^3$ ); blending  $\alpha$  by the difference of the two edges keeps a resistance of about  $w$ , against the jump's 1.5. Both lose the resistance that the jump ring holds fixed as  $s$  grows. The

smooth ring blends the resistivity instead,

$$(24) \quad \frac{1}{\alpha} = \frac{1}{\alpha_{\text{out}}} + \left( \frac{1}{\alpha_{\text{in}}} - \frac{1}{\alpha_{\text{out}}} \right) [\sigma(d_1/\delta) - \sigma(d_2/\delta)],$$

with  $d_1$  and  $d_2$  the signed distances to the inner and outer circles, so that its excess resistance  $\int (1/\alpha - 1/\alpha_{\text{out}}) dn$  is the jump's,  $w(1/\alpha_{\text{in}} - 1/\alpha_{\text{out}})$ , at every  $(s, \delta)$ . As  $s \rightarrow \infty$  at fixed  $\delta$  it tends to a layer of peak resistivity  $R/(2\delta)$ , with  $R$  the contact resistance, as the jump ring tends to the resistance itself. A resistive layer thinner than the grid is an imperfect interface in the limit [SP74, PP13, Has01], and strong-form schemes across such a contact need its jump conditions one derivative higher in two dimensions [LP03, LP06]; the flux seeds of Section 5.4 are that observation for the diffusion operator.

*The flux seeds.* With the fifteen seeds the rows beside the ring stall at fine counts (Section 6.6). Across a kink the flux enters the far side's values multiplied by the distance beyond it; across a thin resistive layer it enters with no such factor, as the jump  $[u] = RF$  with  $F$  the flux through the layer. The seeds of degree four that carry a flux, those with  $b \geq 1$ , have a flux at the ring that is a polynomial of degree three in the offsets; the true flux differs from it by  $O(h^4)$ , which  $R$  turns into an  $O(h^4)$  error in the far side's values and an  $O(h^2)$  error in a row that reads many of them. Fitted by least squares to an exact mode through the ring, the fifteen seeds are  $O(h^5)$  on a stencil's near side alone but  $O(h^4)$  on the rows with 10 to 12 of their 30 nodes across it, where the jump-aware basis is  $O(h^5)$ . On a ring the block therefore adds the five seeds of degree five that carry a flux, those of  $\xi^4\eta$ ,  $\xi^3\eta^2$ ,  $\xi^2\eta^3$ ,  $\xi\eta^4$  and  $\eta^5$ , from the chain of degree five: twenty seeds on thirty nodes, and with them the fit is  $O(h^5)$ . Their moment conditions vanish at the anchor, so the right-hand side is unchanged. An imperfect contact costs strong-form schemes a derivative in the jump conditions, as Lombard and Piraux observed for elastic waves [LP03, LP06]; the flux seeds add the functions of one degree higher that carry the flux, and nothing else. Why the jump-aware basis, whose continuity conditions also carry the flux only to degree three along the curve, fits at  $O(h^5)$  with fifteen functions we have not established; its Cartesian frame couples the circle's curvature into the flux of its degree-four monomials, which is the likely source.

*The Gaussians on a ring.* Two changes to the Gaussians, both measured on the ring. The warp is the level zero of  $\phi_{01}$  alone: its higher levels carry the ring's resistance, which varies 3 : 1 around it, along  $\xi$  into the far side's warped coordinate, a shear of about 4 per stencil radius at every  $h$ , and Gaussians on that coordinate put a floor under the ring's line,  $3.3 \times 10^{-6}$  at 80,000 nodes against  $2.3 \times 10^{-7}$  with level zero alone. Level zero has the full seed's flux and derivative at the anchor, so (21) is unchanged. And for an anchor inside a smooth ring, where  $\alpha_e$  is a twentieth of the pieces' or less, the warp compresses the neighbours by that factor, so  $\varepsilon$  is taken from the nodes' spacing in the warped coordinates instead of the physical one.

*Rows anchored in an edge.* The warp's slope  $\alpha_e/\alpha$  is 1 across the anchor's side of the stencil only where the anchor sits on its piece. The resistivity tail of a smooth ring reaches several  $\delta$  out, and an anchor in it has  $\alpha_e$  below its piece's value, so the warp squeezes the anchor's own side, by 0.2 to 0.4 on the rows concerned. Their consistency is not the problem, since the truncation error on those rows keeps falling; their stability is: the anchor's share of such a row,  $-w_e/\sum_{j \neq e} |w_j|$ , is 0.08 to 0.13, against 0.25 to 0.27 with plain Gaussians. Plain Gaussians fail the other way where the edge is nearly resolved: some of those rows then get an anchor weight of the wrong sign. So on a ring a row whose anchor is off its piece,  $|\ln(\alpha_e/\alpha_{\text{piece}})| > 0.05$  with  $\alpha_{\text{piece}}$  the jump's value there, is solved with both Gaussian blocks on its one march and keeps the weights with the larger diagonal share, the diagonal dominance of [Mar16, App. B] with its sign; a row on its piece keeps the warp, and at  $\delta = 0$  every anchor is on its piece. Off the ring the rule helps on the flat band where it fires and is mixed on case 2, where the row-by-row choice at near-ties can do worse than either uniform choice, so it stays the ring's safeguard.

## 6. TWO-DIMENSIONAL RESULTS

This section measures the operators of Section 2.3 and the seeds of Section 5 on the three cases, in the order of the one-dimensional study: the references, the naive operator and the  $\delta = 0$  construction, the matrices the seeds make, the seeds on the flat band of case 1, on the curved band of case 2 and on the ring of case 3, the coefficient treatments, and a snapshot. Node counts double from 1250 to 40,000; on case 1 the naive operator and the construction, and the seeds at the jump, run to 160,000. Case 1 has the widths  $\delta = 0.04, 0.01, 0.005$  and  $0.0025$  beside the jump, with  $h = \delta$  near 11,000, 44,000 and 177,000 nodes for the last three and  $h < \delta$  at every count for the first; case 2 has the last three. Errors are RMS errors over all nodes, and orders are least-squares fits in  $h$  over the counts named (Section 2.1). Each count has one node set, and the naive operator's error at one count moves by up to a factor of two from one node set to another (Section 6.2), so an order is a fit and never one pair of counts. The parabolic problem of case 1 takes 3 to 19 BD4 steps over 1250 to 40,000 nodes: it tests the operator's spectrum under BD4 more than a transient through the edge.

**6.1. Test problems and references.** On case 1 the smooth medium blends the band's two edges by (4) in turn, from the outside piece into the band and back out, so that in  $y$  it is the one-dimensional medium of the same construction. At  $\delta = 0.04$  the band is five widths wide and never reaches its plateau: on its midline  $\alpha = 0.2107$ , against the band's 0.2. That width is a resolved edge on a band that is not the jump's, and its lines, like every width's, are read against its own reference.

The references of case 1 are separable. With  $\alpha$  a function of  $y$  alone,  $u = e^{ct} \sin 2\pi x v(y)$ , where

$$(\alpha v')' - (4\pi^2 \alpha + c) v = 0, \quad v(0) = 0, \quad v(1) = 1,$$

with  $c = 0$  at equilibrium and  $c = 1$  for the parabolic mode. The profile  $v$  is the Chebyshev collocation of Section 4.1 on the same elements, with the term in  $4\pi^2 \alpha$  added, and the parabolic reference is exact in  $t$ : BD4 starts from its history, and no time integrator enters the reference. Against a second resolution the equilibrium profiles agree to between  $7 \times 10^{-13}$  and  $2 \times 10^{-11}$  over the widths, the parabolic ones to  $2.4 \times 10^{-11}$  at worst (Table 6), and at the jump the profile is the analytic one to  $3.8 \times 10^{-13}$ . The smooth problem's distance from the jump's is first order in  $\delta$ :  $\sup |v_\delta - v_0|/\delta$  rises towards about 1.6 as  $\delta$  falls.

Case 2 does not separate. Its reference is a product grid on the shear  $y = \eta + 0.02 \sin(2\pi x) \beta(\eta)$ , with  $\beta$  the cubic that vanishes at  $\eta = 0$  and 1 and equals 1 at  $\eta = 0.6$  and 0.8, so that both curves and both Dirichlet rows are coordinate lines. The equation is collocated in flux form on 49 Fourier points in  $x$  times the Chebyshev elements of case 1 in  $\eta$ , with  $u$  and its flux through the lines  $\eta = \text{const}$  matched at every element end and each element reading its own  $\alpha$ , so that a tanh edge stays centred on the element cuts at every  $x$  and a jump is taken by the same solver. Each row of the banded system is scaled by its largest entry before the sparse factorisation. The grid agrees with finer ones in both directions to  $10^{-11}$  at every width and on both problems, and at the jump a 160,000-node run of the jump-aware operator lies  $4.3 \times 10^{-9}$  from it in the RMS. It is case 2's reference at every  $\delta$ , the jump included.

The ring of case 3 has no such reference. At  $\delta = 0$  we read it against the port's 160,000-node jump-aware runs at each  $s$ , whose own error is about  $10^{-7}$ , and at  $\delta > 0$  against a 160,000-node run of the seeds at the same  $s$  and  $\delta$ , on the nodes away from the ring (Section 6.6).

**6.2. Naive baseline.** Figure 6 and Table 7 give the naive operator  $D_x A D_x + D_y A D_y$  on case 1 at every width, and Table 8 the  $\delta = 0$  construction.

*The knee.* At the jump the naive operator is first order: its error falls from  $4.14 \times 10^{-3}$  at 1250 nodes to  $2.07 \times 10^{-4}$  at 160,000, a fit of 1.23, with rates between successive counts from  $-0.5$  to  $3.1$  that belong to the node sets. Every line carries that scatter: at one count and width the naive error spreads by 1.03 to 2.0 times over three node sets, and on one node set the rates alone do not show the knee. Divided by the jump's error on the same nodes, the scatter cancels, and the knee is nearly a function of  $h/\delta$  alone. While  $h \gtrsim 6\delta$  the naive error is 0.86 to 0.94 of the jump's at equilibrium (0.78 to 0.97 on the parabolic problem); it falls to 0.19 to 0.27 at  $h \approx 2\delta$ , stalls at 0.081 to 0.084 at  $h \approx \delta$  for all three widths that reach it, and drops to 0.006 to 0.007 at  $h \approx 0.53\delta$  (0.0026 for  $\delta = 0.04$  at  $0.52\delta$ ). Between

TABLE 6. The references' own accuracy: the largest difference from a finer reference, on case 1 in  $y$  at 4001 points for both problems ( $c = 0$  and  $c = 1$ ), on case 2 in  $x$  and in  $\eta$  at 4000 points of the strip (the larger over the two problems). The fourth column is the distance  $\sup |v_\delta - v_0|$  of case 1's profile from the jump's, and at  $\delta = 0$  from the analytic solution. Last line: the RMS difference, at its nodes, between case 2's grid at  $\delta = 0$  and a 160,000-node jump-aware solution.

| $\delta$                                                                        | case 1: Chebyshev in $y$ |                       |                         |               | case 2: Fourier $\times$ Chebyshev |                       |
|---------------------------------------------------------------------------------|--------------------------|-----------------------|-------------------------|---------------|------------------------------------|-----------------------|
|                                                                                 | finer, $c = 0$           | finer, $c = 1$        | $\sup  v_\delta - v_0 $ | $\div \delta$ | finer in $x$                       | finer in $y$          |
| 0                                                                               | $7.2 \times 10^{-13}$    | $4.1 \times 10^{-13}$ | $3.8 \times 10^{-13}$   | —             | $1.2 \times 10^{-13}$              | $5.7 \times 10^{-13}$ |
| 0.04                                                                            | $1.1 \times 10^{-12}$    | $1.4 \times 10^{-12}$ | $2.82 \times 10^{-2}$   | 0.705         | —                                  | —                     |
| 0.01                                                                            | $3.1 \times 10^{-12}$    | $2.1 \times 10^{-12}$ | $1.17 \times 10^{-2}$   | 1.169         | $1.3 \times 10^{-12}$              | $2.3 \times 10^{-12}$ |
| 0.005                                                                           | $4.4 \times 10^{-12}$    | $9.2 \times 10^{-13}$ | $6.80 \times 10^{-3}$   | 1.359         | $1.8 \times 10^{-12}$              | $3.2 \times 10^{-12}$ |
| 0.0025                                                                          | $1.2 \times 10^{-11}$    | $4.7 \times 10^{-12}$ | $3.68 \times 10^{-3}$   | 1.472         | $5.9 \times 10^{-12}$              | $6.6 \times 10^{-12}$ |
| 0.002                                                                           | $1.3 \times 10^{-11}$    | $1.7 \times 10^{-11}$ | $2.99 \times 10^{-3}$   | 1.496         | —                                  | —                     |
| 0.001                                                                           | $9.4 \times 10^{-12}$    | $1.8 \times 10^{-11}$ | $1.55 \times 10^{-3}$   | 1.549         | —                                  | —                     |
| 0.0005                                                                          | $2.1 \times 10^{-11}$    | $2.4 \times 10^{-11}$ | $7.89 \times 10^{-4}$   | 1.578         | —                                  | —                     |
| case 2, $\delta = 0$ , vs. a 160,000-node jump-aware run: $4.32 \times 10^{-9}$ |                          |                       |                         |               |                                    |                       |

$h \approx 2\delta$  and  $h \approx \delta/2$  the error falls 311 and 211 times at  $\delta = 0.01$  and 0.005, while over the same counts the jump's falls 8.1 and 6.2 times: a knee 38 and 34 times deep in the jump's units. Below it the order is that of the degree-five bulk stencils, not four:  $\delta = 0.04$ , resolved at every count, fits 5.31 over 1250 to 160,000 nodes, and the same operator with  $\alpha \equiv 1$  on the same nodes fits 4.98 from 2500. So the naive line follows the jump's only while  $h \gtrsim 6\delta$ , and its knee spans  $4\delta \gtrsim h \gtrsim 0.75\delta$ .

*The flux at the edge.* Readings on the fixed rows beside each curve say what is wrong. The fixed rows are equispaced in  $x$ , so a row's  $\sin 2\pi x$  coefficient is  $e^{ct}v$  at that row for the separable mode; on each side of a curve the flux  $\alpha \partial_y u$  at the innermost row is read from the quadratic through that side's three rows, without crossing the curve, and the reference's same reading is subtracted. On the first rows off the edge the naive operator's relative flux error is a function of  $h/\delta$  alone to within a factor of 1.7, between runs at  $\delta$  on  $n$  nodes and at  $\delta/2$  on  $4n$ , which share  $h/\delta$ . While  $h \geq 4\delta$  it is 0.31 to 0.86 of the flux itself at every count, and at the jump it does not converge: it stays at 0.34 to 0.37 from 10,000 to 160,000 nodes while the RMS error falls 20 times. It is 3.5 to 5% at  $h \approx \delta$  and 1% at  $0.75\delta$  (Figure 6, bottom left). The naive solution converges on average, not at the edge.

*The parabolic problem.* The parabolic errors repeat the equilibrium's, at 0.67 to 1.09 times them from 2500 nodes on at every width, so the knee belongs to the operator. At 1250 nodes they do not: on that node set the

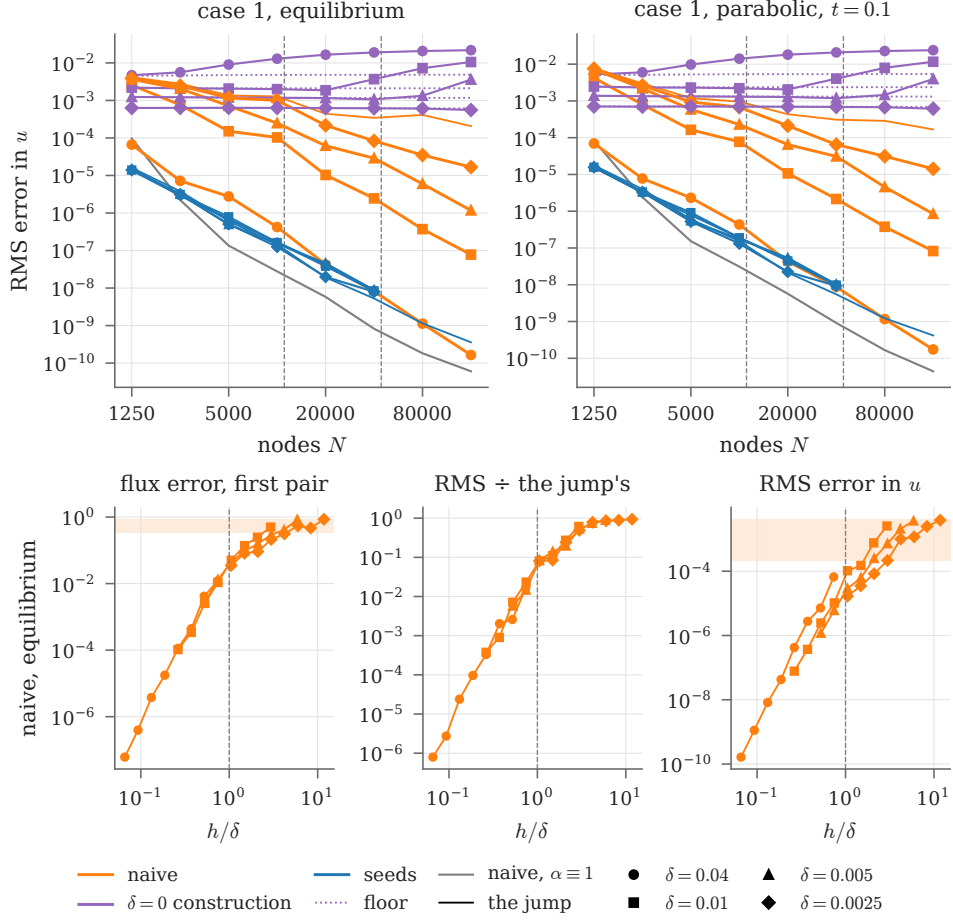


FIGURE 6. Case 1. Top: RMS error against the node count at equilibrium (left) and for the parabolic problem at  $t = 0.1$  (right), for the naive operator (orange), the  $\delta = 0$  construction (purple) with its floor  $u_\delta - u_0$  dotted, and the seeds (blue); markers give  $\delta$ , and the lines without markers are the jump. Grey: the naive operator with  $\alpha \equiv 1$  on the same nodes. Dashed verticals:  $h = \delta$ . Bottom: the naive operator at equilibrium against  $h/\delta$ , as the relative flux error on the first rows off the edge, its RMS error over the jump's on the same nodes, and its RMS error; shaded, the range of the jump's value over the counts.

naive operator has an eigenvalue in the right half-plane at every width, at  $+14.8$  to  $+25.1$ , which BD4 at  $\Delta t = h$  amplifies, and its parabolic error there is 1.7 to 2.0 times the equilibrium's at the jump and for  $\delta \leq 0.01$ . The mode sits on the free nodes in the middle of the band, and none of the naive

operators has it at 2500 nodes; every parabolic naive number at 1250 nodes carries it.

*The  $\delta = 0$  construction.* The jump-aware operator applied at the edge's centre, with the pieces' data as if  $\delta$  were 0, converges to the jump problem's solution. Its *floor* is the RMS at the nodes of the two references' difference,  $u_\delta - u_0$ : 0.12 to  $0.25\delta$  over the four widths. While  $h \geq 2.1\delta$  the construction sits on the floor to three digits (Table 8); at  $h \approx \delta$  it dips 5 to 10% below it, as its rows that do not cross the edge, which read the smooth  $\alpha$ , pull the solution part of the way to the smooth one; and then it grows, to 5.0 times the floor at  $\delta = 0.01$  and 3 times at 0.005 on 160,000 nodes. At  $\delta = 0.04$ , resolved at every count, its error grows with the node count, from  $4.7 \times 10^{-3}$  to  $2.2 \times 10^{-2}$ , a negative order: its rebuilt rows enforce a kink that the solution does not have, as in one dimension (Section 4.3). Its flux reading is the naive operator's mirror image: 0.018 to 0.020 while  $h \geq 4\delta$  at  $\delta = 0.0025$ , where its RMS error is the floor, and 0.81 to 0.89 at  $h \approx \delta$ . The construction beats the naive operator only while  $h \gtrsim 3\delta$ , and by at most 6 times, so that using it means knowing  $\delta$  and switching it off near  $h \approx 3\delta$ .

**6.3. Solvability and spectra.** The seed rows change the global matrix only where they replace rows, and the question is whether it stays as solvable as the two operators the seeds pass between: the direct operator on a resolved edge and the jump-aware operator at  $\delta = 0$ . We measure the matrices on case 1 at  $\delta/h = 8, 1, 1/8, 1/64$  and 0, on the interior system with the Dirichlet values eliminated, at 2500 and 10,000 nodes (Table 9 and Figure 7). The diagonal dominance ratio of a row is  $|a_{ii}| / \sum_{j \neq i} |a_{ij}|$  [Mar16, App. B], read over the rows each method rebuilds: the seeded rows for the seeds, the crossing rows for the others.

*Dominance and conditioning.* The seeded rows' least ratio falls from 0.409 at  $\delta = 8h$  to 0.239 at the jump on 2500 nodes, and from 0.404 to 0.211 on 10,000, towards the jump-aware rows' value, which it reaches at the jump and never goes below; the median hardly moves. A one-norm estimate of the seed matrix's condition number, a lower bound read across operators, rises from  $7.1 \times 10^3$  at  $\delta = 8h$  to  $1.7 \times 10^4$  and is  $1.5 \times 10^4$  at the jump, inside the construction's range over the same widths,  $1.3 \times 10^4$  to  $1.7 \times 10^4$ , and below the naive operator's,  $2.0 \times 10^4$  to  $5.5 \times 10^4$ . GMRES without a preconditioner takes 134, 155, 147, 160 and 161 iterations from  $\delta = 8h$  to the jump, bounded but not monotone, and fewer than the construction's 161 to 208; BiCGSTAB takes 95 to 101 against the construction's 100 to 147. With the preconditioner of [Mar16, App. B] or an incomplete LU factorisation every solve converges at every width, and the direct solves leave residuals of  $10^{-14}$  or below. Nothing breaks down as  $\delta \rightarrow 0$ : the readings move towards the jump's, as they should. This is case 1; the ring's iterative solves were not run.

TABLE 7. The naive operator on case 1: RMS error at equilibrium; that error over the jump's on the same nodes, at equilibrium and for the parabolic problem; the parabolic RMS error over the equilibrium one; and the relative flux error on the first rows off the edge, at equilibrium. Fits over every count.

| $N$               | $h/\delta$ | RMS                    | $\div$ jump          |                      | par./ell. | flux                 |
|-------------------|------------|------------------------|----------------------|----------------------|-----------|----------------------|
|                   |            |                        | ell.                 | par.                 |           |                      |
| <i>the jump</i>   |            |                        |                      |                      |           |                      |
| 1250              | —          | $4.14 \times 10^{-3}$  | —                    | —                    | 1.88      | 0.853                |
| 2500              | —          | $2.78 \times 10^{-3}$  | —                    | —                    | 1.07      | 0.493                |
| 5000              | —          | $1.36 \times 10^{-3}$  | —                    | —                    | 0.876     | 0.553                |
| 10,000            | —          | $1.28 \times 10^{-3}$  | —                    | —                    | 0.746     | 0.370                |
| 20,000            | —          | $4.42 \times 10^{-4}$  | —                    | —                    | 0.984     | 0.351                |
| 40,000            | —          | $3.44 \times 10^{-4}$  | —                    | —                    | 0.899     | 0.345                |
| 80,000            | —          | $4.12 \times 10^{-4}$  | —                    | —                    | 0.696     | 0.337                |
| 160,000           | —          | $2.07 \times 10^{-4}$  | —                    | —                    | 0.809     | 0.364                |
| fit               |            | 1.23                   |                      |                      |           |                      |
| $\delta = 0.04$   |            |                        |                      |                      |           |                      |
| 1250              | 0.74       | $6.66 \times 10^{-5}$  | 0.016                | 0.0090               | 1.05      | 0.012                |
| 2500              | 0.52       | $7.20 \times 10^{-6}$  | 0.0026               | 0.0026               | 1.08      | 0.0042               |
| 5000              | 0.37       | $2.77 \times 10^{-6}$  | 0.0020               | 0.0019               | 0.838     | $4.4 \times 10^{-4}$ |
| 10,000            | 0.26       | $4.22 \times 10^{-7}$  | $3.3 \times 10^{-4}$ | $4.6 \times 10^{-4}$ | 1.03      | $1.1 \times 10^{-4}$ |
| 20,000            | 0.19       | $4.27 \times 10^{-8}$  | $9.7 \times 10^{-5}$ | $9.9 \times 10^{-5}$ | 1.01      | $1.8 \times 10^{-5}$ |
| 40,000            | 0.13       | $8.22 \times 10^{-9}$  | $2.4 \times 10^{-5}$ | $2.8 \times 10^{-5}$ | 1.07      | $3.8 \times 10^{-6}$ |
| 80,000            | 0.09       | $1.13 \times 10^{-9}$  | $2.7 \times 10^{-6}$ | $4.1 \times 10^{-6}$ | 1.04      | $4.0 \times 10^{-7}$ |
| 160,000           | 0.07       | $1.64 \times 10^{-10}$ | $7.9 \times 10^{-7}$ | $1.0 \times 10^{-6}$ | 1.06      | $6.1 \times 10^{-8}$ |
| fit               |            | 5.31                   |                      |                      |           |                      |
| $\delta = 0.01$   |            |                        |                      |                      |           |                      |
| 1250              | 2.94       | $2.55 \times 10^{-3}$  | 0.616                | 0.545                | 1.66      | 0.499                |
| 2500              | 2.08       | $7.62 \times 10^{-4}$  | 0.274                | 0.269                | 1.05      | 0.248                |
| 5000              | 1.49       | $1.51 \times 10^{-4}$  | 0.111                | 0.137                | 1.09      | 0.138                |
| 10,000            | 1.05       | $1.03 \times 10^{-4}$  | 0.081                | 0.081                | 0.748     | 0.050                |
| 20,000            | 0.75       | $1.03 \times 10^{-5}$  | 0.023                | 0.025                | 1.04      | 0.011                |
| 40,000            | 0.53       | $2.45 \times 10^{-6}$  | 0.0071               | 0.0070               | 0.878     | 0.0025               |
| 80,000            | 0.37       | $3.71 \times 10^{-7}$  | $9.0 \times 10^{-4}$ | 0.0013               | 1.02      | $3.4 \times 10^{-4}$ |
| 160,000           | 0.26       | $7.79 \times 10^{-8}$  | $3.8 \times 10^{-4}$ | $4.9 \times 10^{-4}$ | 1.06      | $1.0 \times 10^{-4}$ |
| fit               |            | 4.33                   |                      |                      |           |                      |
| $\delta = 0.005$  |            |                        |                      |                      |           |                      |
| 1250              | 5.88       | $3.66 \times 10^{-3}$  | 0.884                | 0.921                | 1.96      | 0.839                |
| 2500              | 4.17       | $2.05 \times 10^{-3}$  | 0.738                | 0.733                | 1.06      | 0.407                |
| 5000              | 2.99       | $7.18 \times 10^{-4}$  | 0.527                | 0.491                | 0.816     | 0.292                |
| 10,000            | 2.11       | $2.48 \times 10^{-4}$  | 0.194                | 0.239                | 0.921     | 0.157                |
| 20,000            | 1.49       | $6.20 \times 10^{-5}$  | 0.140                | 0.149                | 1.05      | 0.095                |
| 40,000            | 1.05       | $2.89 \times 10^{-5}$  | 0.084                | 0.100                | 1.07      | 0.043                |
| 80,000            | 0.74       | $5.97 \times 10^{-6}$  | 0.014                | 0.016                | 0.762     | 0.013                |
| 160,000           | 0.53       | $1.17 \times 10^{-6}$  | 0.0057               | 0.0051               | 0.721     | 0.0027               |
| fit               |            | 3.33                   |                      |                      |           |                      |
| $\delta = 0.0025$ |            |                        |                      |                      |           |                      |
| 1250              | 11.76      | $3.89 \times 10^{-3}$  | 0.938                | 0.970                | 1.94      | 0.858                |
| 2500              | 8.33       | $2.47 \times 10^{-3}$  | 0.887                | 0.883                | 1.07      | 0.467                |
| 5000              | 5.97       | $1.17 \times 10^{-3}$  | 0.860                | 0.783                | 0.797     | 0.545                |
| 10,000            | 4.21       | $1.00 \times 10^{-3}$  | 0.785                | 0.710                | 0.674     | 0.311                |
| 20,000            | 2.99       | $2.16 \times 10^{-4}$  | 0.490                | 0.486                | 0.976     | 0.211                |
| 40,000            | 2.11       | $8.36 \times 10^{-5}$  | 0.243                | 0.210                | 0.776     | 0.091                |
| 80,000            | 1.49       | $3.50 \times 10^{-5}$  | 0.085                | 0.109                | 0.893     | 0.082                |
| 160,000           | 1.05       | $1.68 \times 10^{-5}$  | 0.081                | 0.085                | 0.840     | 0.035                |

TABLE 8. The  $\delta = 0$  construction on case 1 at equilibrium: its RMS error; the floor, the two references' RMS difference at the nodes; the error over the floor; and the error over the naive operator's on the same nodes. At the jump it is the jump-aware operator.

| $N$                                      | $h/\delta$ | RMS                    | floor                 | $\div$ floor | $\div$ naive         |
|------------------------------------------|------------|------------------------|-----------------------|--------------|----------------------|
| <i>the jump: the jump-aware operator</i> |            |                        |                       |              |                      |
| 1250                                     | –          | $1.60 \times 10^{-5}$  | –                     | –            | 0.0039               |
| 2500                                     | –          | $3.79 \times 10^{-6}$  | –                     | –            | 0.0014               |
| 5000                                     | –          | $6.11 \times 10^{-7}$  | –                     | –            | $4.5 \times 10^{-4}$ |
| 10,000                                   | –          | $1.45 \times 10^{-7}$  | –                     | –            | $1.1 \times 10^{-4}$ |
| 20,000                                   | –          | $1.92 \times 10^{-8}$  | –                     | –            | $4.4 \times 10^{-5}$ |
| 40,000                                   | –          | $5.28 \times 10^{-9}$  | –                     | –            | $1.5 \times 10^{-5}$ |
| 80,000                                   | –          | $1.15 \times 10^{-9}$  | –                     | –            | $2.8 \times 10^{-6}$ |
| 160,000                                  | –          | $3.58 \times 10^{-10}$ | –                     | –            | $1.7 \times 10^{-6}$ |
| $\delta = 0.04$                          |            |                        |                       |              |                      |
| 1250                                     | 0.74       | $4.70 \times 10^{-3}$  | $4.52 \times 10^{-3}$ | 1.040        | 70.5                 |
| 2500                                     | 0.52       | $5.62 \times 10^{-3}$  | $4.63 \times 10^{-3}$ | 1.215        | 781                  |
| 5000                                     | 0.37       | $9.06 \times 10^{-3}$  | $4.73 \times 10^{-3}$ | 1.913        | 3276                 |
| 10,000                                   | 0.26       | $1.29 \times 10^{-2}$  | $4.80 \times 10^{-3}$ | 2.695        | 30627                |
| 20,000                                   | 0.19       | $1.66 \times 10^{-2}$  | $4.82 \times 10^{-3}$ | 3.435        | $3.9 \times 10^5$    |
| 40,000                                   | 0.13       | $1.90 \times 10^{-2}$  | $4.82 \times 10^{-3}$ | 3.943        | $2.3 \times 10^6$    |
| 80,000                                   | 0.09       | $2.07 \times 10^{-2}$  | $4.84 \times 10^{-3}$ | 4.285        | $1.8 \times 10^7$    |
| 160,000                                  | 0.07       | $2.18 \times 10^{-2}$  | $4.85 \times 10^{-3}$ | 4.505        | $1.3 \times 10^8$    |
| $\delta = 0.01$                          |            |                        |                       |              |                      |
| 1250                                     | 2.94       | $2.19 \times 10^{-3}$  | $2.19 \times 10^{-3}$ | 1.000        | 0.858                |
| 2500                                     | 2.08       | $2.10 \times 10^{-3}$  | $2.10 \times 10^{-3}$ | 1.000        | 2.76                 |
| 5000                                     | 1.49       | $2.07 \times 10^{-3}$  | $2.10 \times 10^{-3}$ | 0.985        | 13.7                 |
| 10,000                                   | 1.05       | $2.00 \times 10^{-3}$  | $2.10 \times 10^{-3}$ | 0.950        | 19.3                 |
| 20,000                                   | 0.75       | $1.86 \times 10^{-3}$  | $2.11 \times 10^{-3}$ | 0.885        | 180                  |
| 40,000                                   | 0.53       | $3.69 \times 10^{-3}$  | $2.10 \times 10^{-3}$ | 1.753        | 1505                 |
| 80,000                                   | 0.37       | $7.23 \times 10^{-3}$  | $2.11 \times 10^{-3}$ | 3.432        | 19475                |
| 160,000                                  | 0.26       | $1.06 \times 10^{-2}$  | $2.11 \times 10^{-3}$ | 5.012        | $1.4 \times 10^5$    |
| $\delta = 0.005$                         |            |                        |                       |              |                      |
| 1250                                     | 5.88       | $1.24 \times 10^{-3}$  | $1.23 \times 10^{-3}$ | 1.000        | 0.337                |
| 2500                                     | 4.17       | $1.22 \times 10^{-3}$  | $1.22 \times 10^{-3}$ | 1.000        | 0.593                |
| 5000                                     | 2.99       | $1.20 \times 10^{-3}$  | $1.20 \times 10^{-3}$ | 1.000        | 1.67                 |
| 10,000                                   | 2.11       | $1.18 \times 10^{-3}$  | $1.18 \times 10^{-3}$ | 1.000        | 4.75                 |
| 20,000                                   | 1.49       | $1.16 \times 10^{-3}$  | $1.17 \times 10^{-3}$ | 0.985        | 18.6                 |
| 40,000                                   | 1.05       | $1.07 \times 10^{-3}$  | $1.17 \times 10^{-3}$ | 0.912        | 37.0                 |
| 80,000                                   | 0.74       | $1.33 \times 10^{-3}$  | $1.17 \times 10^{-3}$ | 1.135        | 223                  |
| 160,000                                  | 0.53       | $3.59 \times 10^{-3}$  | $1.17 \times 10^{-3}$ | 3.068        | 3058                 |
| $\delta = 0.0025$                        |            |                        |                       |              |                      |
| 1250                                     | 11.76      | $6.31 \times 10^{-4}$  | $6.31 \times 10^{-4}$ | 1.001        | 0.162                |
| 2500                                     | 8.33       | $6.32 \times 10^{-4}$  | $6.33 \times 10^{-4}$ | 1.000        | 0.256                |
| 5000                                     | 5.97       | $6.35 \times 10^{-4}$  | $6.35 \times 10^{-4}$ | 1.000        | 0.542                |
| 10,000                                   | 4.21       | $6.32 \times 10^{-4}$  | $6.32 \times 10^{-4}$ | 1.000        | 0.630                |
| 20,000                                   | 2.99       | $6.25 \times 10^{-4}$  | $6.25 \times 10^{-4}$ | 1.000        | 2.89                 |
| 40,000                                   | 2.11       | $6.18 \times 10^{-4}$  | $6.19 \times 10^{-4}$ | 0.999        | 7.39                 |
| 80,000                                   | 1.49       | $6.07 \times 10^{-4}$  | $6.17 \times 10^{-4}$ | 0.984        | 17.4                 |
| 160,000                                  | 1.05       | $5.57 \times 10^{-4}$  | $6.16 \times 10^{-4}$ | 0.904        | 33.2                 |

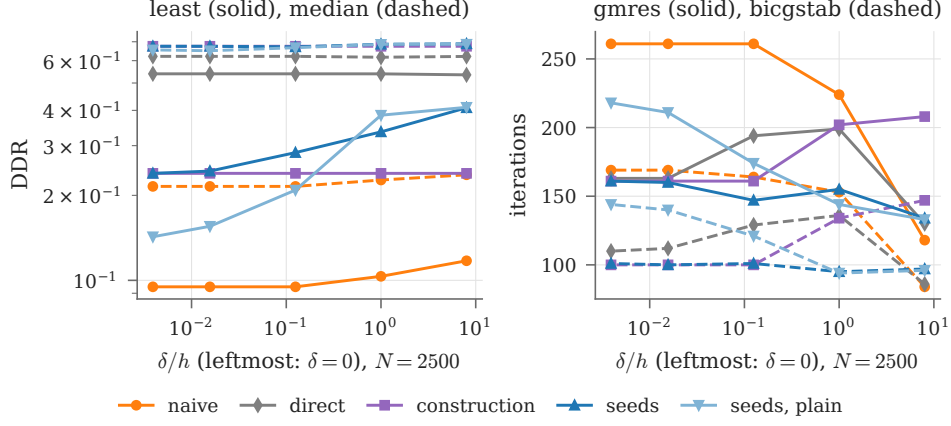


FIGURE 7. Case 1 at 2500 nodes against  $\delta/h$  (leftmost: the jump). Left: the least (solid) and median (dashed) diagonal dominance ratio over the seeded rows for the seeds and over the crossing rows for the other operators. Right: the iterations of GMRES (solid) and BiCGSTAB (dashed) without a preconditioner, to a relative residual of  $10^{-8}$ . Seeds, plain: the seed rows with plain Gaussians.

*Spectra.* The interior spectrum of the seed operator at 1600 nodes is that of the warped jump-aware operator at every width (Table 10 and Figure 8): no eigenvalue in the right half-plane, the rightmost the physical mode,  $-7.27$  at the jump and  $-7.73$  once the edge widens to  $\delta = 0.04$ , and  $h^2 \max |\operatorname{Im} \lambda| \leq 0.23$ . BD4 at  $\Delta t = h$  damps every mode. At 4900 nodes the seed operator at the jump reproduces the port’s warped jump-aware line to the digit: the rightmost eigenvalue  $-7.27$ ,  $h^2 \min \operatorname{Re} \lambda = -13.19$ ,  $h^2 \max |\operatorname{Im} \lambda| = 0.385$ , and BD4’s largest root modulus  $0.897$ . With plain Gaussians the complex loop of the jump-aware operator’s crossing rows returns at the jump,  $h^2 \max |\operatorname{Im} \lambda| = 1.49$  against  $0.385$  at 4900 nodes: a wider spectrum and not an instability, since no eigenvalue crosses the axis and BD4’s largest root is unchanged,  $0.826$  at 1600 nodes. The naive operator keeps the coarse node set’s growing mode at 1600 nodes,  $+1003$  at the jump, where BD4’s root reaches  $1.038$ . On case 2 the tangential chain, the frozen profile of Section 5.3 and the jump-aware operator have no eigenvalue in the right half-plane either, at any width. Our companion paper found degree-four seed stencils unstable across a sharp edge in the wave equation [Mar26]; in diffusion, whose implicit march needs no hyperviscosity, the spectra show no sign of it.

**6.4. The flat sweep.** Figure 9 and Tables 11 to 14 put the seeds on the sweep of Section 6.2: the same node sets, media, references and problems, with the direct operator beside them and, as a control for the stencils’ degree,

TABLE 9. The matrices on case 1, interior system: the seeded rows’ diagonal dominance ratio (least and median), a one-norm condition estimate of the matrix, and the iterations of GMRES and BiCGSTAB without a preconditioner to a relative residual of  $10^{-8}$ . Italic lines: the other operators’ ranges over the same widths, their ratios over the crossing rows.

| $\delta/h$                   | rows   | DDR least   | median      | cond/ $10^3$ | gmres   | bicgstab |
|------------------------------|--------|-------------|-------------|--------------|---------|----------|
| <i>N = 2500, the seeds</i>   |        |             |             |              |         |          |
| 8                            | 2500   | 0.409       | 0.688       | 7.1          | 134     | 97       |
| 1                            | 2164   | 0.336       | 0.686       | 7.8          | 155     | 95       |
| 1/8                          | 1036   | 0.283       | 0.671       | 15           | 147     | 101      |
| 1/64                         | 576    | 0.244       | 0.675       | 17           | 160     | 100      |
| 0                            | 576    | 0.239       | 0.675       | 15           | 161     | 101      |
| <i>construction</i>          | 576    | 0.239       | 0.675       | 13–17        | 161–208 | 100–147  |
| <i>direct</i>                | 576    | 0.534–0.539 | 0.617–0.621 | 8.0–32       | 130–199 | 86–136   |
| <i>naive</i>                 | 576    | 0.095–0.117 | 0.215–0.237 | 20–55        | 118–261 | 84–169   |
| <i>N = 10,000, the seeds</i> |        |             |             |              |         |          |
| 8                            | 10,000 | 0.404       | 0.691       | 32           | 202     | 181      |
| 1                            | 6414   | 0.378       | 0.690       | 40           | 227     | 201      |
| 1/8                          | 2198   | 0.252       | 0.679       | 70           | 237     | 211      |
| 1/64                         | 1134   | 0.215       | 0.676       | 73           | 237     | 209      |
| 0                            | 1134   | 0.211       | 0.676       | 68           | 238     | 213      |
| <i>construction</i>          | 1134   | 0.211       | 0.676       | 58–75        | 238–406 | 211–293  |
| <i>direct</i>                | 1134   | 0.498–0.507 | 0.615–0.620 | 38–130       | 212–390 | 192–287  |
| <i>naive</i>                 | 1134   | 0.098–0.106 | 0.209–0.225 | 140–170      | 223–468 | 200–407  |

the direct operator on the seeds’ own stencil groups, 30 nodes and degree four wherever the seeds are marched and 42 and five elsewhere. One matrix per count and width serves both problems.

*At the jump.* At  $\delta = 0$  the seed operator is the jump-aware operator (Section 5.2), and its line is the port’s at every count run, from  $1.598 \times 10^{-5}$  at 1250 nodes to  $3.58 \times 10^{-10}$  at 160,000, a fit of 4.54 over the eight counts and 4.77 over the six to 40,000. That is the regression check, not a result.

*One line at every width.* At  $\delta > 0$  the seeds are fourth order at every width (Table 11). The equilibrium fits over 1250 to 40,000 nodes are 4.23, 4.31, 4.23 and 4.48 at  $\delta = 0.04, 0.01, 0.005$  and  $0.0025$ , the parabolic ones 4.22, 4.28, 4.18 and 4.44, and over the later windows, from 5000 or from 10,000 nodes, 3.8 to 4.3 (4.6 to 4.8 at the jump); the rates between successive counts wander from 2.6 to 5.9 with the node sets. The five lines, the jump included, lie within 1.2 to 2.4 times of one another at each count: 1.16 at 1250 nodes, 2.38 at 20,000 and 1.67 at 40,000, and 2.53 at most on the parabolic problem, with the jump the lowest at the finest counts. That is no deterioration as  $\delta \rightarrow 0$  on these node sets, and not a proven  $\delta$ -uniform

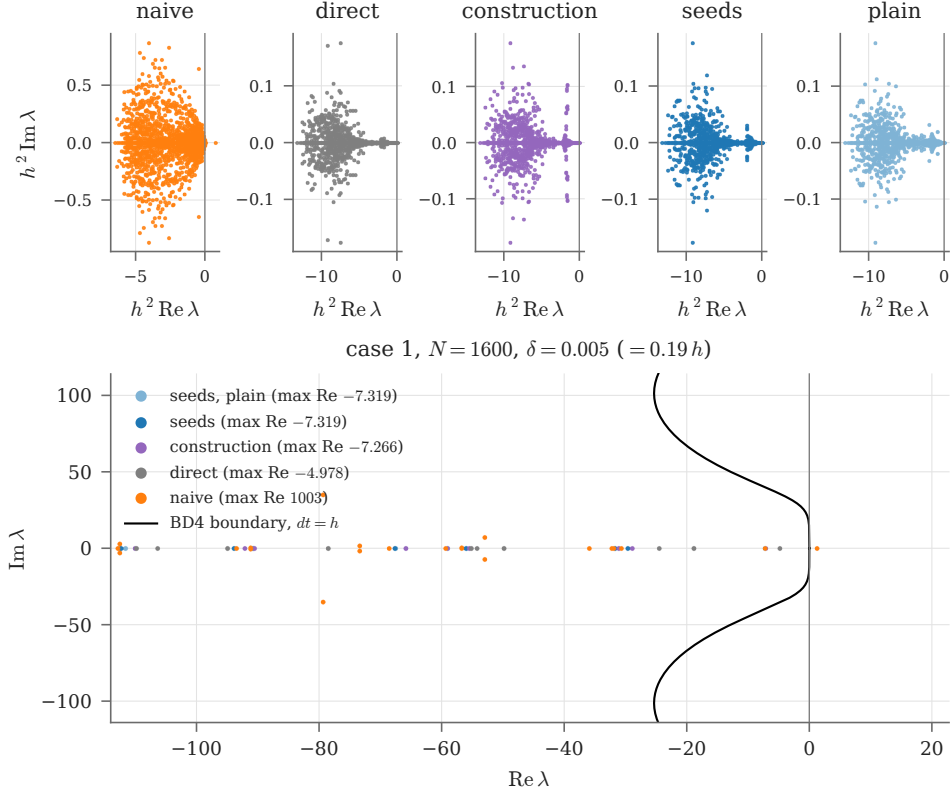


FIGURE 8. Interior eigenvalues  $\lambda$  on case 1 at 1600 nodes and  $\delta = 0.005$ . Top:  $h^2\lambda$  for the naive operator, the direct operator, the  $\delta = 0$  construction, the seeds and the seeds with plain Gaussians (plain). Bottom: all five near the origin, with the boundary of BD4's region of absolute stability at  $\Delta t = h$ , left of which every mode is damped; the naive operator's rightmost eigenvalue lies off the panel.

constant: each count has one node set, and each line holds  $\delta$  fixed while  $h$  falls, so that none holds  $h/\delta$  fixed.

*The elliptic problem.* In one dimension the equilibrium on constant pieces lies in the span of the first two seeds, which are exact there at every width (Remark 4.1). In two dimensions the equilibrium  $\sin 2\pi x v(y)$  is outside the seed span at any resolution (Section 5.1), the seeds' errors above are those of a real test, and the equilibrium ranks the methods. The parabolic errors are 1.04 to 1.17 times the equilibrium's at 40,000 nodes, but that problem is the weaker test of the two.

*The rule.* Every row whose 30 nodes see an edge within  $20\delta$  is seeded (Section 5.2), with no threshold in  $\delta$ . At 40,000 nodes that is 6% of the rows at the jump, 25% at  $\delta = 0.0025$ , 61% at  $\delta = 0.01$  and every row at

TABLE 10. Interior eigenvalues  $\lambda$  on case 1, the seeds with warped Gaussians unless marked plain: the rightmost, the number in the right half-plane, the extremes of  $h^2 \text{Re } \lambda$  and  $h^2 |\text{Im } \lambda|$ , and the largest root modulus of BD4 at  $\Delta t = h$ .

| $\delta$                              | $\delta/h$ | operator     | $\max \text{Re } \lambda$ | $\text{Re} > 0$ | $h^2 \min \text{Re}$ | $h^2 \max  \text{Im} $ | BD4   |
|---------------------------------------|------------|--------------|---------------------------|-----------------|----------------------|------------------------|-------|
| <i>N = 1600, interior eigenvalues</i> |            |              |                           |                 |                      |                        |       |
| 0                                     | 0.00       | seeds        | -7.27                     | 0               | -13.09               | 0.176                  | 0.826 |
| 0                                     | 0.00       | naive        | 1002.83                   | 2               | -6.44                | 0.862                  | 1.038 |
| 0                                     | 0.00       | direct       | -4.36                     | 0               | -13.09               | 0.172                  | 0.892 |
| 0                                     | 0.00       | construction | -7.27                     | 0               | -13.09               | 0.176                  | 0.826 |
| 0                                     | 0.00       | seeds, plain | -7.27                     | 0               | -13.09               | 1.511                  | 0.826 |
| 0.0025                                | 0.10       | seeds        | -7.29                     | 0               | -13.09               | 0.181                  | 0.825 |
| 0.005                                 | 0.19       | seeds        | -7.32                     | 0               | -13.09               | 0.175                  | 0.825 |
| 0.01                                  | 0.38       | seeds        | -7.37                     | 0               | -13.09               | 0.178                  | 0.824 |
| 0.04                                  | 1.52       | seeds        | -7.73                     | 0               | -11.96               | 0.227                  | 0.816 |
| <i>N = 4900, interior eigenvalues</i> |            |              |                           |                 |                      |                        |       |
| 0                                     | 0.00       | seeds        | -7.27                     | 0               | -13.19               | 0.385                  | 0.897 |
| 0                                     | 0.00       | construction | -7.27                     | 0               | -13.19               | 0.385                  | 0.897 |
| 0                                     | 0.00       | seeds, plain | -7.27                     | 0               | -13.19               | 1.489                  | 0.897 |
| 0.005                                 | 0.34       | seeds        | -7.32                     | 0               | -13.19               | 0.385                  | 0.897 |
| 0.005                                 | 0.34       | construction | -7.27                     | 0               | -13.19               | 0.385                  | 0.897 |
| 0.005                                 | 0.34       | seeds, plain | -7.32                     | 0               | -13.19               | 0.385                  | 0.897 |

$\delta = 0.04$ , where  $20\delta$  covers the strip. At a fixed  $\delta > 0$  the seeded share settles to the strip within  $20\delta$  of a curve, so the operator costs a number of marches proportional to the node count: a few milliseconds a row, and at 40,000 nodes and  $\delta = 0.04$  about 170 seconds of marches against under three for the naive operator's assembly. The reach of  $20\delta$  was never varied.

*Against the baselines.* At 40,000 nodes (Table 12) the seeds are four orders below the naive operator at  $\delta = 0.0025$ ,  $7.99 \times 10^{-9}$  against  $8.36 \times 10^{-5}$ , and five below the construction's  $6.18 \times 10^{-4}$ . The readings at the edge say why: the seeds' flux error on the first rows off the edge falls 2000 times from 1250 to 40,000 nodes at the jump and 670 times at  $\delta = 0.0025$ , where the naive operator's does not fall at all while  $h \gtrsim 4\delta$  (Section 6.2). The seeds are the only line whose order does not move with  $\delta$ : over these counts the naive operator walks its knee from a fit of 1.50 at the jump to 5.18 at  $\delta = 0.04$ , the construction is flat at the small widths and of negative order at  $\delta = 0.04$ , and the direct operator, blind to the jump, fits 1.87 at  $\delta = 0.0025$  and rises to 4.24 at  $\delta = 0.04$ .

*A resolved edge.* At  $\delta = 0.04$  every row is seeded, and the whole matrix is marched (Table 13). The seeds' error is 0.10 to 0.22 of the direct operator's at every count, and 0.086 to 0.22 of the direct operator's on their own stencils. A marched row costs nothing in accuracy where the grid resolves the edge: it carries the profile of  $\alpha$  through the stencil, where the direct row carries  $\alpha$  and  $\nabla\alpha$  at the evaluation node and pays for the rest in powers

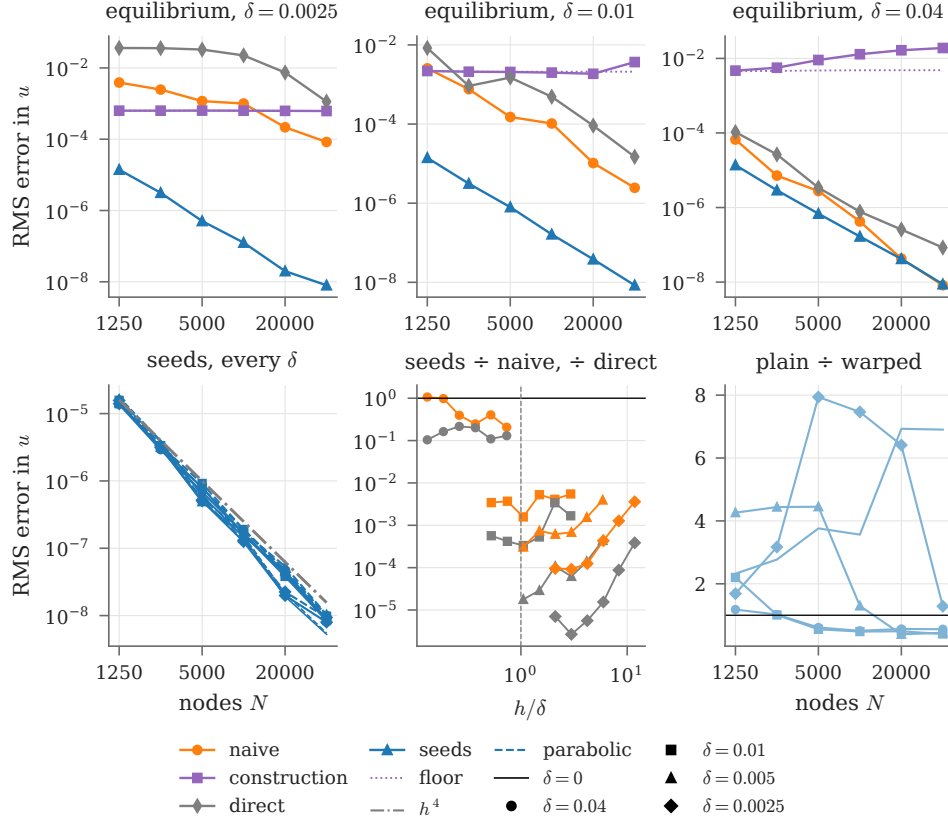


FIGURE 9. The seeds on case 1. Top: RMS error at equilibrium against the node count at  $\delta = 0.0025$ ,  $0.01$  and  $0.04$ , for the naive operator (orange), the  $\delta = 0$  construction (purple) with its floor dotted, the direct operator (grey) and the seeds (blue). Bottom left: the seeds at every width, at equilibrium (solid) and for the parabolic problem (dashed); markers give  $\delta$ , the unmarked lines are the jump, and the dash-dotted line has the slope of  $h^4$ . Bottom centre: the seeds' error over the naive operator's (orange) and the direct operator's (grey) against  $h/\delta$ . Bottom right: the seed rows with plain Gaussians over the warped ones (Table 14).

of  $h$ . The one line that catches the seeds is the naive operator, at 20,000 and 40,000 nodes, where they are 0.98 and 1.07 of it (1.15 for the parabolic problem at 40,000). That is the stencils' degree: the naive operator's bulk rows have degree five and fit 5.2 here, against the seeded rows' degree four and 4.2, so it will pass them on finer node sets, while on the seeds' own stencil groups its error at 10,000 to 40,000 nodes is  $1.30 \times 10^{-6}$ ,  $8.40 \times 10^{-6}$  and  $1.76 \times 10^{-7}$ , 8 to 200 times the seeds'. At  $\delta = 0.01$ , which the grid

TABLE 11. The seeds on case 1: RMS error at each count and width, at equilibrium and for the parabolic problem at  $t = 0.1$ ; max/min, the largest of the five widths' errors over the smallest at one count; fit, the least-squares order over 1250 to 40,000 nodes, and at the jump over all eight counts.

| $N$                                                            | $\delta = 0$           | $\delta = 0.04$       | 0.01                  | 0.005                 | 0.0025                | max/min |
|----------------------------------------------------------------|------------------------|-----------------------|-----------------------|-----------------------|-----------------------|---------|
| <i>equilibrium, RMS error of the seeds</i>                     |                        |                       |                       |                       |                       |         |
| 1250                                                           | $1.60 \times 10^{-5}$  | $1.37 \times 10^{-5}$ | $1.40 \times 10^{-5}$ | $1.44 \times 10^{-5}$ | $1.40 \times 10^{-5}$ | 1.16    |
| 2500                                                           | $3.79 \times 10^{-6}$  | $2.91 \times 10^{-6}$ | $3.13 \times 10^{-6}$ | $3.16 \times 10^{-6}$ | $3.14 \times 10^{-6}$ | 1.31    |
| 5000                                                           | $6.11 \times 10^{-7}$  | $6.85 \times 10^{-7}$ | $7.95 \times 10^{-7}$ | $4.91 \times 10^{-7}$ | $5.06 \times 10^{-7}$ | 1.62    |
| 10,000                                                         | $1.45 \times 10^{-7}$  | $1.67 \times 10^{-7}$ | $1.63 \times 10^{-7}$ | $1.52 \times 10^{-7}$ | $1.26 \times 10^{-7}$ | 1.33    |
| 20,000                                                         | $1.92 \times 10^{-8}$  | $4.18 \times 10^{-8}$ | $3.82 \times 10^{-8}$ | $4.58 \times 10^{-8}$ | $1.99 \times 10^{-8}$ | 2.38    |
| 40,000                                                         | $5.28 \times 10^{-9}$  | $8.78 \times 10^{-9}$ | $8.42 \times 10^{-9}$ | $8.82 \times 10^{-9}$ | $7.99 \times 10^{-9}$ | 1.67    |
| 80,000                                                         | $1.15 \times 10^{-9}$  | —                     | —                     | —                     | —                     | —       |
| 160,000                                                        | $3.58 \times 10^{-10}$ | —                     | —                     | —                     | —                     | —       |
| fit                                                            | 4.77                   | 4.23                  | 4.31                  | 4.23                  | 4.48                  |         |
| fit, 8 $N$                                                     | 4.54                   |                       |                       |                       |                       |         |
| <i>parabolic, <math>t = 0.1</math>, RMS error of the seeds</i> |                        |                       |                       |                       |                       |         |
| 1250                                                           | $1.76 \times 10^{-5}$  | $1.55 \times 10^{-5}$ | $1.58 \times 10^{-5}$ | $1.63 \times 10^{-5}$ | $1.57 \times 10^{-5}$ | 1.13    |
| 2500                                                           | $3.99 \times 10^{-6}$  | $3.21 \times 10^{-6}$ | $3.35 \times 10^{-6}$ | $3.34 \times 10^{-6}$ | $3.33 \times 10^{-6}$ | 1.24    |
| 5000                                                           | $6.01 \times 10^{-7}$  | $7.87 \times 10^{-7}$ | $9.17 \times 10^{-7}$ | $5.44 \times 10^{-7}$ | $5.22 \times 10^{-7}$ | 1.76    |
| 10,000                                                         | $1.54 \times 10^{-7}$  | $1.90 \times 10^{-7}$ | $1.90 \times 10^{-7}$ | $1.75 \times 10^{-7}$ | $1.32 \times 10^{-7}$ | 1.44    |
| 20,000                                                         | $2.13 \times 10^{-8}$  | $4.77 \times 10^{-8}$ | $4.43 \times 10^{-8}$ | $5.37 \times 10^{-8}$ | $2.25 \times 10^{-8}$ | 2.53    |
| 40,000                                                         | $5.51 \times 10^{-9}$  | $1.01 \times 10^{-8}$ | $9.80 \times 10^{-9}$ | $1.03 \times 10^{-8}$ | $9.37 \times 10^{-9}$ | 1.87    |
| 80,000                                                         | $1.22 \times 10^{-9}$  | —                     | —                     | —                     | —                     | —       |
| 160,000                                                        | $4.18 \times 10^{-10}$ | —                     | —                     | —                     | —                     | —       |
| fit                                                            | 4.77                   | 4.22                  | 4.28                  | 4.18                  | 4.44                  |         |
| fit, 8 $N$                                                     | 4.52                   |                       |                       |                       |                       |         |

TABLE 12. Every operator on case 1 at 40,000 nodes ( $h = 0.0053$ ), RMS error at equilibrium. Direct, 30/4: the direct operator on the seeds' own stencil groups; plain: the seed rows with plain Gaussians.

| $\delta$ | $h/\delta$ | naive                 | constr.               | direct                | direct, 30/4          | seeds                 | plain                 |
|----------|------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| jump     | —          | $3.44 \times 10^{-4}$ | $5.28 \times 10^{-9}$ | $3.63 \times 10^{-2}$ | $3.63 \times 10^{-2}$ | $5.28 \times 10^{-9}$ | $3.64 \times 10^{-8}$ |
| 0.04     | 0.13       | $8.22 \times 10^{-9}$ | $1.90 \times 10^{-2}$ | $8.43 \times 10^{-8}$ | $1.02 \times 10^{-7}$ | $8.78 \times 10^{-9}$ | $4.88 \times 10^{-9}$ |
| 0.01     | 0.53       | $2.45 \times 10^{-6}$ | $3.69 \times 10^{-3}$ | $1.48 \times 10^{-5}$ | $7.65 \times 10^{-6}$ | $8.42 \times 10^{-9}$ | $3.40 \times 10^{-9}$ |
| 0.005    | 1.05       | $2.89 \times 10^{-5}$ | $1.07 \times 10^{-3}$ | $4.96 \times 10^{-4}$ | $4.02 \times 10^{-4}$ | $8.82 \times 10^{-9}$ | $3.89 \times 10^{-9}$ |
| 0.0025   | 2.11       | $8.36 \times 10^{-5}$ | $6.18 \times 10^{-4}$ | $1.14 \times 10^{-3}$ | $6.47 \times 10^{-4}$ | $7.99 \times 10^{-9}$ | $1.03 \times 10^{-8}$ |

resolves at 20,000 and 40,000 nodes, the seeds are 0.003 to 0.004 of the naive operator's error.

*The warp.* Table 14 sets the seed rows with plain Gaussians against the warped ones. At the jump the factor is the jump-aware operator's, since the seed rows are the jump-aware rows with either block of Gaussians: 2.3 at

TABLE 13. The resolved edge, case 1 at  $\delta = 0.04$ , where every row is seeded: the seeds' RMS error at equilibrium, and its ratio to the direct operator's (on the degree-five stencils and on the seeds' own) and to the naive operator's, at equilibrium and for the parabolic problem. Fits over 1250 to 40,000 nodes.

| $N$                          | $\delta/h$ | seeds                 | $\div$ direct | $\div$ direct, 30/4 | $\div$ naive | $\div$ naive, par. |
|------------------------------|------------|-----------------------|---------------|---------------------|--------------|--------------------|
| 1250                         | 1.36       | $1.37 \times 10^{-5}$ | 0.130         | 0.196               | 0.206        | 0.223              |
| 2500                         | 1.92       | $2.91 \times 10^{-6}$ | 0.110         | 0.114               | 0.404        | 0.414              |
| 5000                         | 2.68       | $6.85 \times 10^{-7}$ | 0.201         | 0.224               | 0.248        | 0.340              |
| 10,000                       | 3.80       | $1.67 \times 10^{-7}$ | 0.216         | 0.135               | 0.396        | 0.437              |
| 20,000                       | 5.36       | $4.18 \times 10^{-8}$ | 0.163         | 0.116               | 0.981        | 1.11               |
| 40,000                       | 7.60       | $8.78 \times 10^{-9}$ | 0.104         | 0.086               | 1.07         | 1.15               |
| fits: seeds 4.23, naive 5.18 |            |                       |               |                     |              |                    |

1250 nodes to 6.9 at 40,000 on case 1, and 5.5 to 11.7 from 5000 nodes for the tangential chain on case 2. That is a regression check, not a measurement. At  $\delta > 0$  the warp wins while the edge is unresolved, by 1.7 to 7.9 at  $\delta = 0.0025$  while  $h \geq 3\delta$ , and loses by up to 2.6 once the grid resolves it. It turns at  $h \approx 2\delta$  on case 1, at  $h \approx \delta/2$  for  $\delta = 0.04$ , where  $\alpha$  varies over the whole stencil, and once  $h \lesssim 2\delta$  for the tangential chain on case 2, where plain over warped is then 0.55 to 1.1. Where the warp loses, the errors are not negligible,  $10^{-7}$  to  $10^{-6}$  on the intermediate node sets: at  $\delta = 0.01$  on 5000 nodes,  $7.95 \times 10^{-7}$  warped against  $4.39 \times 10^{-7}$  plain. The seeds stay warped, for the unresolved regime that is this study's subject and for the spectrum at the jump (Section 6.3), and pay up to 2.6 times once the grid resolves the edge.

**6.5. The curved feature.** Case 2 bends both curves and varies  $\alpha$  along them. Section 5.3 argued that the straight-feature seeds with a profile frozen along the normal are inconsistent at  $O(1)$  through an unresolved edge, for either of two reasons alone, and built the tangential chain in the curve's own coordinates. Here both run on case 2 against the product grid, beside two geometries that split the reasons: *A*, the sine pair with  $\alpha = 0.2$  in the band, carries the curvature alone, and *B*, flat lines at  $y = 0.6$  and  $0.8$  with case 2's  $\alpha$  in the band, the variation of  $\alpha$  along the edge alone. Figures 10 and 11 are the flat sweep's figure for the two constructions, and Tables 15 to 17 give the numbers. A further reading needs nothing but the reference: the *probe* applies each row to the equilibrium reference at the nodes, where  $Lu = 0$ , so that the row's value is its truncation error, read as an RMS over the crossing rows.

*The frozen profile.* At the jump the frozen profile is first order on case 2, a fit of 1.33 over 1250 to 40,000 nodes, and 24 to 6742 times the flat seeds' error on case 1 at the same count (17 to 6742 over every width and

TABLE 14. The warp: the RMS error of the seed rows with plain Gaussians over that with warped ones (above 1, the warp wins), for the seeds on case 1 and for the tangential chain on case 2.

| $\delta$                                         | 1250 | 2500 | 5000  | 10,000 | 20,000 | 40,000 |
|--------------------------------------------------|------|------|-------|--------|--------|--------|
| <i>case 1: the seeds, equilibrium</i>            |      |      |       |        |        |        |
| jump                                             | 2.32 | 2.77 | 3.76  | 3.56   | 6.93   | 6.90   |
| 0.04                                             | 1.18 | 1.02 | 0.610 | 0.506  | 0.563  | 0.556  |
| 0.01                                             | 2.20 | 1.01 | 0.553 | 0.482  | 0.490  | 0.403  |
| 0.005                                            | 4.26 | 4.44 | 4.45  | 1.30   | 0.395  | 0.440  |
| 0.0025                                           | 1.69 | 3.17 | 7.94  | 7.47   | 6.41   | 1.28   |
| <i>case 1: the seeds, parabolic</i>              |      |      |       |        |        |        |
| jump                                             | 2.81 | 3.09 | 4.96  | 4.43   | 8.12   | 8.66   |
| 0.04                                             | 1.17 | 1.00 | 0.601 | 0.489  | 0.550  | 0.541  |
| 0.01                                             | 2.36 | 1.02 | 0.529 | 0.463  | 0.469  | 0.380  |
| 0.005                                            | 4.48 | 4.69 | 4.63  | 1.27   | 0.384  | 0.408  |
| 0.0025                                           | 1.59 | 3.09 | 8.43  | 8.16   | 6.61   | 1.32   |
| <i>case 2: the tangential chain, equilibrium</i> |      |      |       |        |        |        |
| jump                                             | 1.06 | 1.85 | 6.30  | 5.47   | 10.1   | 11.7   |
| 0.01                                             | 1.79 | 1.17 | 0.877 | 1.09   | 1.02   | 1.06   |
| 0.005                                            | 4.00 | 3.55 | 1.85  | 0.844  | 0.546  | 0.826  |
| 0.0025                                           | 1.56 | 3.02 | 5.17  | 6.11   | 3.50   | 0.820  |

count). Its crossing rows stall on the probe, a fit of 0.38, where the jump-aware operator's rows on the same nodes converge at 2.86. Each reason alone is enough. On A at the jump, with constant pieces, the frozen profile is the jump-aware construction without its curvature terms, whose line in [MF17b, Fig. 10] is first order, and its crossing rows stall at 0.36; on B, where nothing is curved, it is first order to two digits, a fit of 1.00, and its rows stall at 0.18, while the jump-aware rows converge at 3.0 on both geometries. A resolved edge forgives both terms, which are then smooth perturbations on the stencil, but not their constant: at  $\delta = 0.01$ , where  $h$  falls through  $\delta$  at 10,000 nodes, the last two rates are 4.87 and 3.66, and the error is still 130 times the flat seeds' at 40,000 nodes. The frozen profile's order follows the resolution of the edge, not the seeds: its fits fall from 3.07 at  $\delta = 0.01$  through 2.40 to 1.71 at  $\delta = 0.0025$ , where  $h/\delta$  never falls below 2.1. Through a smooth edge it is still below the naive operator, the direct operator and the construction at every width and count, at 0.04 to 0.39 of the naive operator's error, at most 0.05 of the direct operator's and at most 0.46 of the construction's, which reads the edge as a jump and sits on its floor.

*The tangential chain.* In the curve's own coordinates the order comes back at every width (Table 15). From 5000 nodes on the tangential chain's error on case 2 is 1.4 to 3.9 times the flat seeds' on case 1 at the same width and count in the RMS norm, and 1.9 to 4.8 times in the maximum norm; on A

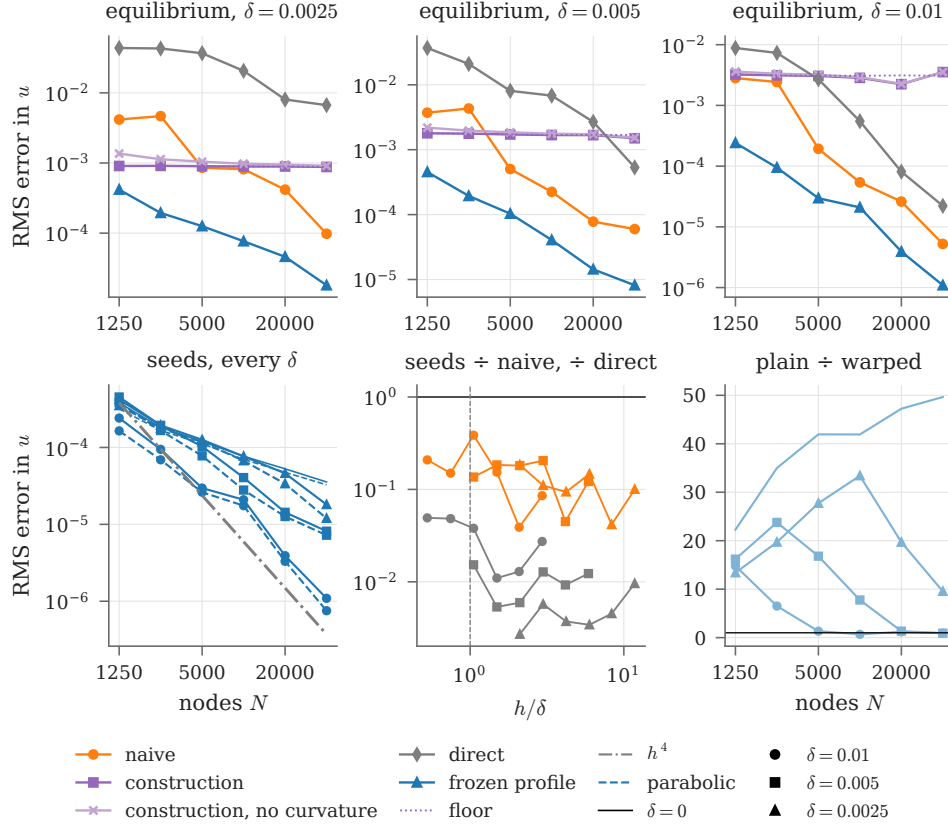


FIGURE 10. The straight-feature seeds with a frozen profile on case 2, in the layout of Figure 9. Top: RMS error at equilibrium against the node count at  $\delta = 0.0025, 0.005$  and  $0.01$ , for the naive operator (orange), the  $\delta = 0$  construction (purple), the same without its curvature terms (light purple), the direct operator (grey) and the frozen profile (blue). Bottom: the frozen profile at every width, its ratios to the naive and direct operators against  $h/\delta$ , and plain over warped Gaussians.

and B it is 1.1 to 4.4 times (Table 16). At 40,000 nodes it is  $7.18 \times 10^{-9}$ ,  $3.01 \times 10^{-8}$ ,  $3.00 \times 10^{-8}$  and  $2.69 \times 10^{-8}$  at  $\delta = 0, 0.01, 0.005$  and  $0.0025$ , and its fits from 5000 nodes are 4.96, 4.23, 3.86 and 4.13. At the jump it lies below the jump-aware operator with its curvature terms from 5000 nodes on, at 0.14 to 0.45 times its error, and on B, run at the jump to 160,000 nodes, it fits 4.49 against the jump-aware operator's 4.21. Its crossing rows converge on the probe at 2.7 to 3.6 on every geometry and width, the jump-aware rows' rate or better, where the frozen profile's rows stall (Table 17).

TABLE 15. Case 2 at 40,000 nodes, RMS error at equilibrium against the product grid, and each line’s fit over 1250 to 40,000 nodes: the naive operator; the  $\delta = 0$  construction (at the jump, the jump-aware operator with its curvature terms); the frozen profile; the tangential chain with warped and plain Gaussians; and, last, the tangential chain over the flat seeds on case 1 at the same width and count.

| $\delta$                                                  | $h/\delta$ | naive                 | constr.               | frozen                | tangential            | tang., plain          | $\div$ flat |
|-----------------------------------------------------------|------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-------------|
| <i>case 2, equilibrium, RMS error at the finest count</i> |            |                       |                       |                       |                       |                       |             |
| jump                                                      | –          | $6.26 \times 10^{-4}$ | $5.29 \times 10^{-8}$ | $3.56 \times 10^{-5}$ | $7.18 \times 10^{-9}$ | $8.37 \times 10^{-8}$ | 1.36        |
| 0.01                                                      | 0.53       | $5.23 \times 10^{-6}$ | $3.54 \times 10^{-3}$ | $1.09 \times 10^{-6}$ | $3.01 \times 10^{-8}$ | $3.18 \times 10^{-8}$ | 3.58        |
| 0.005                                                     | 1.05       | $5.99 \times 10^{-5}$ | $1.50 \times 10^{-3}$ | $8.15 \times 10^{-6}$ | $3.00 \times 10^{-8}$ | $2.48 \times 10^{-8}$ | 3.40        |
| 0.0025                                                    | 2.11       | $9.80 \times 10^{-5}$ | $8.78 \times 10^{-4}$ | $1.81 \times 10^{-5}$ | $2.69 \times 10^{-8}$ | $2.21 \times 10^{-8}$ | 3.37        |
| <i>fitted order, 1250–40,000</i>                          |            |                       |                       |                       |                       |                       |             |
| jump                                                      |            | 1.32                  | 3.89                  | 1.33                  | 5.72                  | 4.32                  |             |
| 0.01                                                      |            | 3.85                  | 0.05                  | 3.07                  | 4.24                  | 4.48                  |             |
| 0.005                                                     |            | 2.78                  | 0.09                  | 2.40                  | 4.36                  | 5.54                  |             |
| 0.0025                                                    |            | 2.16                  | 0.02                  | 1.71                  | 5.03                  | 5.25                  |             |

On two concentric circles, where the coupling along the curve vanishes, its rows converge at 3.45 and lie 4.6 to 7.3 times below the jump-aware rows.

Three things do not come back. On the two coarsest node sets the chain is 2.8 to 8.0 times the flat seeds, which steepens its fits over 1250 to 40,000 nodes to 4.24 to 5.72. At  $\delta = 0.0025$  its probe slows to 1.7 to 2.2 between 20,000 and 40,000 nodes on all three geometries, where  $h/\delta$  passes from 3 to 2.1, the transition at which the warp turns on the flat band; the errors keep falling there, at 2.5 to 3.0. And a row costs 2.5 to 3.8 times a row of the frozen profile.

**6.6. The ring and the extremizing sweep.** The ring of case 3 is the extreme case of Section 5.4: a resistive layer thinner than  $h$  at every count, of contrast about 1500 : 1 at  $s = 10^3$ , thinned at a fixed resistance up to  $s = 10^{11}$  [MF17b]. The seeds run with the flux seeds and the ring’s Gaussians of Section 5.4, against the jump-aware operator, whose lines here are the port’s.

*Exactness and conditioning.* The matched radial profile through the ring’s constant part lies in the span of every construction, so each seeded row applied to it should vanish (Figure 12, the twin of [MF17b, Fig. 20]). At 10,000 nodes the seeds’ worst relative residual is  $1.3 \times 10^{-15}$  to  $2.1 \times 10^{-15}$  at every  $s$  from  $10^3$  to  $10^{11}$ , where the jump-aware rows lose digits in proportion to  $s$ , to  $1.36 \times 10^{-7}$  at  $10^{11}$ ; with the stored radii in place of the ring’s carried width the seeds lose digits too, to  $1.2 \times 10^{-10}$  at  $10^{11}$ . The seeds’ stencil systems condition independently of  $s$ : from  $10^5$  to  $10^{11}$  the mean 2-norm condition numbers of their block are  $7.1 \times 10^3$  to  $7.2 \times 10^3$ , or  $5.0 \times 10^3$  to  $5.1 \times 10^3$  with its columns scaled, and of the saddle-point system  $1.0 \times 10^6$

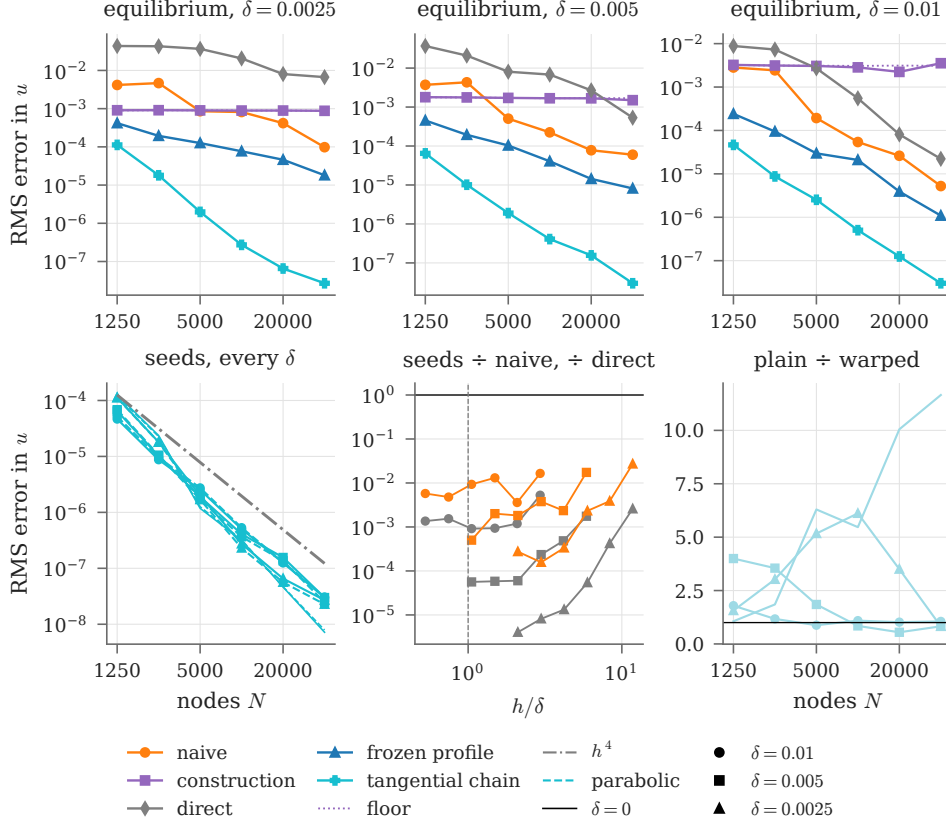


FIGURE 11. The tangential chain on case 2 (cyan), in the layout of Figure 9, with the frozen profile (blue) and the lines of Figure 10. Bottom: the tangential chain at every width, its ratios to the naive and direct operators against  $h/\delta$ , and plain over warped Gaussians.

to  $1.1 \times 10^6$ , where the jump-aware block and system reach  $4.6 \times 10^{10}$  and  $2.7 \times 10^{13}$  at  $10^{11}$ . The seeds carry the ring's resistance, which the sweep holds fixed, and not its width. The global matrix's conditioning was not measured.

*The extremizing sweep.* Figure 13 is the twin of [MF17b, Fig. 19] with the seeds, and Table 18 gives its numbers. The seeds are fourth order at every  $s$ , with fits of 4.43 to 4.71 over 1250 to 40,000 nodes against the jump-aware operator's 4.08 to 4.13, and lie at 0.52 to 0.97 times its error from 5000 to 40,000 nodes; the two coarsest counts sit 1.03 to 1.46 times above it. At 80,000 nodes the seeds are at  $1.4 \times 10^{-7}$  to  $1.6 \times 10^{-7}$  for  $s \geq 10^8$ , the size of the references' own error, and that count is not read as a ratio. Without the five flux seeds the line stalls: at  $s = 10^3$  it ends 8.2 and 11.8 times the jump-aware operator's error at 40,000 and 80,000 nodes, and 2.2 and 3.2

TABLE 16. Case 2 and the split geometries over the flat seeds of case 1 at the same width and count, RMS error at equilibrium unless marked: the frozen profile and the tangential chain on case 2; the tangential chain on A (the sine pair,  $\alpha = 0.2$  in the band) and on B (flat lines, case 2's  $\alpha$ ); the chain in the maximum norm; and, at the jump, the chain over the jump-aware operator on case 2.

| $\delta$                                                                              | 1250 | 2500 | 5000  | 10,000 | 20,000 | 40,000 |
|---------------------------------------------------------------------------------------|------|------|-------|--------|--------|--------|
| <i>case 2: the frozen profile <math>\div</math> flat seeds</i>                        |      |      |       |        |        |        |
| jump                                                                                  | 23.8 | 48.1 | 195   | 529    | 2741   | 6742   |
| 0.01                                                                                  | 17.2 | 30.2 | 37.3  | 128    | 102    | 130    |
| 0.005                                                                                 | 31.4 | 61.3 | 210   | 266    | 313    | 924    |
| 0.0025                                                                                | 29.8 | 61.8 | 248   | 610    | 2314   | 2262   |
| <i>case 2: tangential <math>\div</math> flat seeds</i>                                |      |      |       |        |        |        |
| jump                                                                                  | 7.84 | 6.24 | 1.93  | 2.18   | 2.40   | 1.36   |
| 0.01                                                                                  | 3.32 | 2.79 | 3.18  | 3.09   | 3.27   | 3.58   |
| 0.005                                                                                 | 4.48 | 3.21 | 3.87  | 2.68   | 3.42   | 3.40   |
| 0.0025                                                                                | 8.05 | 5.74 | 3.93  | 2.17   | 3.30   | 3.37   |
| <i>A, the curvature alone: tangential <math>\div</math> flat seeds</i>                |      |      |       |        |        |        |
| jump                                                                                  | 2.29 | 1.54 | 1.99  | 1.39   | 3.56   | 1.92   |
| 0.0025                                                                                | 2.56 | 1.60 | 1.73  | 1.10   | 1.86   | 1.87   |
| <i>B, <math>\alpha</math> along the edge: tangential <math>\div</math> flat seeds</i> |      |      |       |        |        |        |
| jump                                                                                  | 3.89 | 2.13 | 3.29  | 3.40   | 4.41   | 3.26   |
| 0.0025                                                                                | 4.13 | 1.98 | 1.58  | 1.45   | 4.30   | 3.79   |
| <i>case 2: tangential <math>\div</math> flat seeds, max norm</i>                      |      |      |       |        |        |        |
| jump                                                                                  | 5.06 | 6.63 | 2.68  | 3.58   | 4.05   | 2.74   |
| 0.01                                                                                  | 3.26 | 2.26 | 4.36  | 4.08   | 3.08   | 3.42   |
| 0.005                                                                                 | 2.90 | 3.45 | 3.02  | 2.87   | 4.78   | 2.96   |
| 0.0025                                                                                | 4.99 | 5.79 | 2.80  | 1.90   | 3.08   | 4.71   |
| <i>case 2, the jump: tangential <math>\div</math> jump-aware</i>                      |      |      |       |        |        |        |
| jump                                                                                  | 4.09 | 1.90 | 0.189 | 0.451  | 0.258  | 0.136  |

times at  $10^{11}$ . On the exact mode through the ring at its constant part, the seeds' crossing rows lie 43 to 144 times below the jump-aware rows. The seed operator has no eigenvalue in the right half-plane at 5000 nodes at any  $s$  or  $\delta$ , its slowest mode the physical one at  $-14.1$  to  $-14.3$ . The loss of convergence of the 2016 sweep near  $s = 10^{11}$  is not reproduced by either construction (Section 2.5).

*The smooth ring.* With  $\delta = 0.0025, 0.001$  and  $0.00025$ , that is 2.5, 1 and  $1/4$  times the ring's width at  $s = 10^3$ , the smooth ring (24) keeps the jump's contact resistance at every  $s$  and  $\delta$ , and both its width and its edges lie below  $h$  at every count. It has no independent reference, so each line is read against a 160,000-node run of the seeds at the same  $s$  and  $\delta$ , on the nodes away from the ring, those whose nearest fine node's stencil is not seeded: 74

TABLE 17. The probe: the RMS over the crossing rows of each operator applied to the equilibrium reference at the nodes, at the first and the finest count, and its fitted order in  $h$ . A: the sine pair with  $\alpha = 0.2$  in the band (the curvature alone); B: flat lines with case 2's  $\alpha$  (its variation along the edge alone). At  $\delta > 0$  the jump-aware rows read the smooth edge as a jump and are not listed.

| $\delta$ |        |            | $N = 1250$            | 40,000                | fit  |
|----------|--------|------------|-----------------------|-----------------------|------|
| case 2   | jump   | tangential | $1.51 \times 10^{-2}$ | $6.35 \times 10^{-5}$ | 3.22 |
| case 2   | jump   | frozen     | $6.44 \times 10^{-2}$ | $3.33 \times 10^{-2}$ | 0.38 |
| case 2   | jump   | jump-aware | $1.10 \times 10^{-2}$ | $9.24 \times 10^{-5}$ | 2.86 |
| case 2   | 0.01   | tangential | $1.13 \times 10^{-2}$ | $2.81 \times 10^{-5}$ | 3.58 |
| case 2   | 0.01   | frozen     | $5.68 \times 10^{-2}$ | $1.81 \times 10^{-3}$ | 1.99 |
| case 2   | 0.005  | tangential | $1.23 \times 10^{-2}$ | $4.43 \times 10^{-5}$ | 3.19 |
| case 2   | 0.005  | frozen     | $6.71 \times 10^{-2}$ | $9.60 \times 10^{-3}$ | 1.09 |
| case 2   | 0.0025 | tangential | $1.44 \times 10^{-2}$ | $6.50 \times 10^{-5}$ | 3.24 |
| case 2   | 0.0025 | frozen     | $6.58 \times 10^{-2}$ | $2.24 \times 10^{-2}$ | 0.61 |
| A        | jump   | tangential | $9.48 \times 10^{-3}$ | $3.85 \times 10^{-5}$ | 3.23 |
| A        | jump   | frozen     | $4.54 \times 10^{-2}$ | $2.36 \times 10^{-2}$ | 0.36 |
| A        | jump   | jump-aware | $6.45 \times 10^{-3}$ | $4.03 \times 10^{-5}$ | 2.99 |
| A        | 0.0025 | tangential | $8.92 \times 10^{-3}$ | $4.01 \times 10^{-5}$ | 3.25 |
| A        | 0.0025 | frozen     | $4.69 \times 10^{-2}$ | $1.37 \times 10^{-2}$ | 0.64 |
| B        | jump   | tangential | $9.19 \times 10^{-3}$ | $3.75 \times 10^{-5}$ | 3.07 |
| B        | jump   | frozen     | $4.17 \times 10^{-2}$ | $3.03 \times 10^{-2}$ | 0.18 |
| B        | jump   | jump-aware | $8.43 \times 10^{-3}$ | $3.48 \times 10^{-5}$ | 3.00 |
| B        | 0.0025 | tangential | $8.50 \times 10^{-3}$ | $7.03 \times 10^{-5}$ | 2.71 |
| B        | 0.0025 | frozen     | $4.25 \times 10^{-2}$ | $2.14 \times 10^{-2}$ | 0.38 |

to 96% of the nodes. The error near the ring is not scored at  $\delta > 0$ , and no ring line has a maximum-norm error. Figure 14 and Table 19 give the lines.

On that far field the seeds converge at fourth order: fits of 4.0 to 4.8 over 2500 to 20,000 nodes at every  $s$  and  $\delta$ , and of 4.4 to 4.8 to 40,000 on the five lines that do not stop on their reference (below). The  $\delta = 0$  construction reads the two edges as jumps and sits at  $10^{-4}$  to  $7 \times 10^{-4}$ , the smooth ring's distance from the jump ring, and the naive and direct operators do not see the ring until the grid samples it: at  $\delta = 0.00025$  both sit near  $5 \times 10^{-3}$ , the error of ignoring the ring. Two caveats bound the seeds' numbers. At  $\delta \leq 0.001$  their errors at 40,000 nodes,  $7.1 \times 10^{-7}$  to  $1.0 \times 10^{-6}$ , are the size of the differences between two rebuilds of the fine runs,  $1.3 \times 10^{-6}$  to  $3.9 \times 10^{-6}$ ; and the steep last rates at  $s = 10^3$ , 6.7 and 6.9, may be cancellation against a fine run's own error. Only one  $s$  and  $\delta$  was read against a second, independent fine run.

*Rows anchored in the ring.* With the warp on every row the seeds had two outliers; the worse, at  $\delta = 0.001$  on 20,000 nodes, put the far-field error at  $6.2 \times 10^{-5}$  while the probe kept falling. Both are rows anchored in the

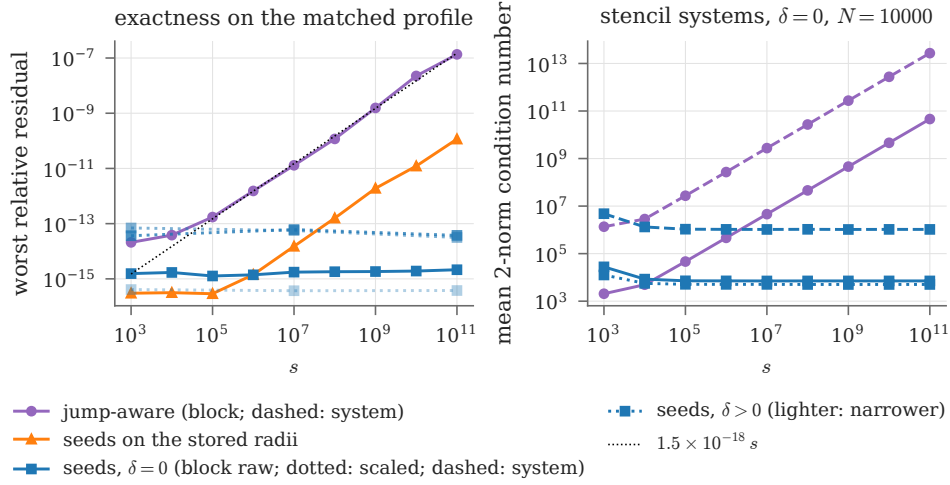


FIGURE 12. Case 3 at 10,000 nodes, the twin of [MF17b, Fig. 20], against  $s$ . Left: the worst relative residual of the seeded rows on the matched radial profile, for the jump-aware operator, the seeds on the stored radii, and the seeds at the jump and, lighter for narrower edges, at  $\delta > 0$ ; dotted, the port's fit  $1.5 \times 10^{-18} s$  to the jump-aware rows. Right: the mean 2-norm condition numbers of the stencil systems at the jump, the jump-aware operator's polynomial block (solid) and saddle-point system (dashed), and the seeds' block raw (solid) and column-scaled (dotted) and their system (dashed).

smooth ring's resistivity tail, which the warp leaves consistent but not stable (Section 5.4). The diagonal rule removes both, and where it keeps rows plain it is within 0.91 to 1.25 times the better of the two uniform choices, warped or plain Gaussians on every row, at every  $s$ ,  $\delta$  and count through 40,000 nodes. At  $s = 10^{11}$  and  $\delta = 0.001$  the rule's own 160,000-node solution is the discrepant one: a second fine run built with plain Gaussians differs from it by  $3.55 \times 10^{-6}$  in the RMS, every coarse line converges towards the second run, the seeds to  $8.0 \times 10^{-7}$  at 40,000 nodes with a fit of 4.77, and against the first they stop at  $3.6 \times 10^{-6}$ . So at that  $s$  and  $\delta$  the rule's convergence past 40,000 nodes is unresolved.

**6.7. Coefficient treatments.** On scattered nodes there is no cell, and in place of the cell means of Section 4.5 we take discs: the harmonic mean of  $\alpha$  over a disc of radius  $h/2$  or  $h$  about each node (T1), the natural scattered-node analogue of the harmonic-mean conductance [Pat80], and the arithmetic mean over the same discs (T2), each sampled into  $A$  in the naive operator. Averaging the diffusivity over a node cloud [KKMS23] or over a pair of particles [CM99] is in use, and with pairwise means Kraus, Kuhnert

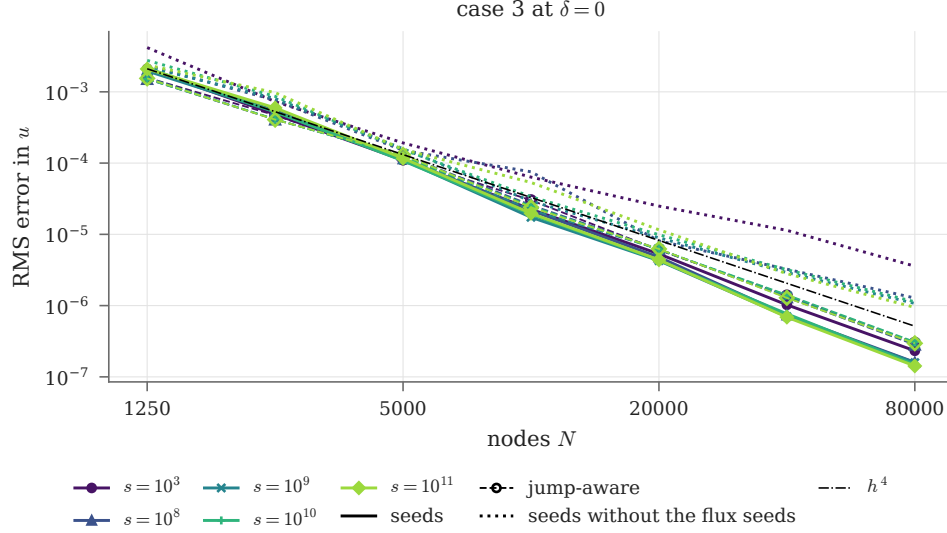


FIGURE 13. Case 3 at  $\delta = 0$ , the twin of [MF17b, Fig. 19]: RMS error against the node count at  $s = 10^3$  and  $10^8$  to  $10^{11}$ , each against the port’s 160,000-node jump-aware reference at the same  $s$ , for the seeds (solid), the jump-aware operator (dashed, open markers) and the seeds without the flux seeds (dotted); dash-dotted, the slope of  $h^4$ .

TABLE 18. Case 3 at  $\delta = 0$ : the RMS error of the seeds, with and without the flux seeds, over the jump-aware operator’s at each count, all against the port’s reference at each  $s$ ; the two fits over 1250 to 40,000 nodes; and the line’s own RMS error at 80,000 nodes.

|                                                                  | ratio to jump-aware, $N =$ |      |       |        |        |        |        | fit to 40,000 |      | RMS at               |
|------------------------------------------------------------------|----------------------------|------|-------|--------|--------|--------|--------|---------------|------|----------------------|
| $s$                                                              | 1250                       | 2500 | 5000  | 10,000 | 20,000 | 40,000 | 80,000 | j.-aware      | line | 80,000               |
| <i>the seeds, <math>\delta = 0</math></i>                        |                            |      |       |        |        |        |        |               |      |                      |
| $10^3$                                                           | 1.33                       | 1.03 | 0.973 | 0.745  | 0.882  | 0.738  | 0.769  | 4.12          | 4.43 | $2.3 \times 10^{-7}$ |
| $10^8$                                                           | 1.34                       | 1.34 | 0.878 | 0.840  | 0.774  | 0.564  | 0.566  | 4.11          | 4.61 | $1.6 \times 10^{-7}$ |
| $10^9$                                                           | 1.24                       | 1.27 | 0.886 | 0.785  | 0.691  | 0.524  | 0.540  | 4.13          | 4.65 | $1.6 \times 10^{-7}$ |
| $10^{10}$                                                        | 1.43                       | 1.27 | 0.872 | 0.763  | 0.705  | 0.541  | 0.480  | 4.08          | 4.64 | $1.5 \times 10^{-7}$ |
| $10^{11}$                                                        | 1.36                       | 1.46 | 0.896 | 0.823  | 0.703  | 0.535  | 0.481  | 4.13          | 4.71 | $1.4 \times 10^{-7}$ |
| <i>the seeds without the flux seeds, <math>\delta = 0</math></i> |                            |      |       |        |        |        |        |               |      |                      |
| $10^3$                                                           | 2.64                       | 1.55 | 1.70  | 2.11   | 4.14   | 8.20   | 11.8   | 4.12          | 3.39 | $3.6 \times 10^{-6}$ |
| $10^8$                                                           | 1.54                       | 1.87 | 1.16  | 2.93   | 1.31   | 2.49   | 4.52   | 4.11          | 3.92 | $1.3 \times 10^{-6}$ |
| $10^9$                                                           | 1.42                       | 1.96 | 1.27  | 1.32   | 1.45   | 2.17   | 3.66   | 4.13          | 4.02 | $1.1 \times 10^{-6}$ |
| $10^{10}$                                                        | 1.83                       | 2.09 | 1.25  | 1.36   | 1.61   | 2.25   | 3.65   | 4.08          | 4.05 | $1.1 \times 10^{-6}$ |
| $10^{11}$                                                        | 1.54                       | 2.39 | 1.19  | 2.23   | 1.86   | 2.19   | 3.19   | 4.13          | 3.99 | $9.4 \times 10^{-7}$ |

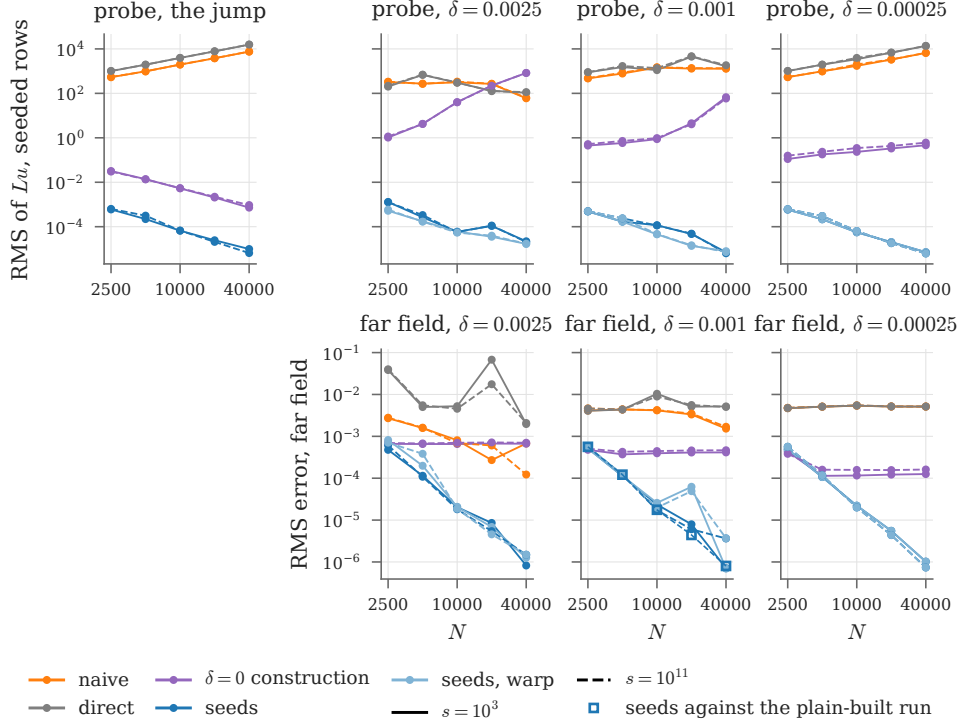


FIGURE 14. The smooth ring,  $s = 10^3$  (solid) and  $10^{11}$  (dashed). Top: the probe, the RMS over the seeded rows of each operator applied to the exact radial mode through the ring at its constant part, at the jump and at  $\delta = 0.0025$ ,  $0.001$  and  $0.00025$ . Bottom: the RMS error on the far field against the 160,000-node seed run at the same  $s$  and  $\delta$ . The seeds (with the diagonal rule), the seeds with the warp on every row, the  $\delta = 0$  construction (at the jump, the jump-aware operator), and the naive and direct operators; open squares, the seeds against the fine run built with plain Gaussians at  $s = 10^{11}$ ,  $\delta = 0.001$ .

and Suchde observe first order at an interface [KKS25]; we found no source that averages over a disc about each node. T0 widens the band's edges to  $\max(\delta, mh)$ ,  $m = 1$  or  $2$ . The disc integrals are exact to rounding, taken in the coordinates of case 2's shear, in which both curves are coordinate lines. No conservative scheme on scattered nodes, a twin of T1-FV with control volumes and face conductances, was built, so that the two-dimensional comparators are the naive operator and the  $\delta = 0$  construction, and a statement about the strongest low-order comparator is a statement about these six; as in one dimension, the band-limited coefficient is left out.

TABLE 19. The smooth ring: RMS error on the far field against the fine seed run at 160,000 nodes, for the seeds with the diagonal rule and with the warp on every row; the rule over the better of warped and plain Gaussians on every row (–: the rule keeps no row plain and is the warp); the other operators at 40,000 nodes, the construction being the  $\delta = 0$  construction; and at  $s = 10^{11}$ ,  $\delta = 0.001$ , the seeds against a second fine run built with plain Gaussians.

| $\delta$                                                                      | 0.0025               | 0.0025               | 0.001                | 0.001                | 0.00025              | 0.00025              |
|-------------------------------------------------------------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| $s$                                                                           | $10^3$               | $10^{11}$            | $10^3$               | $10^{11}$            | $10^3$               | $10^{11}$            |
| <i>the seeds (the diagonal rule), RMS error away from the ring</i>            |                      |                      |                      |                      |                      |                      |
| 2500                                                                          | $4.8 \times 10^{-4}$ | $6.1 \times 10^{-4}$ | $5.1 \times 10^{-4}$ | $5.6 \times 10^{-4}$ | $4.9 \times 10^{-4}$ | $5.6 \times 10^{-4}$ |
| 5000                                                                          | $1.1 \times 10^{-4}$ | $1.1 \times 10^{-4}$ | $1.1 \times 10^{-4}$ | $1.2 \times 10^{-4}$ | $1.1 \times 10^{-4}$ | $1.2 \times 10^{-4}$ |
| 10,000                                                                        | $2.0 \times 10^{-5}$ | $1.8 \times 10^{-5}$ | $2.4 \times 10^{-5}$ | $1.7 \times 10^{-5}$ | $2.2 \times 10^{-5}$ | $2.0 \times 10^{-5}$ |
| 20,000                                                                        | $8.5 \times 10^{-6}$ | $5.4 \times 10^{-6}$ | $7.9 \times 10^{-6}$ | $6.0 \times 10^{-6}$ | $5.5 \times 10^{-6}$ | $4.4 \times 10^{-6}$ |
| 40,000                                                                        | $8.2 \times 10^{-7}$ | $1.5 \times 10^{-6}$ | $7.1 \times 10^{-7}$ | $3.6 \times 10^{-6}$ | $1.0 \times 10^{-6}$ | $7.4 \times 10^{-7}$ |
| fit to 20,000                                                                 | 4.02                 | 4.65                 | 4.11                 | 4.53                 | 4.38                 | 4.76                 |
| fit to 40,000                                                                 | 4.45                 | 4.36                 | 4.60                 | 3.79                 | 4.44                 | 4.81                 |
| <i>the warp on every row</i>                                                  |                      |                      |                      |                      |                      |                      |
| 2500                                                                          | $8.2 \times 10^{-4}$ | $7.1 \times 10^{-4}$ | $5.1 \times 10^{-4}$ | $5.6 \times 10^{-4}$ | $4.9 \times 10^{-4}$ | $5.6 \times 10^{-4}$ |
| 5000                                                                          | $2.0 \times 10^{-4}$ | $3.8 \times 10^{-4}$ | $1.1 \times 10^{-4}$ | $1.2 \times 10^{-4}$ | $1.1 \times 10^{-4}$ | $1.2 \times 10^{-4}$ |
| 10,000                                                                        | $2.1 \times 10^{-5}$ | $1.9 \times 10^{-5}$ | $2.6 \times 10^{-5}$ | $2.0 \times 10^{-5}$ | $2.2 \times 10^{-5}$ | $2.0 \times 10^{-5}$ |
| 20,000                                                                        | $6.9 \times 10^{-6}$ | $4.6 \times 10^{-6}$ | $6.2 \times 10^{-5}$ | $4.9 \times 10^{-5}$ | $5.5 \times 10^{-6}$ | $4.4 \times 10^{-6}$ |
| 40,000                                                                        | $1.3 \times 10^{-6}$ | $1.5 \times 10^{-6}$ | $7.0 \times 10^{-7}$ | $3.7 \times 10^{-6}$ | $1.0 \times 10^{-6}$ | $7.4 \times 10^{-7}$ |
| <i>the rule over the better of warp and plain Gaussians</i>                   |                      |                      |                      |                      |                      |                      |
| 2500                                                                          | 1.00                 | 0.999                | –                    | –                    | –                    | –                    |
| 5000                                                                          | 0.912                | 1.04                 | –                    | –                    | –                    | –                    |
| 10,000                                                                        | 0.982                | 0.973                | 1.00                 | 0.965                | –                    | –                    |
| 20,000                                                                        | 1.25                 | 1.19                 | 0.993                | 0.975                | –                    | –                    |
| 40,000                                                                        | 0.938                | 1.00                 | 1.02                 | 0.986                | –                    | –                    |
| <i>the others at <math>N = 40,000</math></i>                                  |                      |                      |                      |                      |                      |                      |
| constr.                                                                       | $6.7 \times 10^{-4}$ | $7.0 \times 10^{-4}$ | $4.1 \times 10^{-4}$ | $4.7 \times 10^{-4}$ | $1.3 \times 10^{-4}$ | $1.6 \times 10^{-4}$ |
| naive                                                                         | $6.8 \times 10^{-4}$ | $1.2 \times 10^{-4}$ | $1.5 \times 10^{-3}$ | $1.7 \times 10^{-3}$ | $5.1 \times 10^{-3}$ | $5.2 \times 10^{-3}$ |
| direct                                                                        | $2.0 \times 10^{-3}$ | $2.1 \times 10^{-3}$ | $5.2 \times 10^{-3}$ | $5.0 \times 10^{-3}$ | $5.1 \times 10^{-3}$ | $5.2 \times 10^{-3}$ |
| vs. plain-built                                                               | –                    | –                    | –                    | $8.0 \times 10^{-7}$ | –                    | –                    |
| its fit                                                                       | –                    | –                    | –                    | 4.77                 | –                    | –                    |
| fine runs at $s = 10^{11}$ , $\delta = 0.001$ differ by $3.55 \times 10^{-6}$ |                      |                      |                      |                      |                      |                      |

The comparison is against coefficient sampling on the same nodes and against our implementation of the cell harmonic and arithmetic means, a widened edge and, in one dimension, the conservative scheme with exact face conductances, on one contrast and one test problem per dimension. On cases 1 and 2 (Figures 15 and 16, Tables 20 and 21) no treatment improves on sampling by more than 1.75 times in the RMS norm anywhere in either sweep, the parabolic problem on case 1's 1250 nodes aside, whose growing mode every treatment carries (Section 6.2). In the maximum norm the

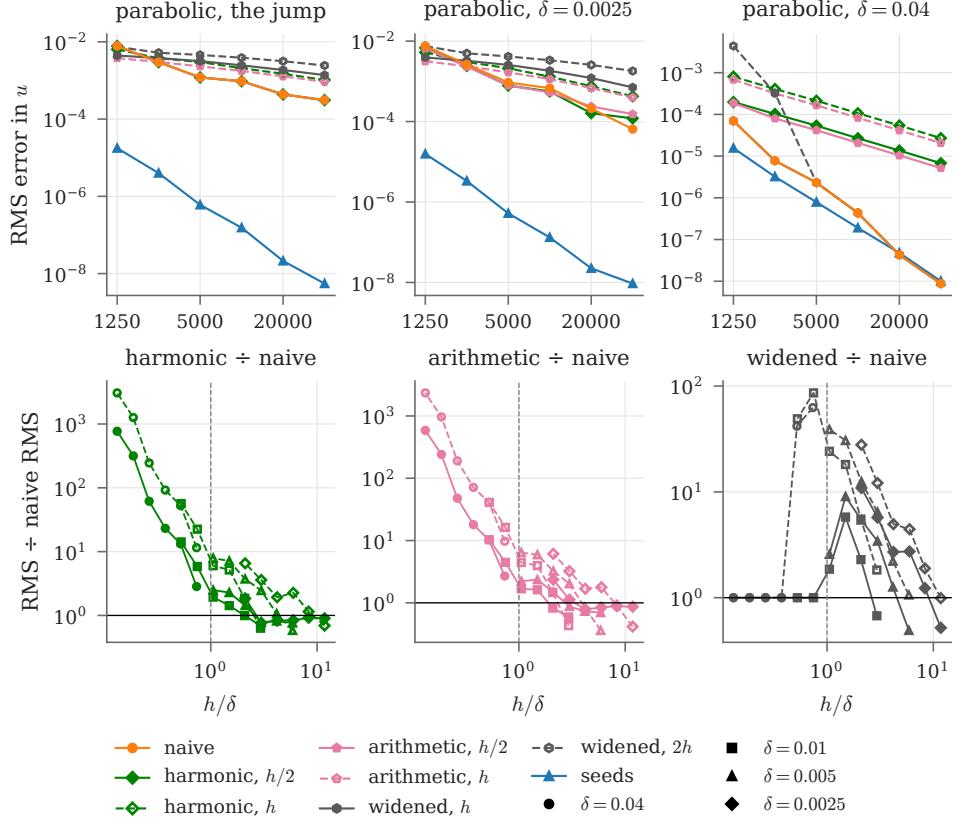


FIGURE 15. The coefficient treatments on case 1, parabolic problem, RMS error. Top: every line against the node count at the jump and at  $\delta = 0.0025$  and  $0.04$ : the naive operator (orange); the harmonic (green) and arithmetic (pink) means of  $\alpha$  over discs of radius  $h/2$  (solid) and  $h$  (dashed, open markers); the edge widened to  $\max(\delta, h)$  and  $\max(\delta, 2h)$  (dark grey, the second dashed); and the seeds (blue). Bottom: each family over the naive operator against  $h/\delta$ , a marker per width.

coarsest node sets differ more, by up to 3.1 times on case 1 at 1250 nodes and 2.4 on case 2 at 2500, where the sampled line is itself an outlier; from 5000 nodes on no treatment improves on sampling by more than 1.60 in the RMS norm or 1.55 in the maximum norm. On case 1's parabolic problem at the jump the seeds lie 2.3 orders below the best of the treatments at 1250 nodes and 4.7 orders below at 40,000.

*Where they help and where they hurt.* A disc of half a spacing helps a little while the edge is two to three spacings narrower than the grid, and hurts from there on: the half-spacing means cross sampling at  $h/\delta = 1.8$  to

TABLE 20. The treatments at 40,000 nodes, RMS error at equilibrium: the naive operator’s error, each treatment’s over it (H, A: the harmonic and arithmetic means of  $\alpha$  over a disc of radius  $h/2$  or  $h$ ; W: the edge widened to  $\max(\delta, mh)$ ), and the seeds’ error (the tangential chain on case 2).

| $\delta$                                            | $h/\delta$ | treatment RMS $\div$ naive RMS |         |       |         |       |       |        |                       | seeds |
|-----------------------------------------------------|------------|--------------------------------|---------|-------|---------|-------|-------|--------|-----------------------|-------|
|                                                     |            | naive                          | H $h/2$ | H $h$ | A $h/2$ | A $h$ | W $h$ | W $2h$ |                       |       |
| <i>case 1, equilibrium, <math>N = 40,000</math></i> |            |                                |         |       |         |       |       |        |                       |       |
| jump                                                | –          | $3.44 \times 10^{-4}$          | 1.00    | 2.90  | 1.00    | 2.42  | 3.60  | 6.37   | $5.28 \times 10^{-9}$ |       |
| 0.04                                                | 0.13       | $8.22 \times 10^{-9}$          | 775     | 3099  | 584     | 2333  | 1.00  | 1.00   | $8.78 \times 10^{-9}$ |       |
| 0.01                                                | 0.53       | $2.45 \times 10^{-6}$          | 12.1    | 48.1  | 8.15    | 32.3  | 1.00  | 38.8   | $8.42 \times 10^{-9}$ |       |
| 0.005                                               | 1.05       | $2.89 \times 10^{-5}$          | 2.58    | 8.22  | 2.16    | 6.19  | 2.49  | 37.5   | $8.82 \times 10^{-9}$ |       |
| 0.0025                                              | 2.11       | $8.36 \times 10^{-5}$          | 1.48    | 4.94  | 1.69    | 4.28  | 7.60  | 19.4   | $7.99 \times 10^{-9}$ |       |
| <i>case 2, equilibrium, <math>N = 40,000</math></i> |            |                                |         |       |         |       |       |        |                       |       |
| jump                                                | –          | $6.26 \times 10^{-4}$          | 1.00    | 1.88  | 1.00    | 1.76  | 2.80  | 5.15   | $7.18 \times 10^{-9}$ |       |
| 0.01                                                | 0.53       | $5.23 \times 10^{-6}$          | 7.03    | 26.5  | 5.30    | 21.6  | 1.00  | 27.3   | $3.01 \times 10^{-8}$ |       |
| 0.005                                               | 1.05       | $5.99 \times 10^{-5}$          | 1.67    | 4.68  | 1.26    | 3.83  | 1.46  | 26.9   | $3.00 \times 10^{-8}$ |       |
| 0.0025                                              | 2.11       | $9.80 \times 10^{-5}$          | 1.19    | 4.99  | 1.79    | 4.68  | 9.12  | 24.5   | $2.69 \times 10^{-8}$ |       |

3.5, at every width and on both cases and problems, as in one dimension they help only while  $h \gtrsim \delta$  (Section 4.5). At the jump the half-spacing disc is sampling by construction on these node sets, whose innermost rows lie exactly  $h/2$  from the curve, so “worse than sampling at the jump” is a statement about the radius- $h$  means and T0, not about averaging  $\alpha$  as such: from 5000 nodes on the radius- $h$  harmonic mean is 1.6 to 3.4 times worse than sampling there, the arithmetic one 1.3 to 3.0 times, and T0 1.7 to 8.2 times. Once the grid resolves the edge the disc means cap the naive operator at second order: at  $\delta = 0.04$  they fit 1.95 to 2.05 and lie 584 to 3100 times above it at 40,000 nodes, the radius- $h$  line 4.00 times the half-spacing one, which is the  $r^2/8$  term of a disc mean’s expansion. T0 there is the naive operator itself wherever  $mh \leq \delta$ ; while the edge is unresolved it solves the widened problem, and its error is that problem’s distance from the true one, 0.996 to 1.000 of it for  $m = 2$  on case 1. At the edge nothing converges: at the jump on 40,000 nodes of case 1, the flux error on the first rows off the edge is 0.30 to 0.59 under the treatments that change  $\alpha$  there, against sampling’s 0.34.

**6.8. A snapshot.** Figure 17 and Table 22 put the four operators on one node set, as Section 4.6 did in one dimension: case 1 at 2500 nodes and  $\delta = 0.0025$ , so that  $h = 8.3\delta$ , for the parabolic problem at  $t = 0.1$ . It is the count at which the lines are ordered and apart: the naive operator at  $2.63 \times 10^{-3}$ , 3.8 times the construction’s  $6.97 \times 10^{-4}$ , the direct operator at  $3.71 \times 10^{-2}$ , and the seeds at  $3.33 \times 10^{-6}$ , 210 times below the construction

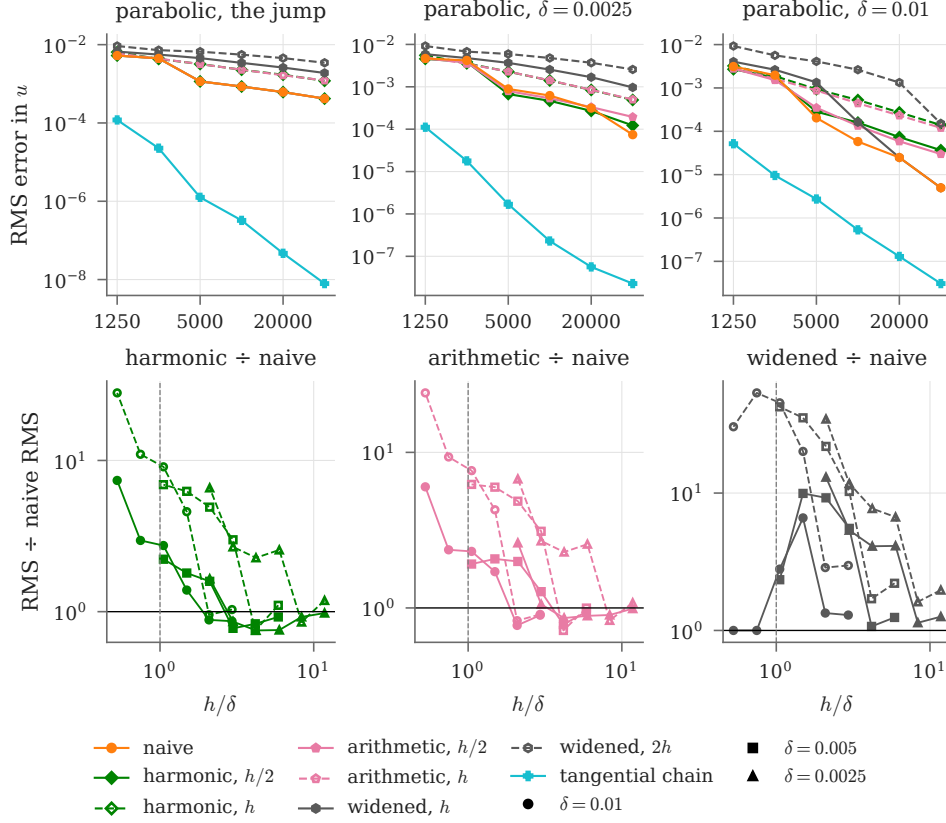


FIGURE 16. The coefficient treatments on case 2, in the layout of Figure 15, at the jump and at  $\delta = 0.0025$  and  $0.01$ , with the tangential chain (cyan) as the seeds.

and 790 times below the naive operator. At 5000 nodes the naive operator and the construction are 1.3 times apart, and at 10,000 they cross.

The naive error is the band's, not the edge's. It is speckled over the whole strip on the scale of the node spacing and largest in the middle of the band, at  $y = 0.710$ . A quarter of its energy  $\|e\|_2^2$  lies within  $2h$  of a curve, where 15% of the nodes are, so that it leans on the edges only 1.6 times and three quarters of it lie elsewhere: the flux defect at the edge of Section 6.2, spread by the operator's reach to the neighbours of the neighbours into every row the band's rows couple to. In one dimension too, 84% of the naive error's energy lay away from the edge (Section 4.6). The construction's error is a smooth shift with the shape  $\sin 2\pi x$  of the solution, largest beside the upper curve, with 65% of its energy within  $2h$  of a curve, four times the nodes' share: it is the floor, at 0.98 of it. The direct operator, blind to the edge, is the worst line, 14 times the naive operator. The seeds' error does not see the edge: 12% of its energy lies within  $2h$  of a curve, less than the nodes'

TABLE 21. The best treatment against sampling at each count: the largest ratio of the naive operator’s error to a treatment’s, over every width and all six treatments (above 1, some treatment beats sampling), for both cases and problems, in the RMS and maximum norms; case 1’s parabolic entry at 1250 nodes carries the naive operator’s growing mode. Last column:  $\log_{10}$  of the best treatment’s RMS error over the seeds’, case 1, parabolic problem, at the jump.

| $N$    | case 1 |      |           |           | case 2 |      |           |           | seeds  |
|--------|--------|------|-----------|-----------|--------|------|-----------|-----------|--------|
|        | ell.   | par. | ell., max | par., max | ell.   | par. | ell., max | par., max | orders |
| 1250   | 1.51   | 2.77 | 3.08      | 4.90      | 1.15   | 1.16 | 1.30      | 1.32      | 2.3    |
| 2500   | 1.37   | 1.37 | 1.61      | 1.64      | 1.75   | 1.40 | 2.38      | 2.29      | 2.9    |
| 5000   | 1.42   | 1.27 | 1.14      | 1.13      | 1.33   | 1.32 | 1.52      | 1.54      | 3.3    |
| 10,000 | 1.60   | 1.27 | 1.00      | 1.01      | 1.55   | 1.33 | 1.36      | 1.43      | 3.8    |
| 20,000 | 1.30   | 1.31 | 1.34      | 1.34      | 1.34   | 1.16 | 1.16      | 1.20      | 4.3    |
| 40,000 | 1.00   | 1.00 | 1.00      | 1.00      | 1.00   | 1.00 | 1.00      | 1.00      | 4.7    |

TABLE 22. The snapshot of Figure 17: the RMS error over all nodes, the largest nodal error and where it lies, and the share of  $\|e\|_2^2$  from the nodes within  $2h$  of a curve, where 15% of the nodes lie.

| operator                                              | RMS                   | max $ e $             | at $(x, y)$    | share within $2h$ |
|-------------------------------------------------------|-----------------------|-----------------------|----------------|-------------------|
| naive $D_x A D_x + D_y A D_y$                         | $2.63 \times 10^{-3}$ | $1.82 \times 10^{-2}$ | (0.708, 0.710) | 0.25              |
| $\delta = 0$ construction                             | $6.97 \times 10^{-4}$ | $4.05 \times 10^{-3}$ | (0.750, 0.790) | 0.65              |
| direct $\alpha \nabla^2 + \nabla \alpha \cdot \nabla$ | $3.71 \times 10^{-2}$ | $1.52 \times 10^{-1}$ | (0.250, 0.790) | 0.58              |
| seeds                                                 | $3.33 \times 10^{-6}$ | $2.27 \times 10^{-5}$ | (0.259, 0.980) | 0.12              |

share, and its largest value lies beside the top Dirichlet row, at  $y = 0.980$ , not in the band.

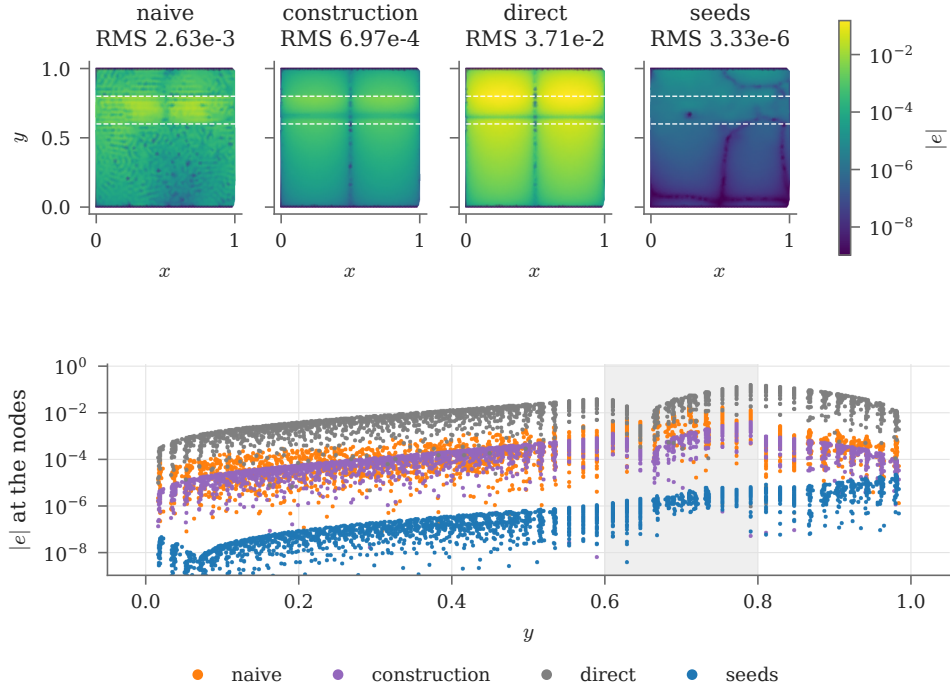


FIGURE 17. Case 1 at 2500 nodes and  $\delta = 0.0025$  ( $h = 8.3\delta$ ), the parabolic problem at  $t = 0.1$ . Top:  $|e|$  over the strip on one colour scale for the naive operator, the  $\delta = 0$  construction, the direct operator and the seeds; dashed, the curves  $y = 0.6$  and  $0.8$ . Bottom:  $|e|$  at every node against  $y$ , the band shaded.

## 7. DISCUSSION

**7.1. Limitations.** *[Stub, #50: corners, extreme contrasts, the cost in three dimensions, what the experiments leave open.]*

**7.2. Future work.** *[Stub, #50: in the order to do it.]*

**7.3. Conclusions.** *[Stub, #50: LITERATURE.md section 6b, verbatim.]*

## ACKNOWLEDGMENTS

*[Stub, #52: acknowledgments; where the code and data live.]*

## TOOL AND COMPUTATIONAL RESOURCE DISCLOSURE

*[Stub, #52.]*

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